R.D. & S. H. National College & S. W.A. Science College

Bandra, Mumbai - 400050.

Department of Computer Science

CERTIFICATE

This is to certify that Mr. **Sujal Sunil Singh** of **TY-BSC-CS** class (5 Semester) has satisfactorily completed **10** Practicals, in the subject of **SOFTWARE TESTING & QUALITY ASSURANCE** as a part of B.Sc. in Computer Science Program during the academic year 2024 / 2025.

Date of Certification:

Subject Incharge **Co-ordinator,**
Department Computer Science

Signature of Examiner

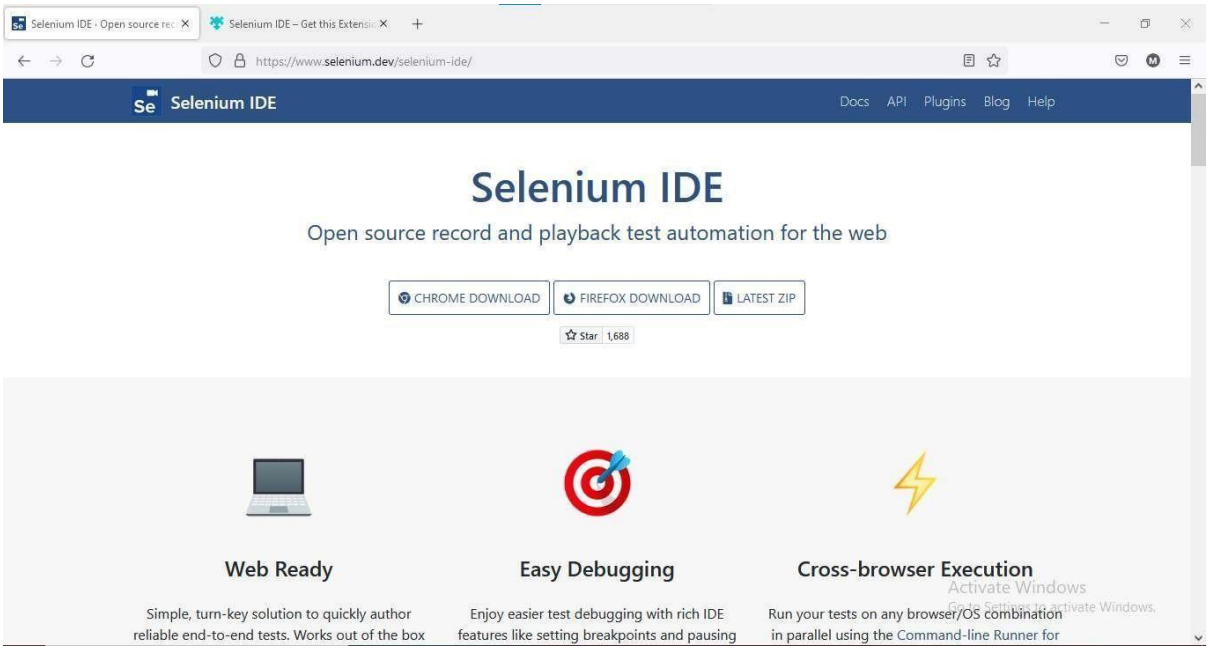
Sr no	Title	Date	Sign
1	Install Selenium IDE and create a test suite containing a minimum of 4 test cases for different web page formats (e.g., HTML, XML, JSON, etc.).		
2	Conduct a test suite for two different websites using Selenium IDE. Perform variousactions like clicking links, filling forms, and verifying content.		
3	Install Selenium Server (Selenium RC) and demonstrate its usage by executing a scriptin Java or PHP to automate browser actions.		
4	Write a program using Selenium WebDriver to automate the login process on a specificweb page. Verify successful login with appropriate assertions.		
5	Write a program using Selenium WebDriver to update 10 student records in an Excel file.Perform data manipulation and verification.		
6	Write a program using Selenium WebDriver to select the number of students who have scored more than 60 in any one subject (or all subjects). Perform data extraction and analysis.		
7	Write a program using Selenium WebDriver to provide the total number of objectspresent or available on a web page. Perform object identification and counting.		
8	Write a program using Selenium WebDriver to get the number of items in a list or combobox on a web page. Perform element identification and counting.		
9	Write a program using Selenium WebDriver to count the number of checkboxes on a web page, including checked and unchecked counts. Perform checkbox identification and counting.		
10	Perform load testing on a web application using JMeter. Generate and analyze load scenarios. Additionally, explore bug tracking using Bugzilla as a tool for logging and tracking software defects.		

NAME: Sujal Singh
ROLL NO: CS22075

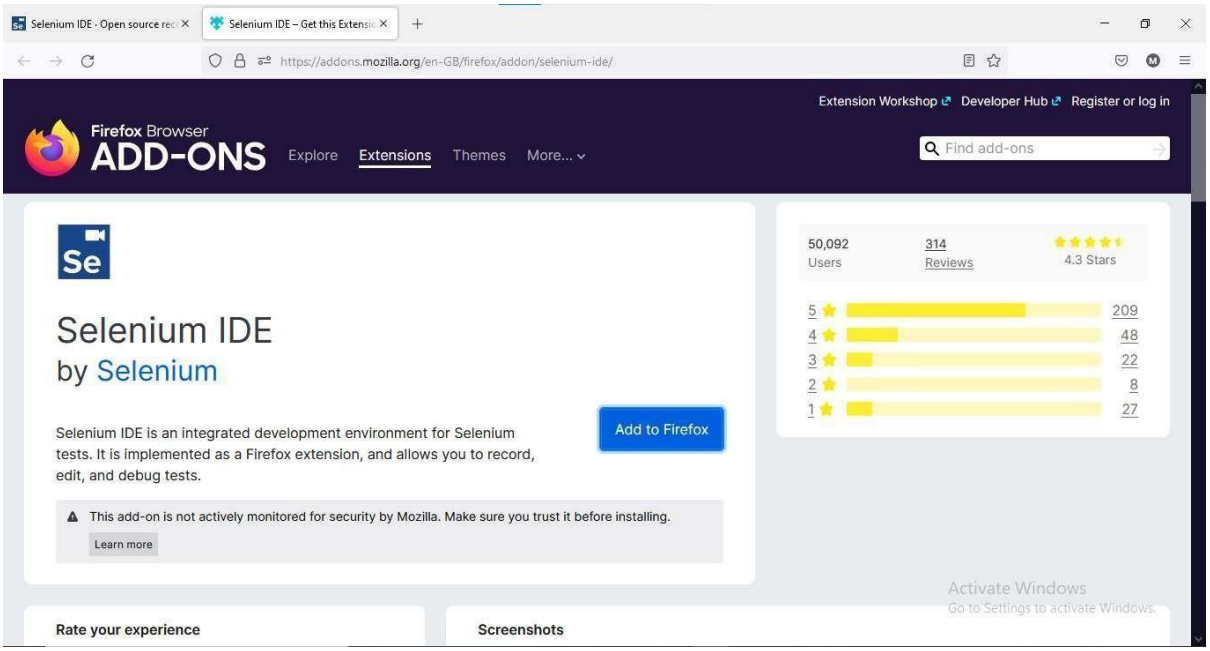
PRACTICAL NO. 1

Aim: Install Selenium IDE; Write a test suite containing minimum 4 test cases for different formats.

Following are the steps for installing Selenium IDE on Mozilla Firefox.

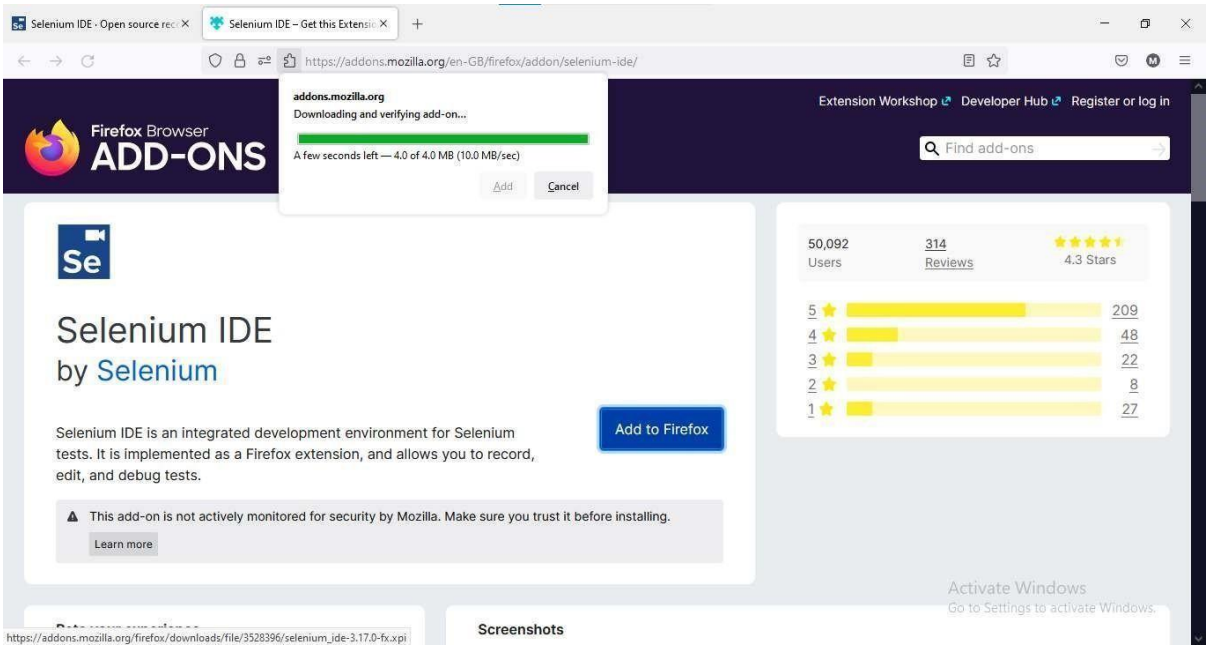


1. Click on ' +Add to Firefox '

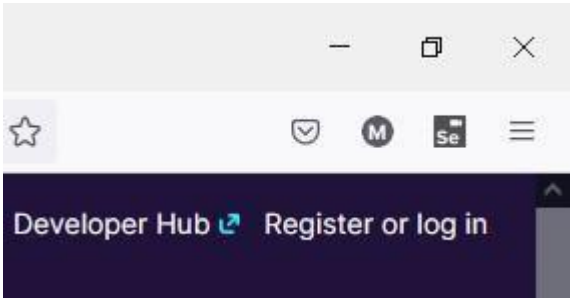


2. Wait for the download to finish

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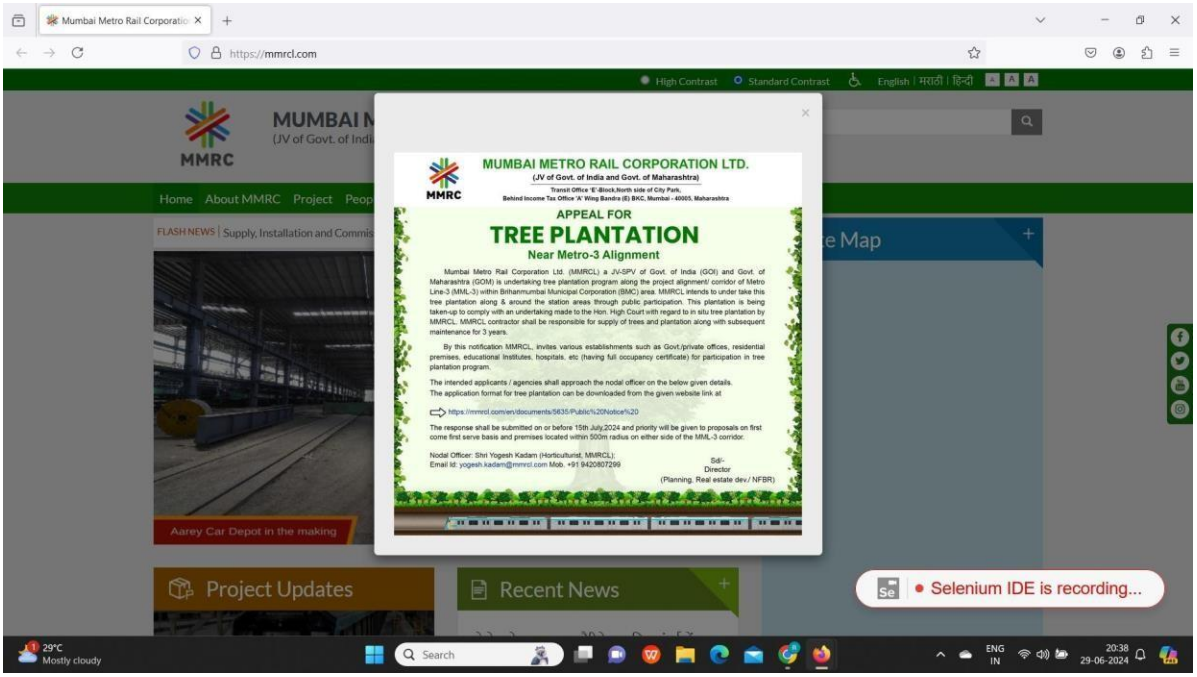
3. Click on the ‘Se’ icon which is on the Firefox toolbar



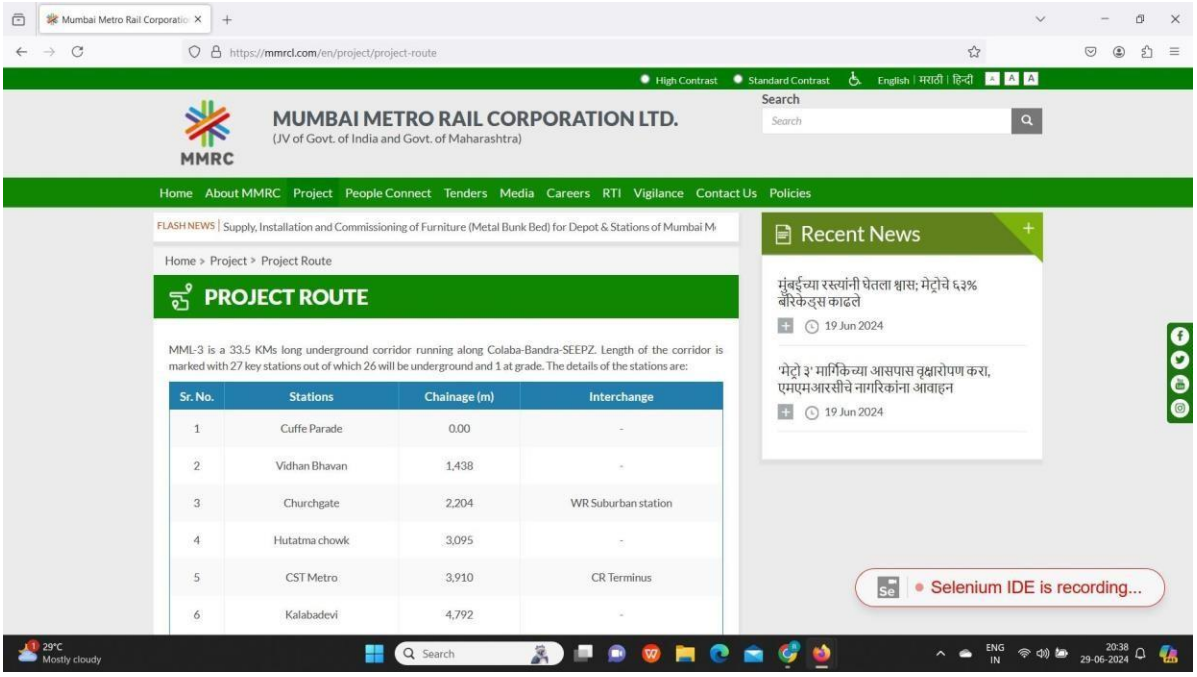
For https://mmrcl.com/ →

Step 1: Open Mozilla Firefox and search for https://mmrcl.com/ Copy and paste the same URL into Selenium IDE, start recording subsequently.

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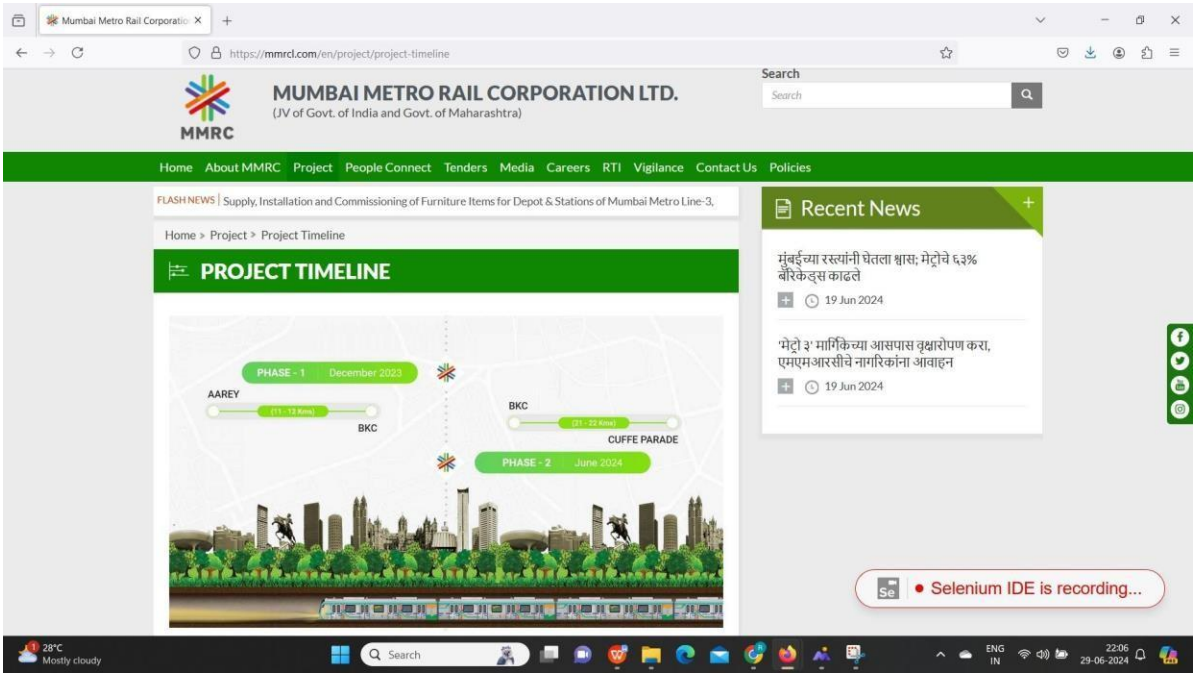
Step 2: Navigate through the mmrcl website by choosing options like project route, project timeline etc.



Sr. No.	Stations	Chainage (m)	Interchange
1	Cuffe Parade	0.00	-
2	Vidhan Bhavan	1,438	-
3	Churchgate	2,204	WR Suburban station
4	Hutatma chowk	3,095	-
5	CST Metro	3,910	CR Terminus
6	Kalabadevi	4,792	-

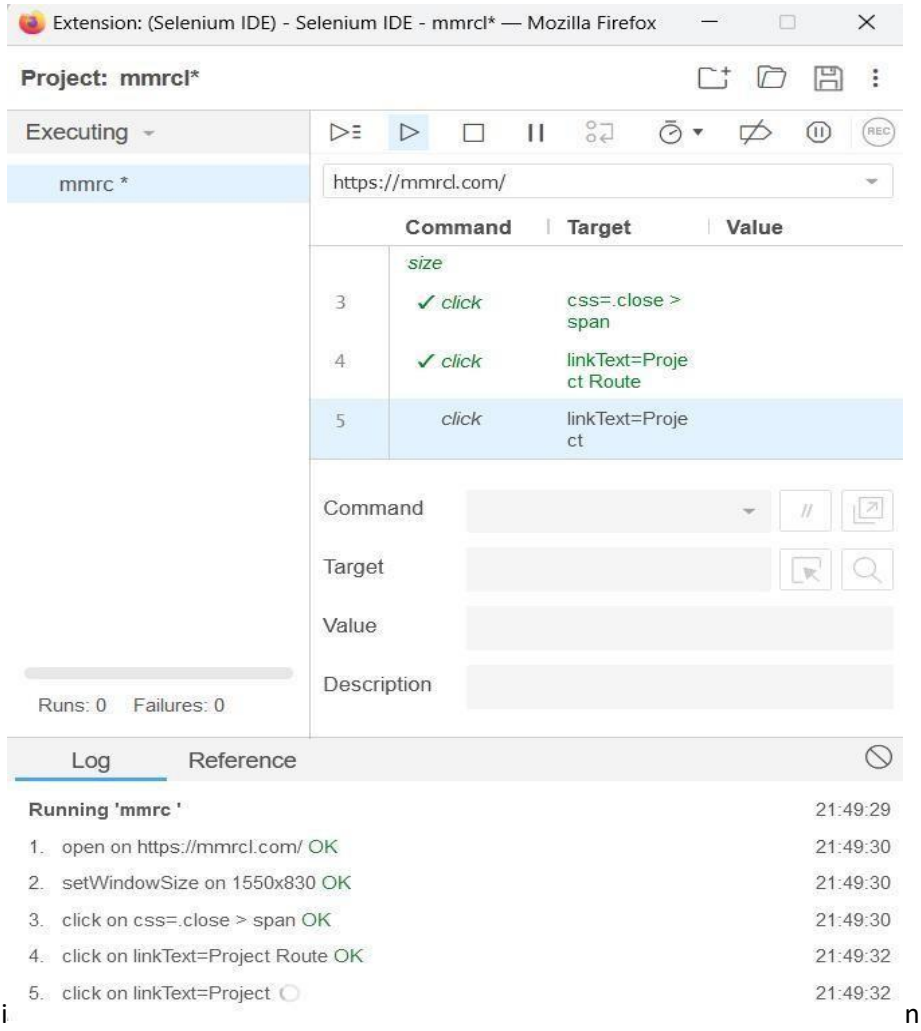
Step 3:

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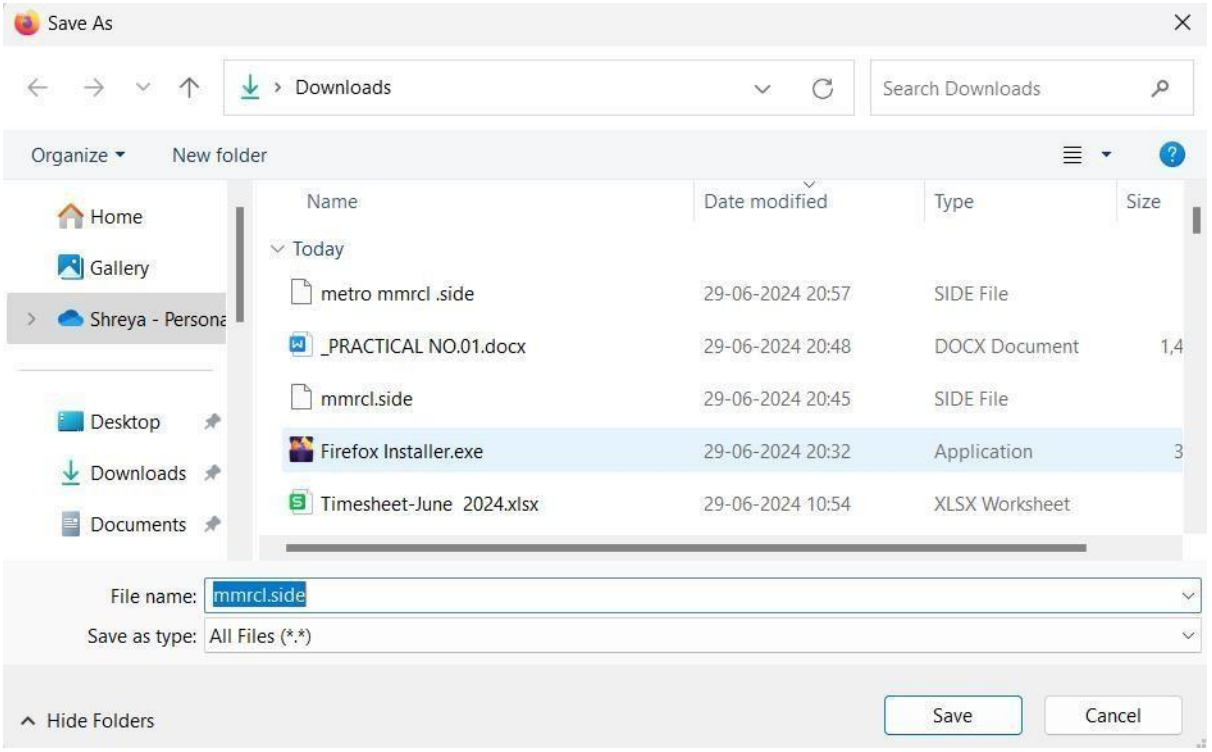
Step 4: Abreast, you will see the commands



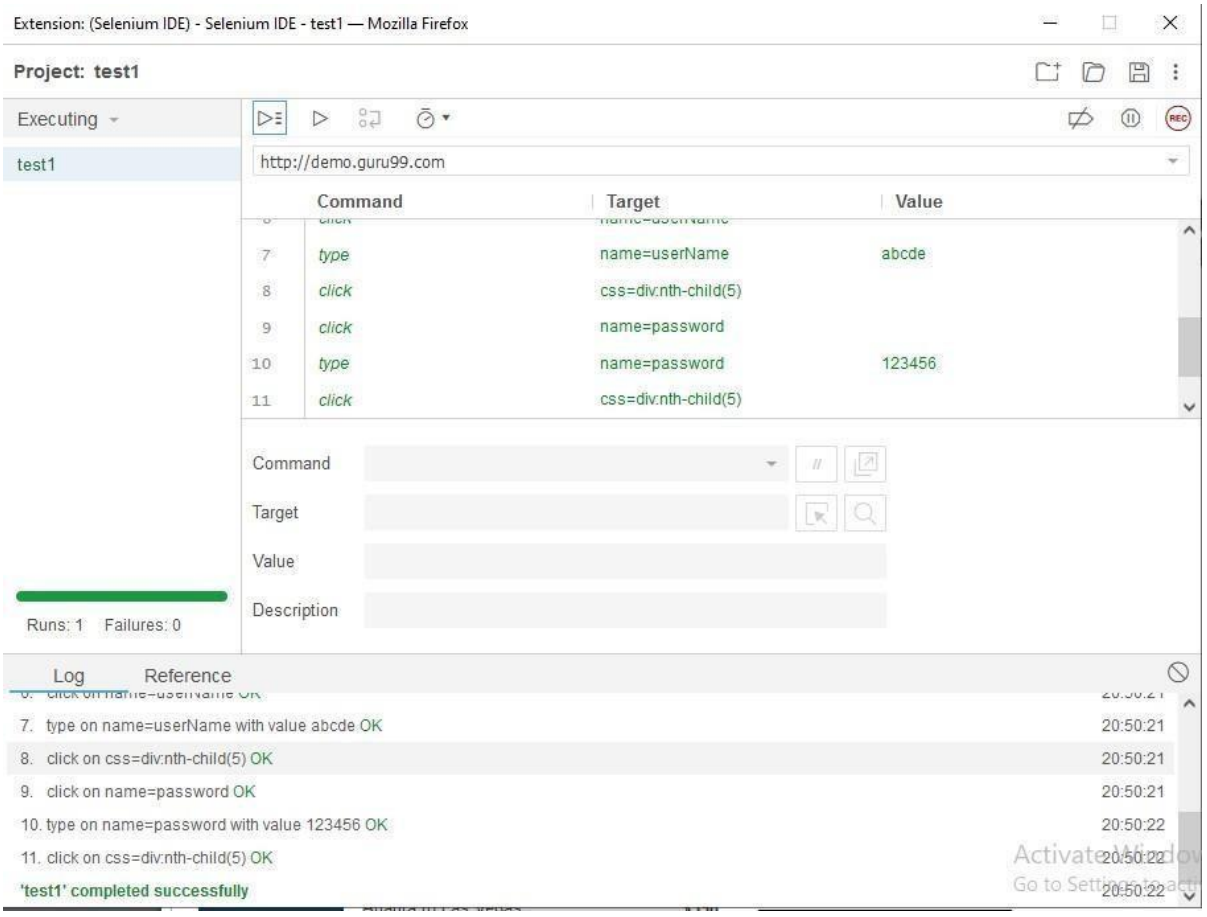
in Selenium IDE which you had performed on website.

Step 5: Save your test case with a name followed by the .side extension.

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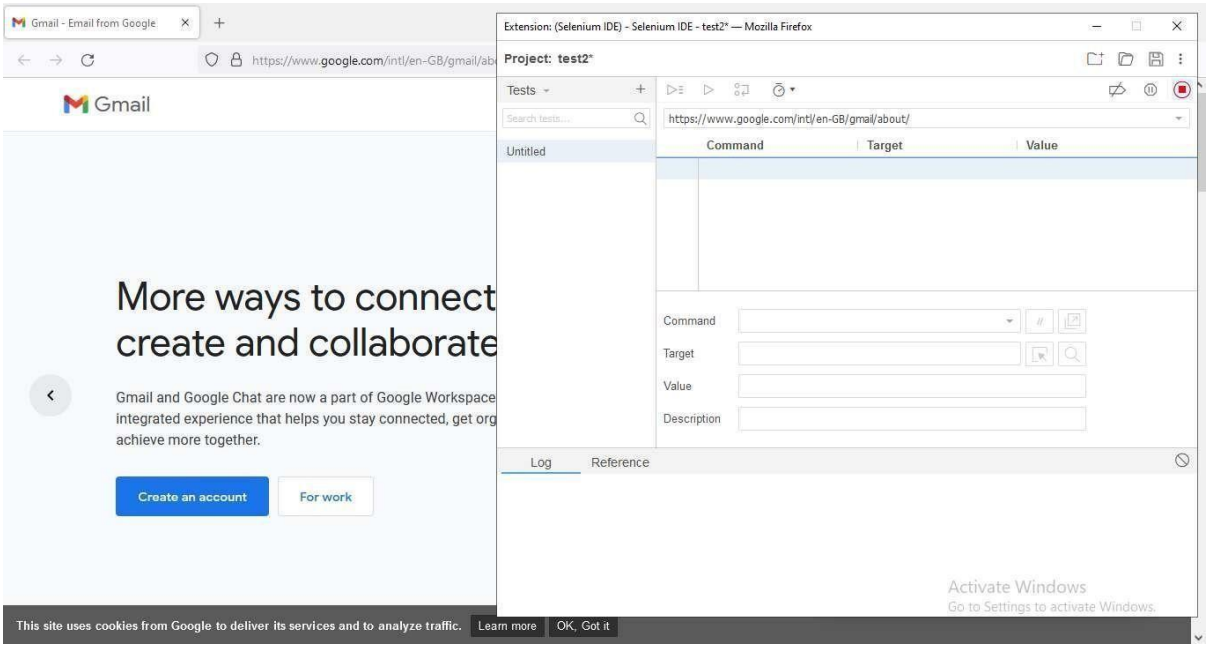


Step 6: Go to the Selenium IDE and click onto Run current test then you will see the 'test case has completed successfully' in the Log section.

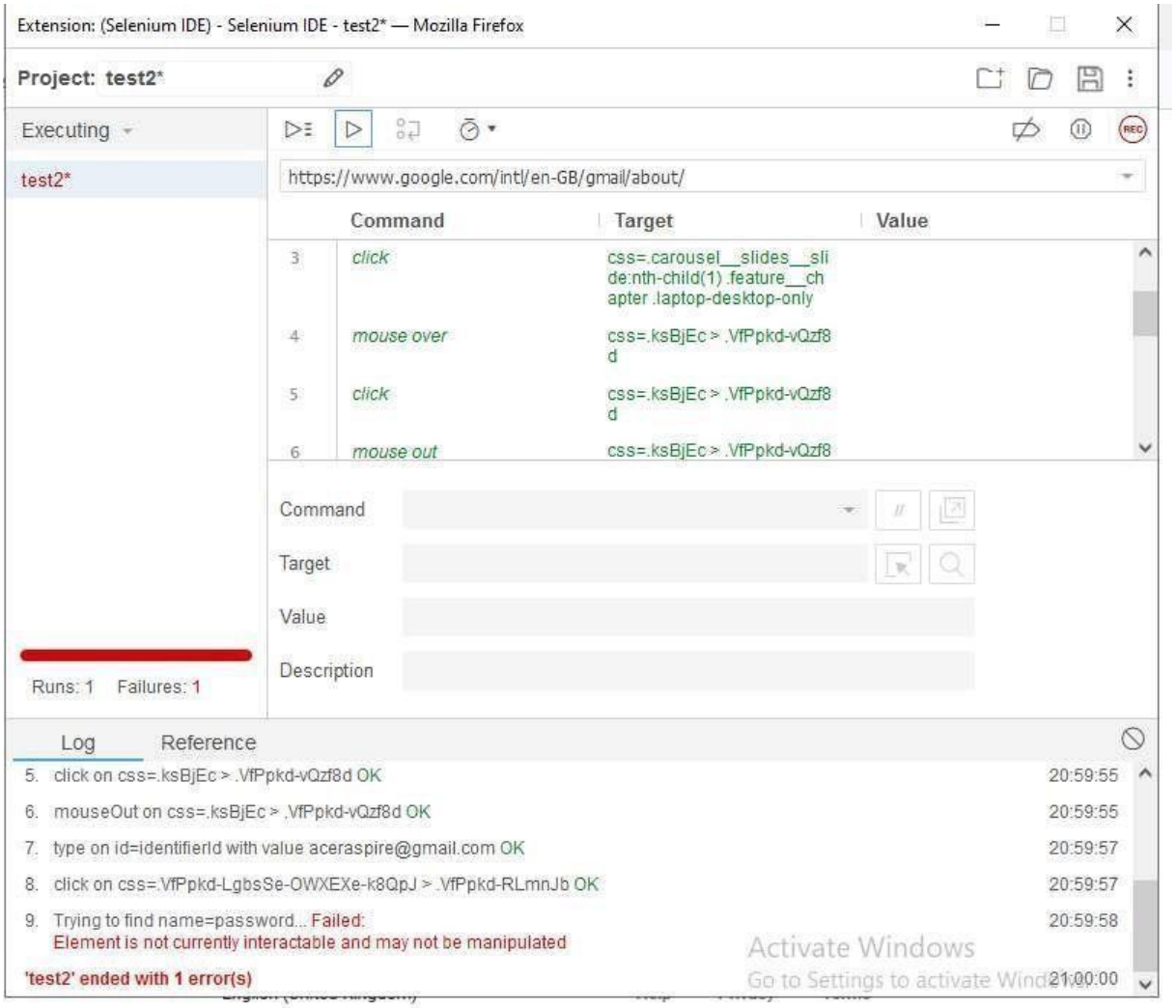


For Gmail→ 1. Go to www.google.com and search for Gmail.

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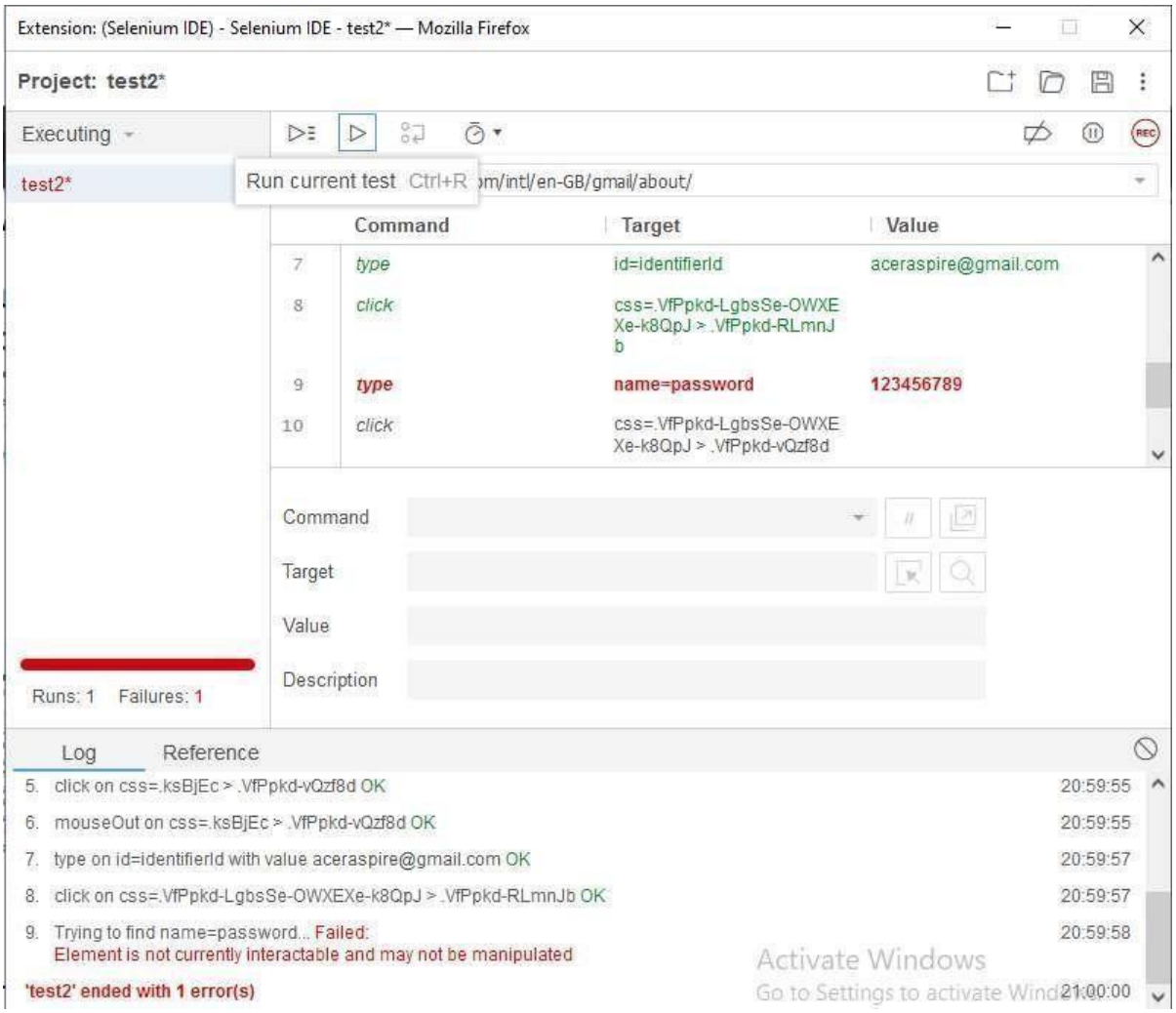


2. Click on Gmail icon



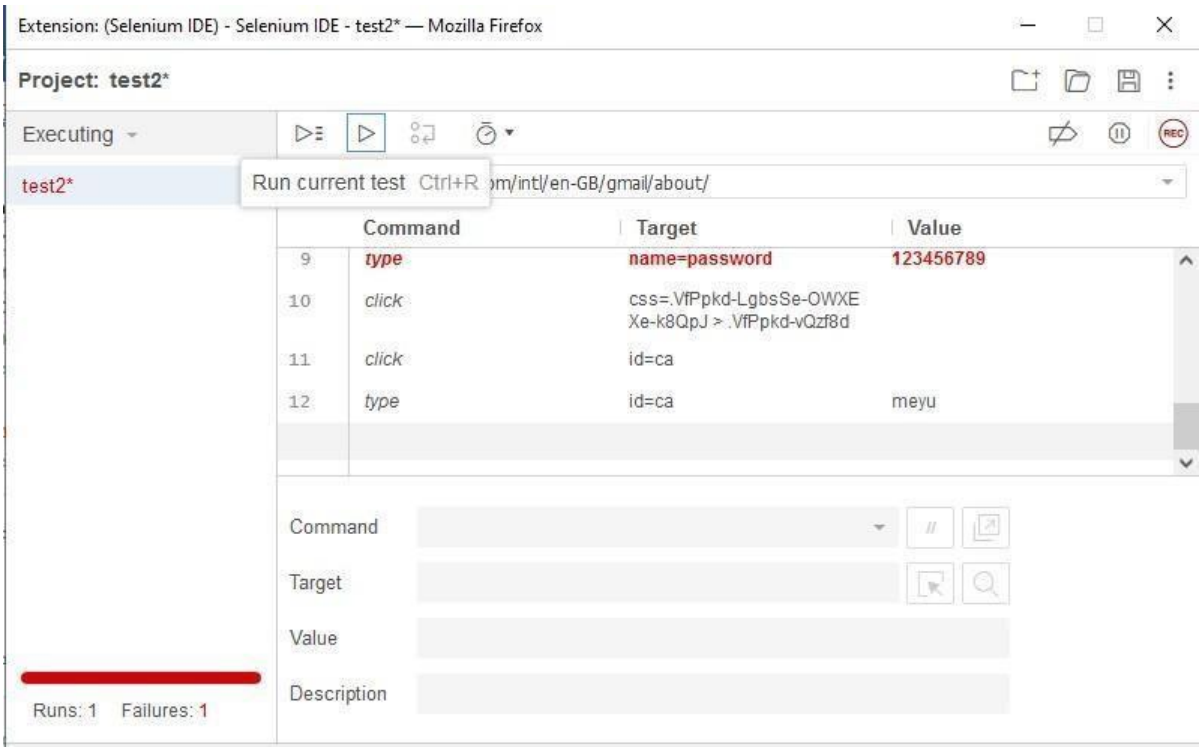
3. Enter some ID and Password

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4. Stop recording and then click on play

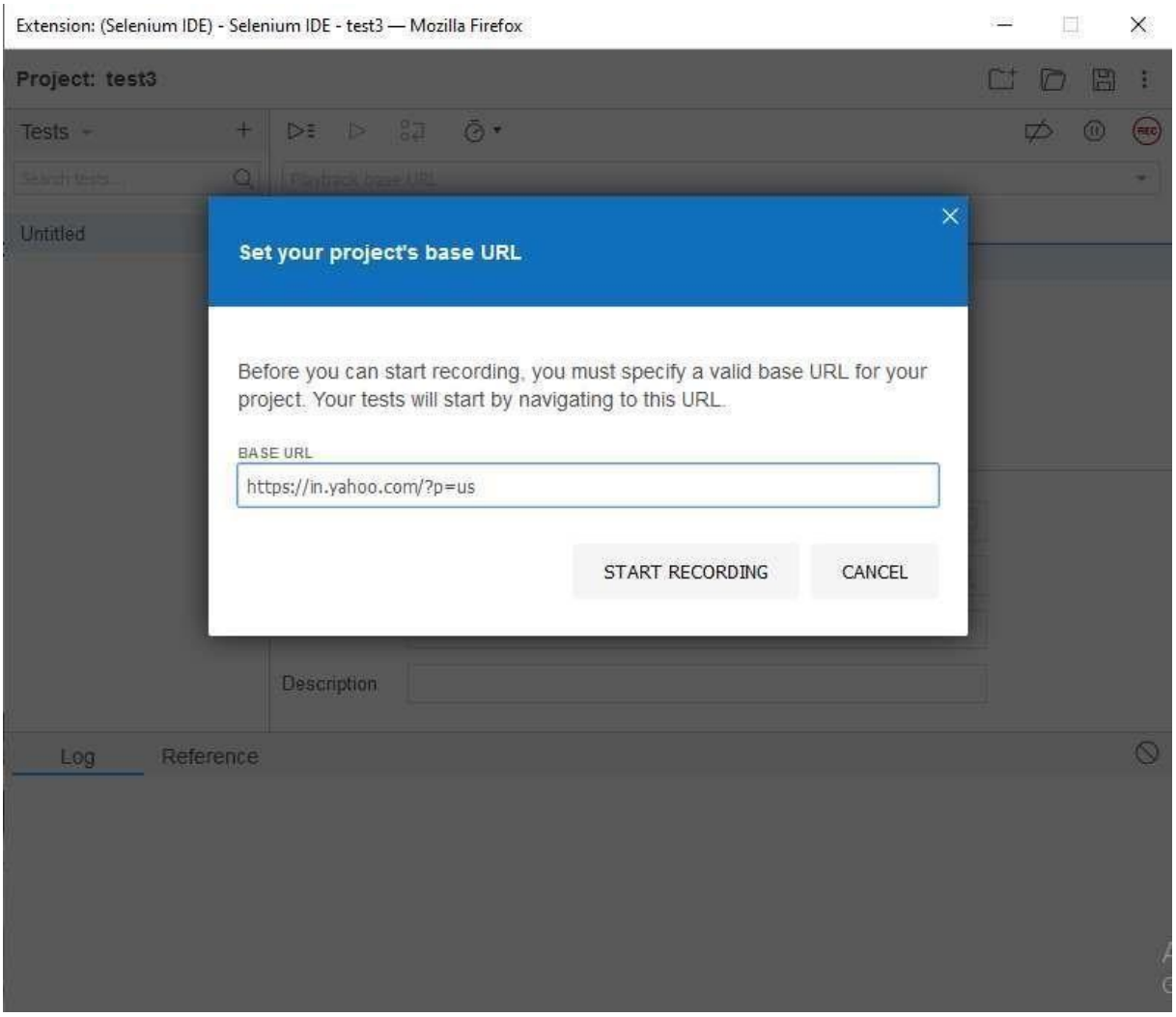
NAME: Sujal Singh
ROLL NO: CS22075



For Yahoo→

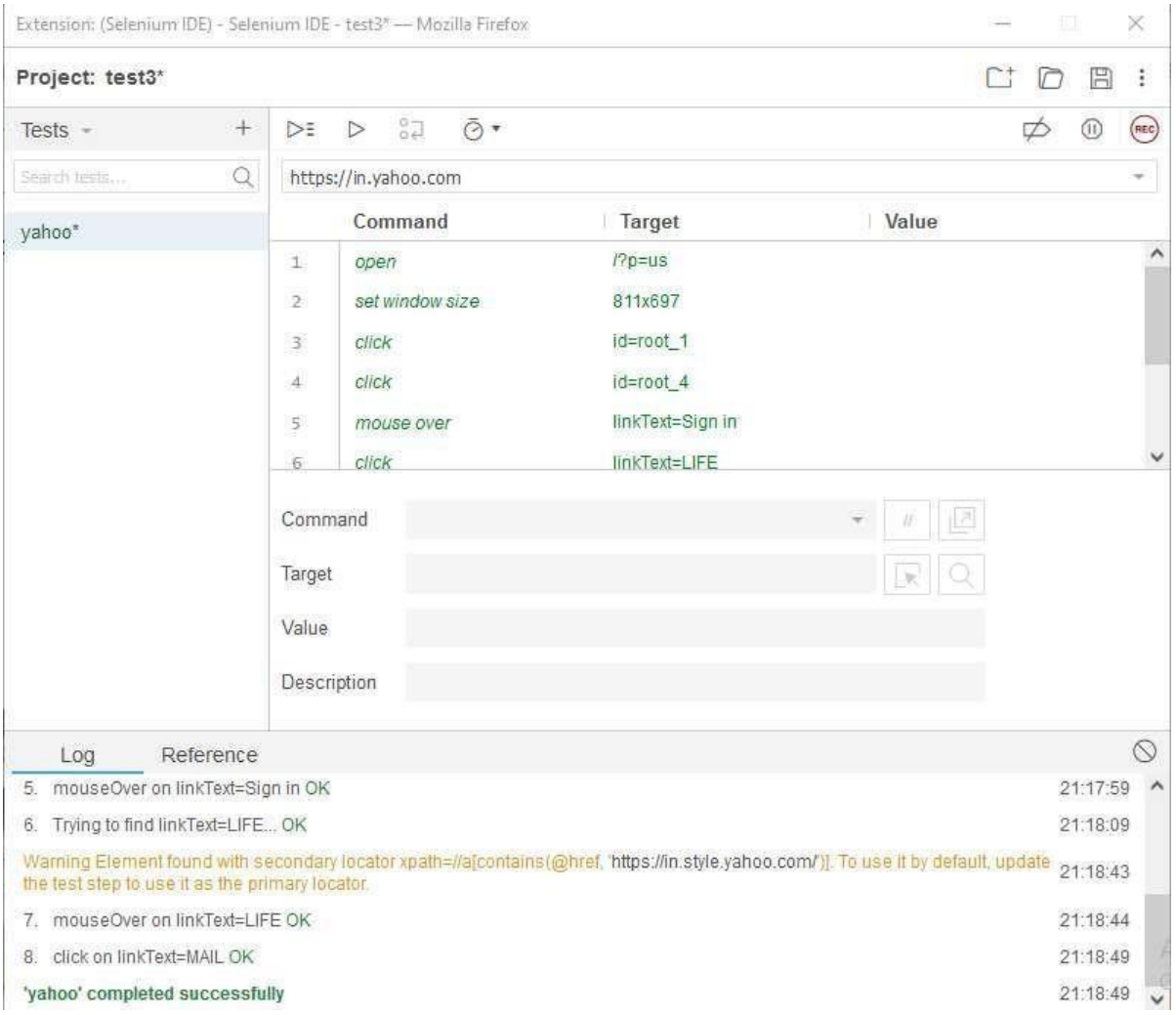
1. Go to www.yahoo.com and start recording.

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2. Perform some actions and then click on play button.

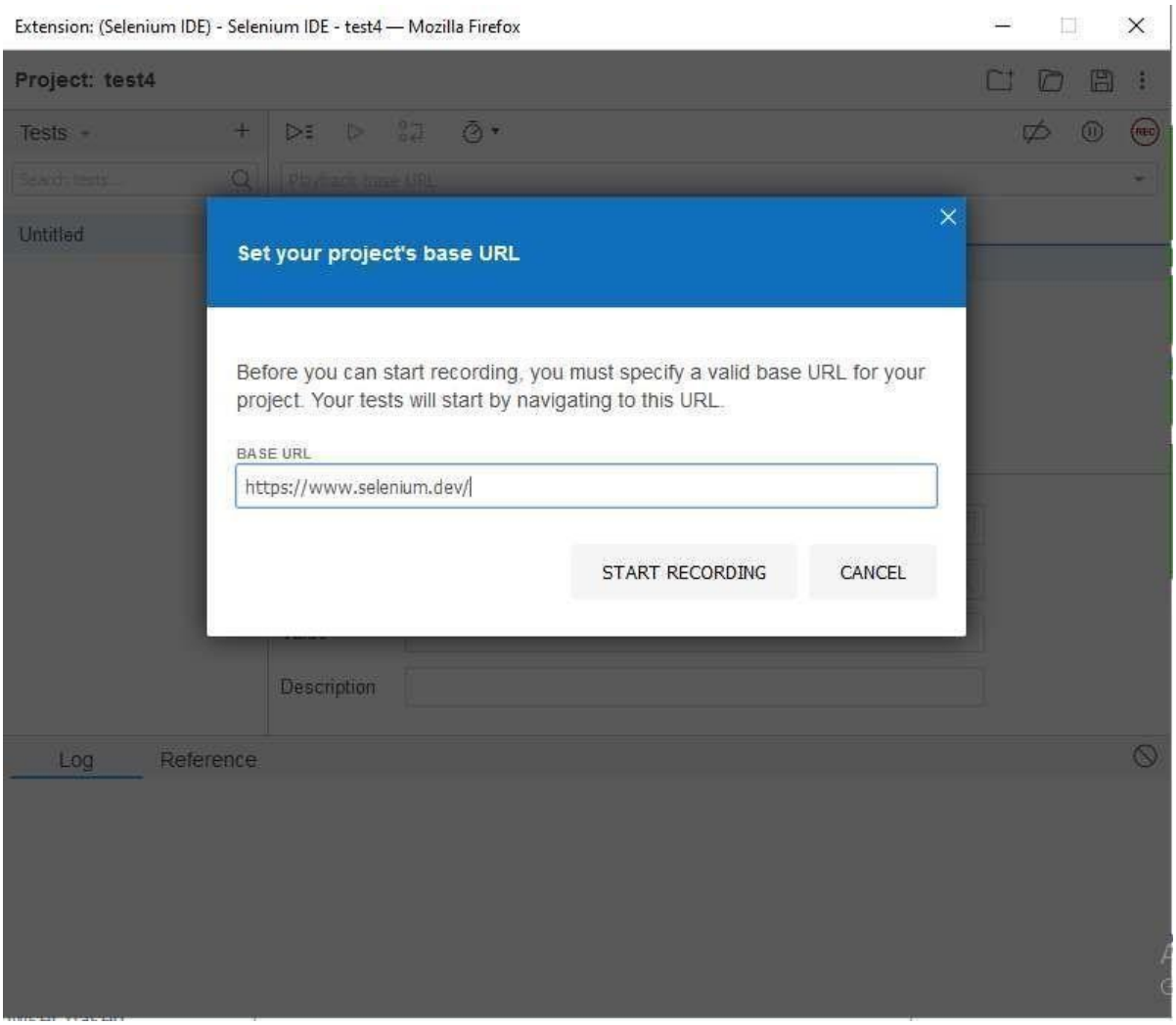
NAME: Sujal Singh
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For Selenium→

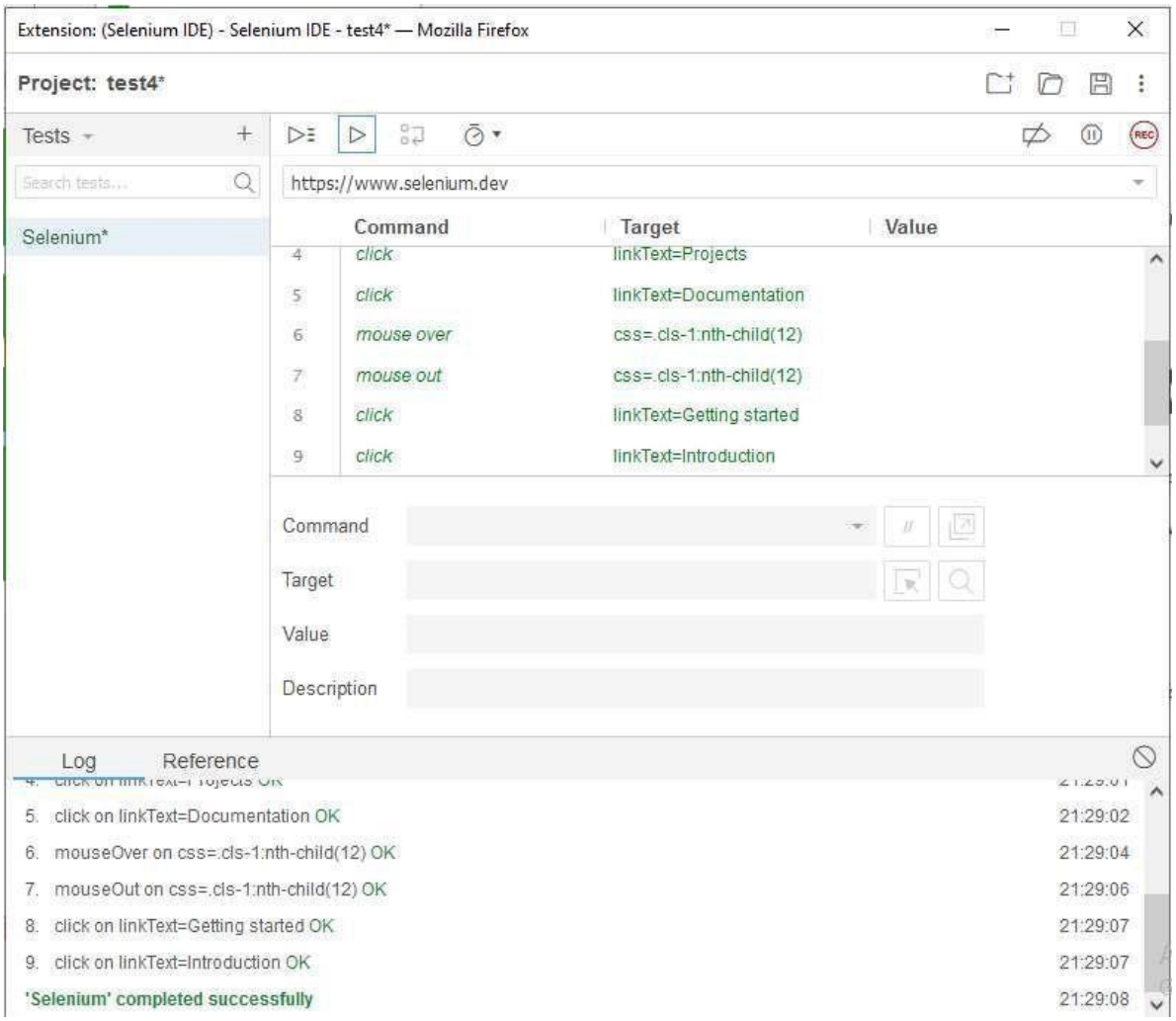
1. Go to https://www.seleniumhq.org and start recording .

NAME: Sujal Singh
ROLL NO: CS22075



2. Perform some actions and then click on play button.

NAME: Sujal Singh
ROLL NO: CS22075



Name: Sujal Singh

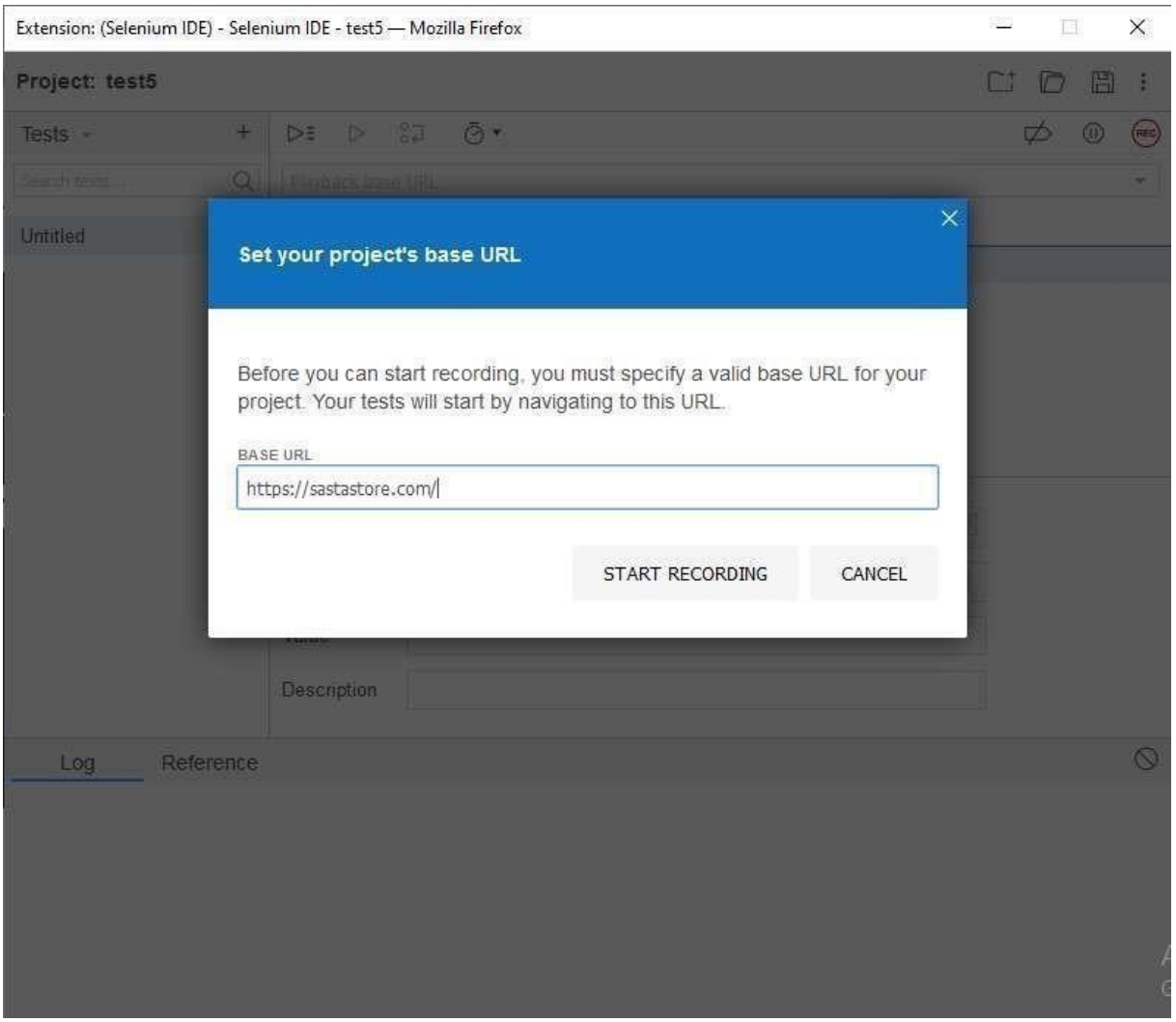
Roll No : CS022075

PRACTICAL NO.2

Aim: **Conduct a test suite for two different websites using Selenium IDE. Perform various actions like clicking links, filling forms, and verifying content.**

A. For www.sastastore.com

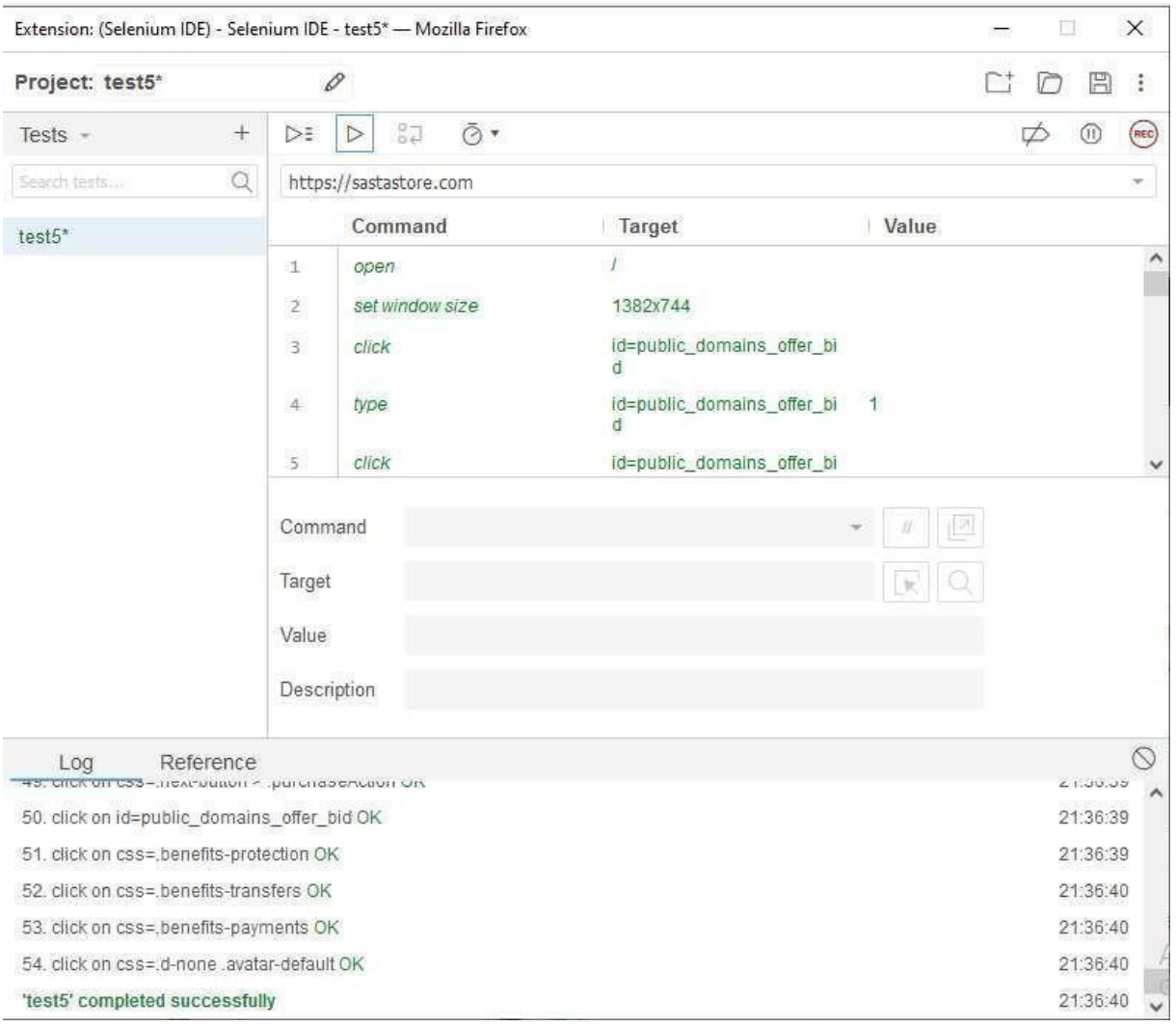
1. Open the website and open selenium IDE. Copy the URL and paste it in the given space for further test. Then save & run the test to see log files.



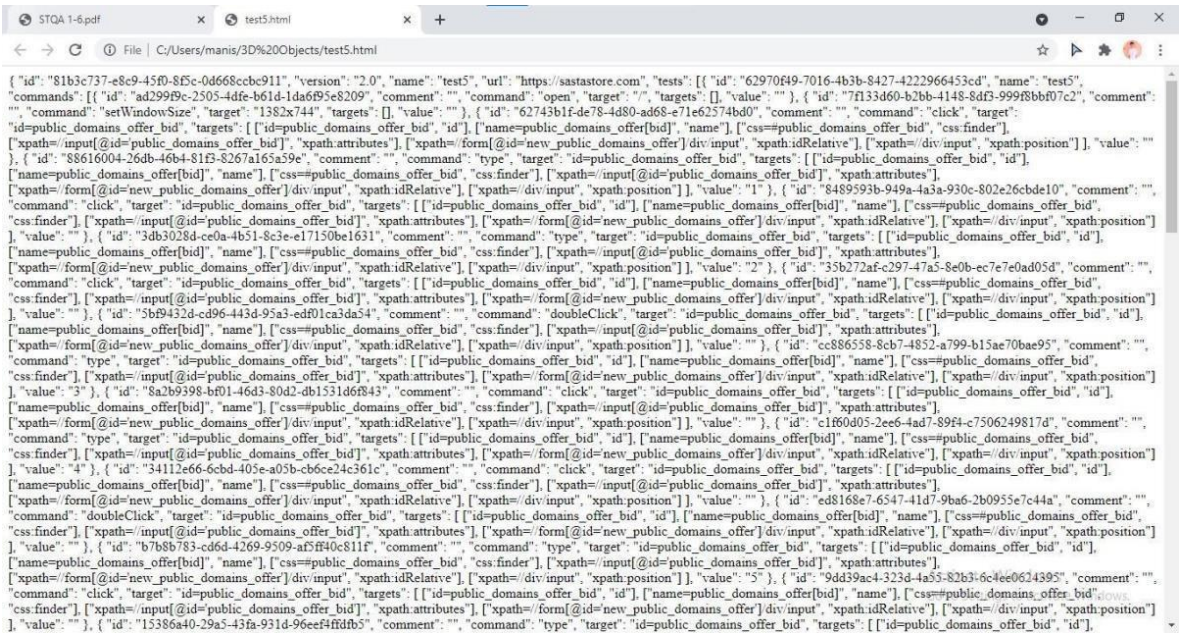
2. After every successful run it gives message “completed successfully” with green text highlighted.

Name: Sujal Singh

Roll No : CS022075



3. Save it in the HTML file and open it on your browser webpage.

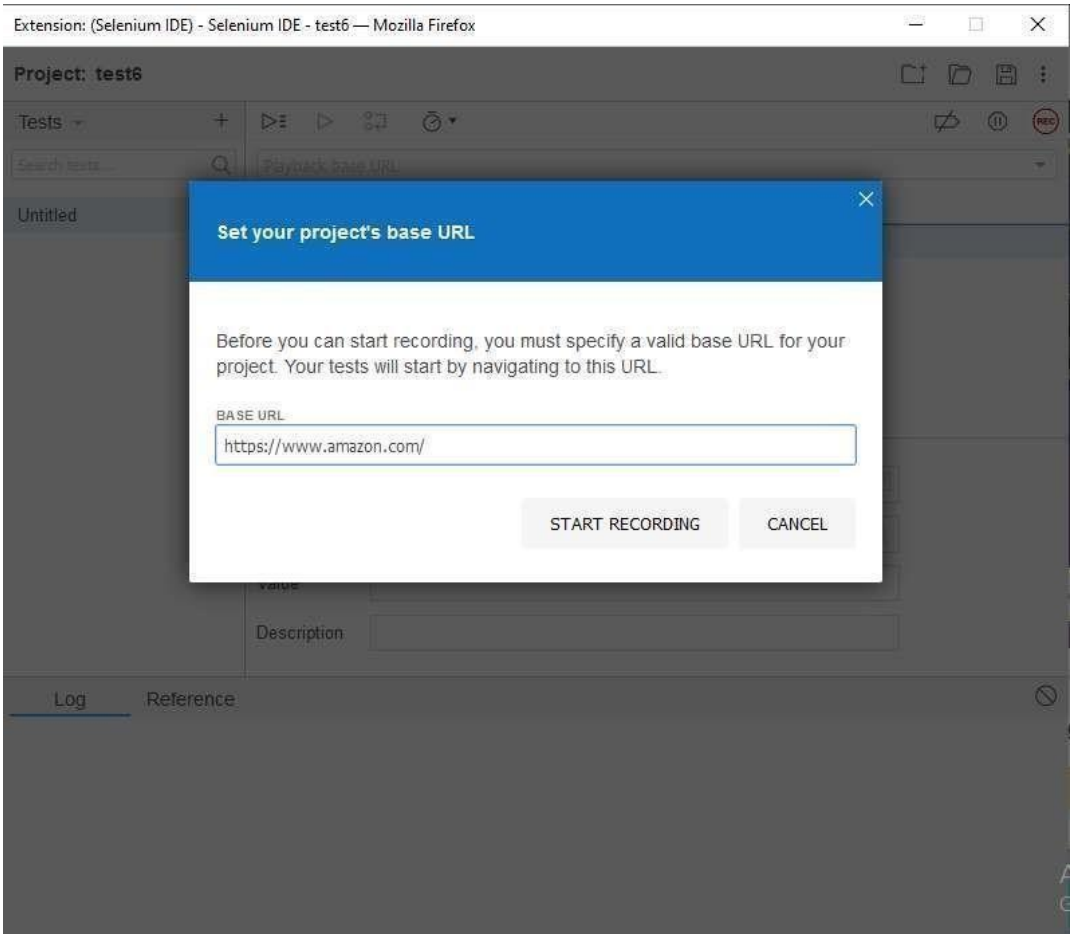


Name: Sujal Singh

Roll No : CS022075

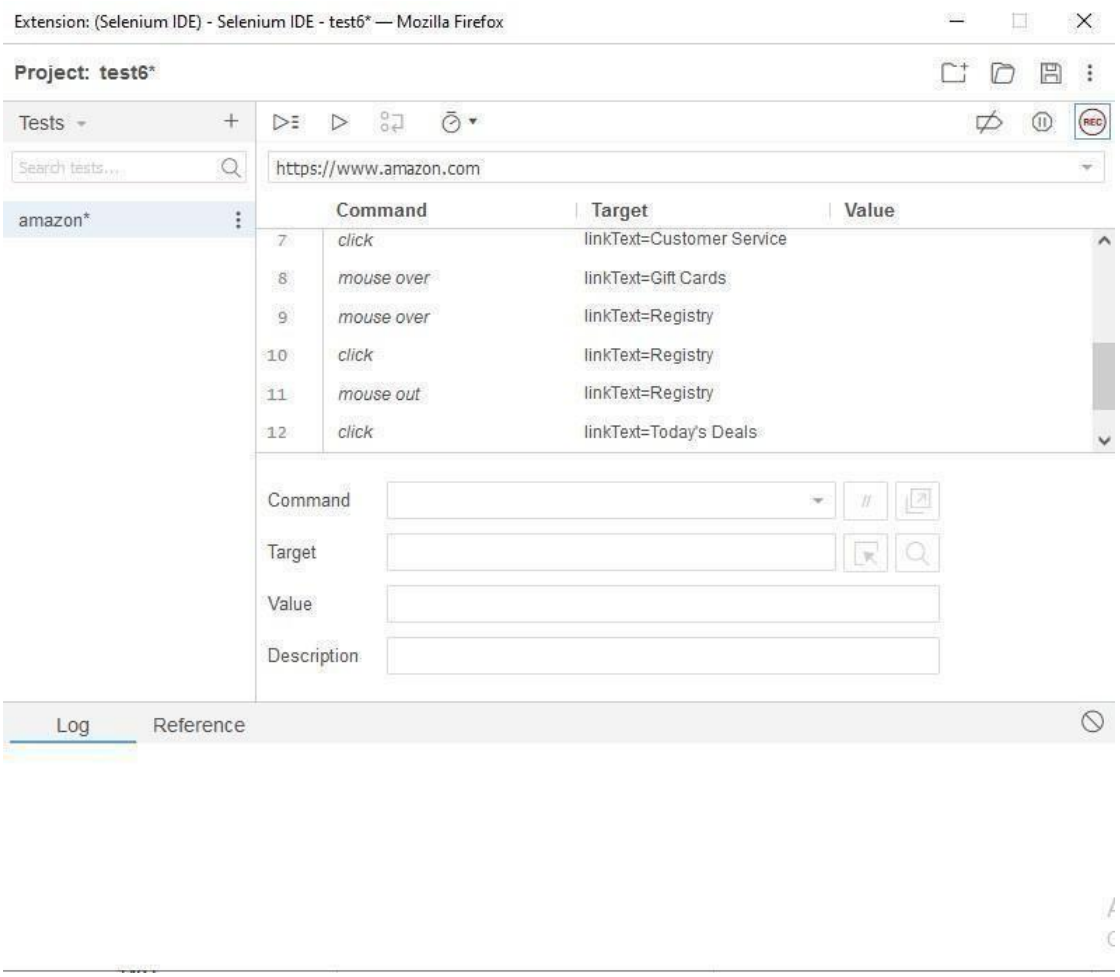
B. For www.amazon.com

1. Open the website and open selenium IDE. Copy the URL and paste it in the given space for further test. Then save & run the test to see log files.



Name: Sujal Singh

Roll No : CS022075

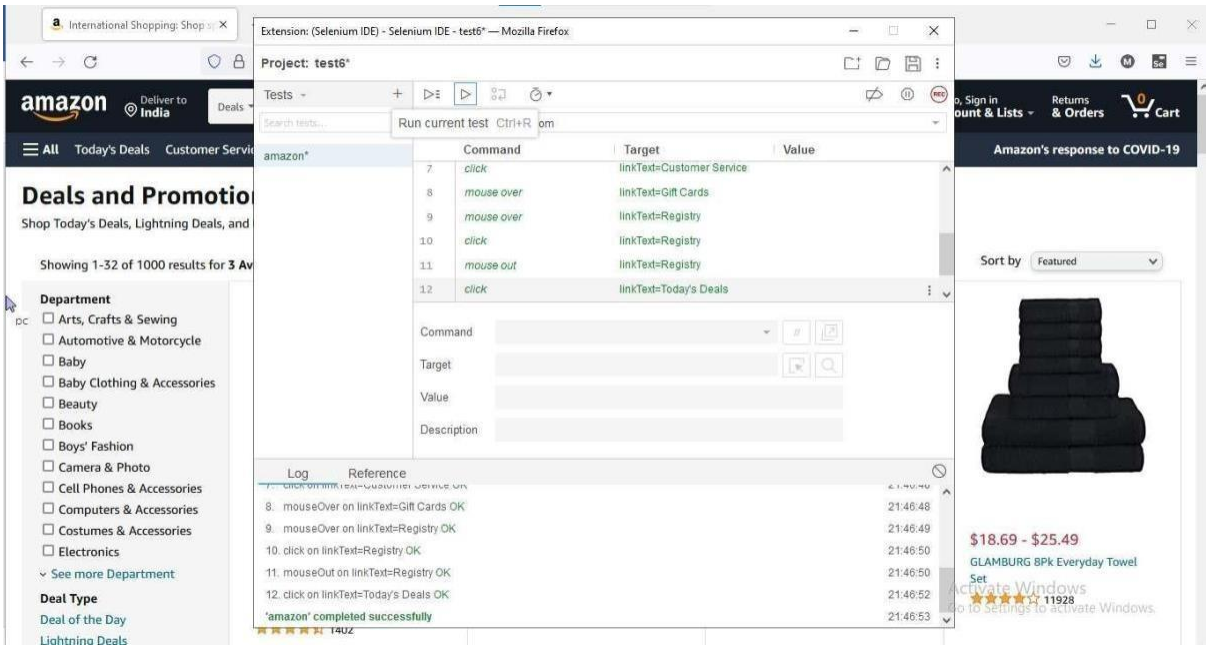


2. After saving the project, click on play and see how commands execute...

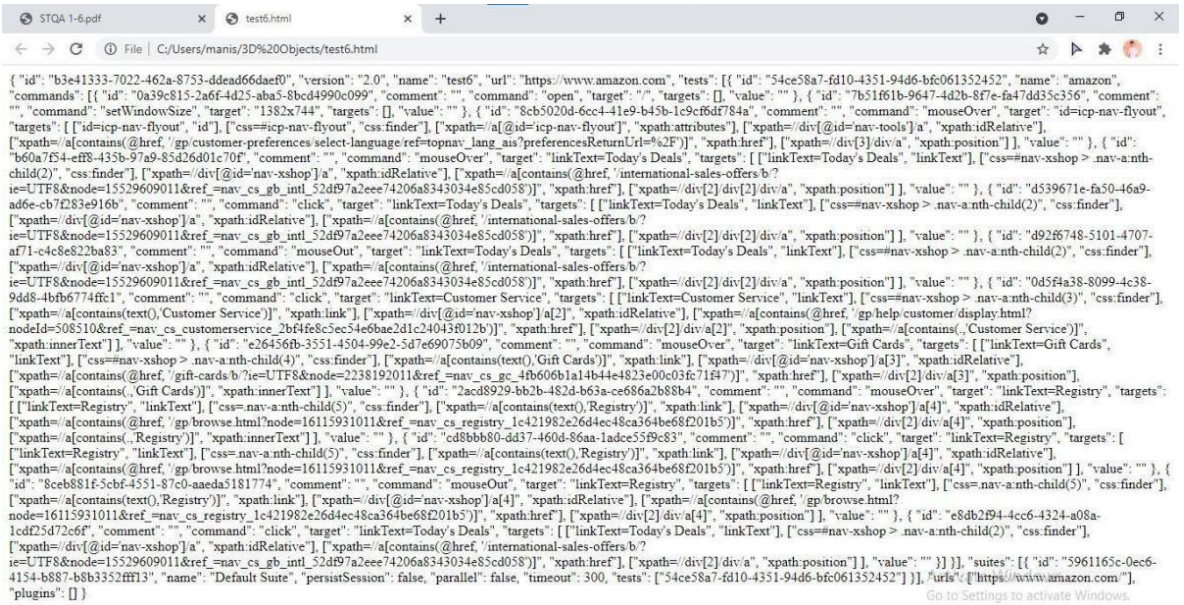
3. After successfully completing you will see message “completed successfully” with green text highlighted with log .

Name: Sujal Singh

Roll No : CS022075



4. Save it in the HTML file and open it on your browser webpage.



{ "id": "b3e41333-7022-462a-8753-ddead66daef0", "version": "2.0", "name": "test6", "url": "https://www.amazon.com", "tests": [{ "id": "54ce58a7-fd10-4351-94d6-bfc061352452", "name": "amazon", "commands": [{ "id": "0a39c815-2a6f-4d25-aba5-8bcd4990c099", "comment": "", "command": "open", "target": "/", "targets": [], "value": "" }, { "id": "7b51f61b-9647-4d2b-8f7e-fa47dd35c356", "comment": "", "command": "setWindowSize", "target": "1382x744", "targets": [], "value": "" }, { "id": "8cb5020d-6cc4-41e9-b45b-1c9cf6df784a", "comment": "", "command": "mouseOver", "target": "id=icp-nav-flyout", "targets": [["id=icp-nav-flyout", "id"], ["css=#icp-nav-flyout", "css:finder"], ["xpath=//a[@id='icp-nav-flyout']", "xpath:attributes"], ["xpath=//div[@id='nav-tools']/a", "xpath:idRelative"], ["xpath=//a[contains(@href, '/gp/customer-preferences/select-language/ref=topnav_lang_ais?preferencesReturnUrl=%2F')]", "xpath:href"], ["xpath=//div[3]/div/a", "xpath:position"]], "value": "" }, { "id": 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[["linkText=Registry", "linkText"], ["css=.nav-a:nth-child(5)", "css:finder"], ["xpath=//a[contains(text(),'Registry')]", "xpath:link"], ["xpath=//div[@id='nav-xshop']/a[4]", "xpath:idRelative"], ["xpath=//a[contains(@href, '/gp/browse.html?node=16115931011&ref_=nav_cs_registry_1c421982e26d4ec48ca364be68f201b5')]", "xpath:href"], ["xpath=//div[2]/div/a[4]", "xpath:position"], ["xpath=//a[contains(.,'Registry')]", "xpath:innerText"]], "value": "" }, { "id": "cd8bbb80-dd37-460d-86aa-1adce55f9c83", "comment": "", "command": "click", "target": "linkText=Registry", "targets": [["linkText=Registry", "linkText"], ["css=.nav-a:nth-child(5)", "css:finder"], ["xpath=//a[contains(text(),'Registry')]", "xpath:link"], ["xpath=//div[@id='nav-xshop']/a[4]", "xpath:idRelative"], ["xpath=//a[contains(@href, '/gp/browse.html?node=16115931011&ref_=nav_cs_registry_1c421982e26d4ec48ca364be68f201b5')]", "xpath:href"], ["xpath=//div[2]/div/a[4]", "xpath:position"]], "value": "" }, { "id": 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"value": "" }] }], "suites": [{ "id": "5961165c-0ec6-4154-b887-b8b3352fff13", "name": "Default Suite", "persistSession": false, "parallel": false, "timeout": 300, "tests": ["54ce58a7-fd10-4351-94d6-bfc061352452"] }], "urls": ["https://www.amazon.com/"], "plugins": [] }

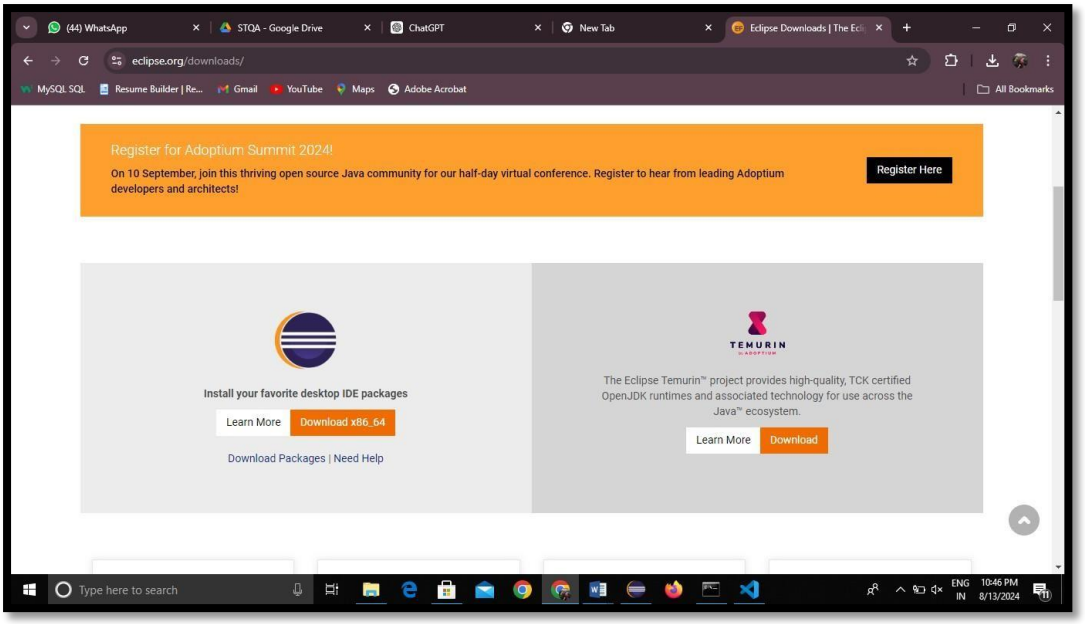
Name: Sujal Singh
Roll No : CS022075

Practical 3

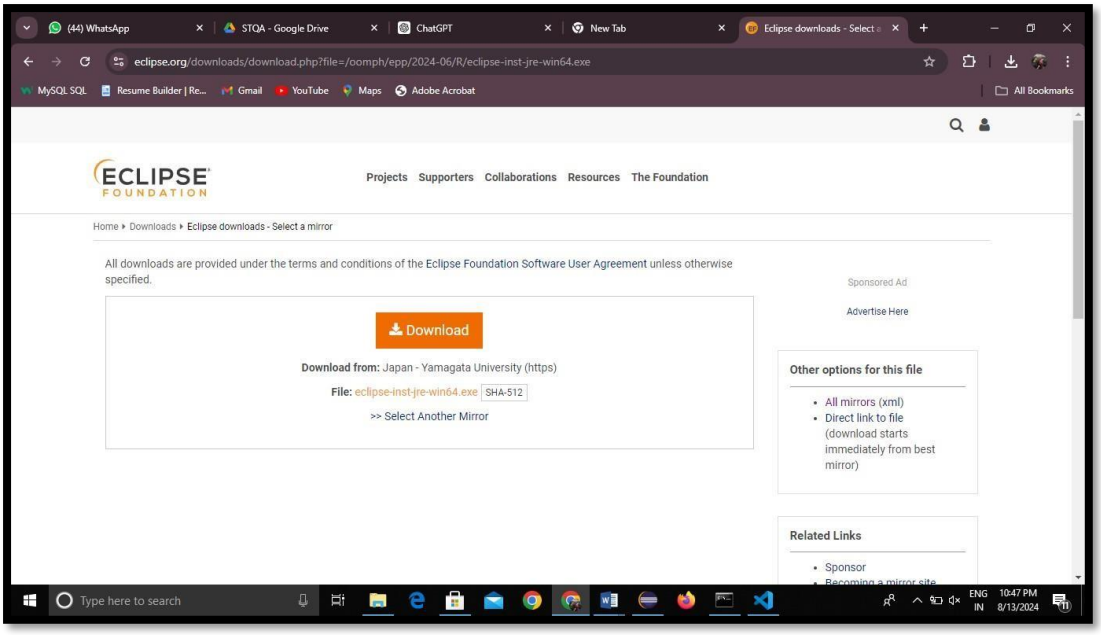
Aim: Install Selenium server and demonstrate it using a script in Java/PHP.

Steps:

Go to URL – http://www.eclipse.org/downloads/



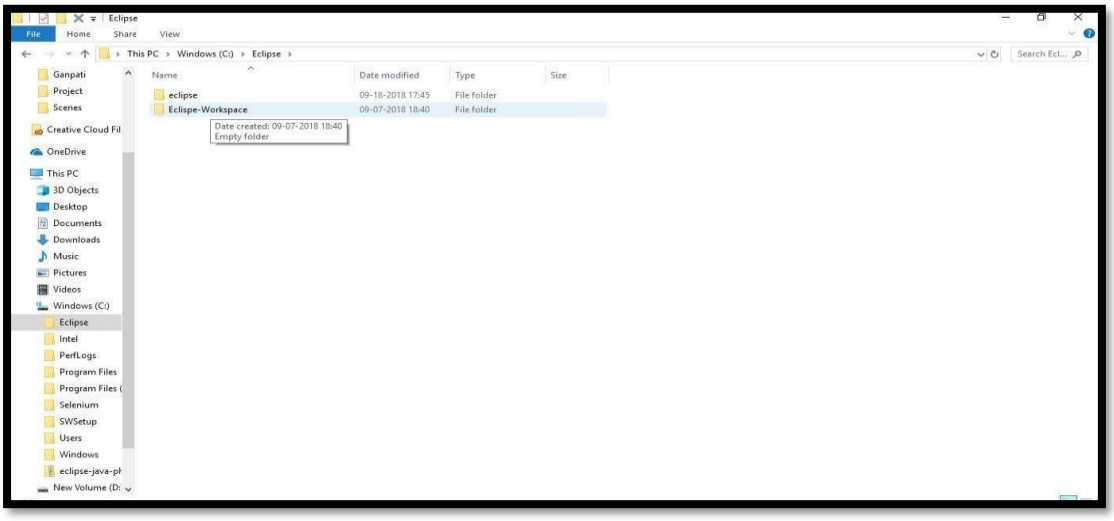
2. Select **Eclipse IDE for Java Developers** (Click on Windows 64 bit platform)



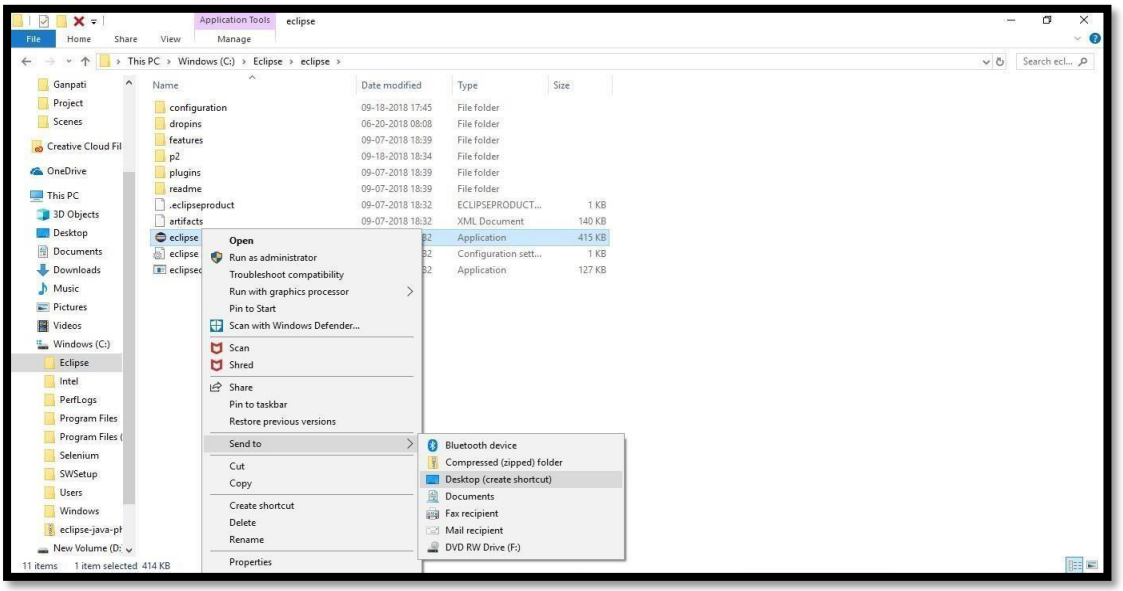
Name: Sujal Singh
Roll No : CS022075

3. Click on OK button and save to a local drive (i.e. C:)

Unzip the downloaded zip file and rename that to Eclipse.



4. Create one more folder “Eclipse-Workspace” (i.e. C:Eclipse-Workspace)in the same drive where Eclipse is unzipped and renamed.



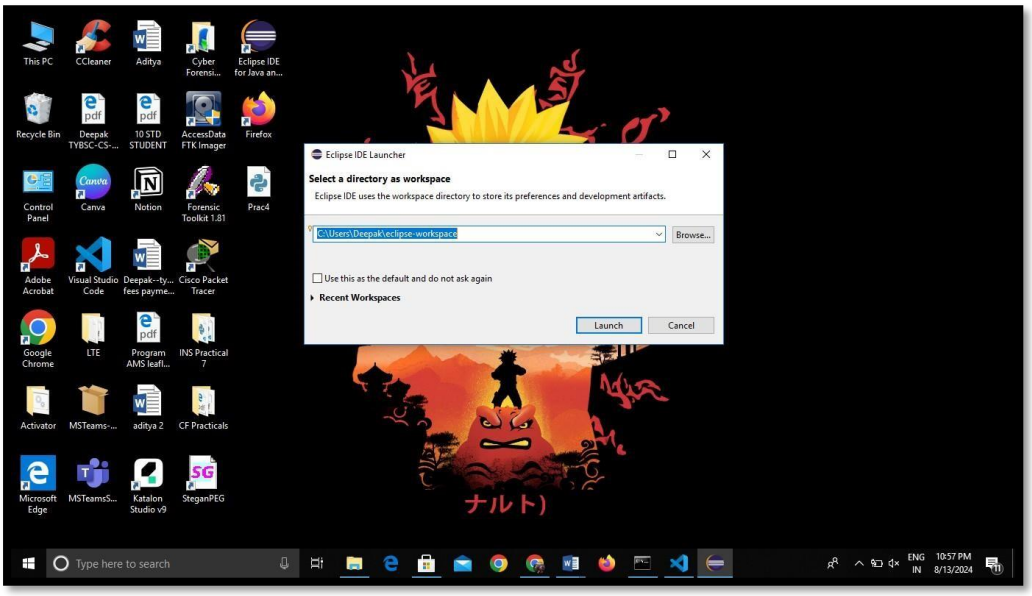
5. Create Eclipse desktop shortcut (go to C:Eclipse folder –> right click Eclipse.exe and then click on “desktop create shortcut”).

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6. Now we need to create a workspace folder –> C:Eclipse WorkspaceSeleniumTests .

Double click on “Eclipse shortcut on Desktop”



Download Selenium server: http://seleniumhq.org/download/

Download Selenium Client driver for Java (from Selenium Client Drivers section)

Selenium Standalone Server

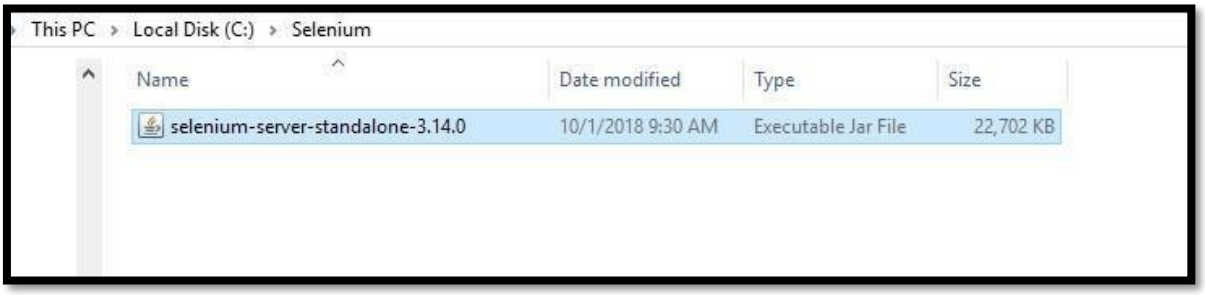
The Selenium Server is needed in order to run Remote Selenium WebDriver. Selenium 3.X is no longer capable of running Selenium RC directly, rather it does it through emulation and the WebDriverBackedSelenium interface.

Download version 3.14.0

To run Selenium tests exported from the legacy IDE, use the Selenium Html Runner.

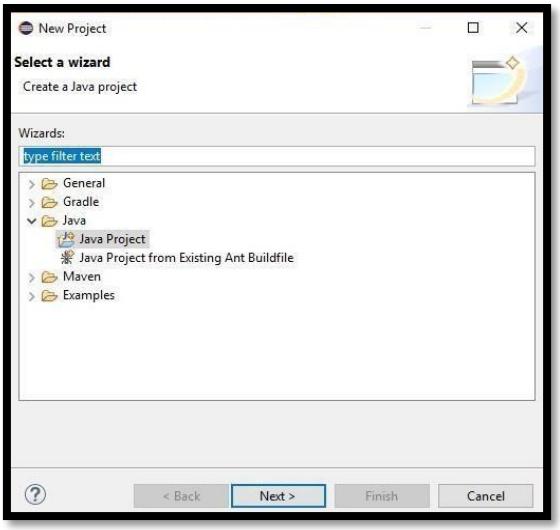
To use the Selenium Server in a Grid configuration see the wiki page.

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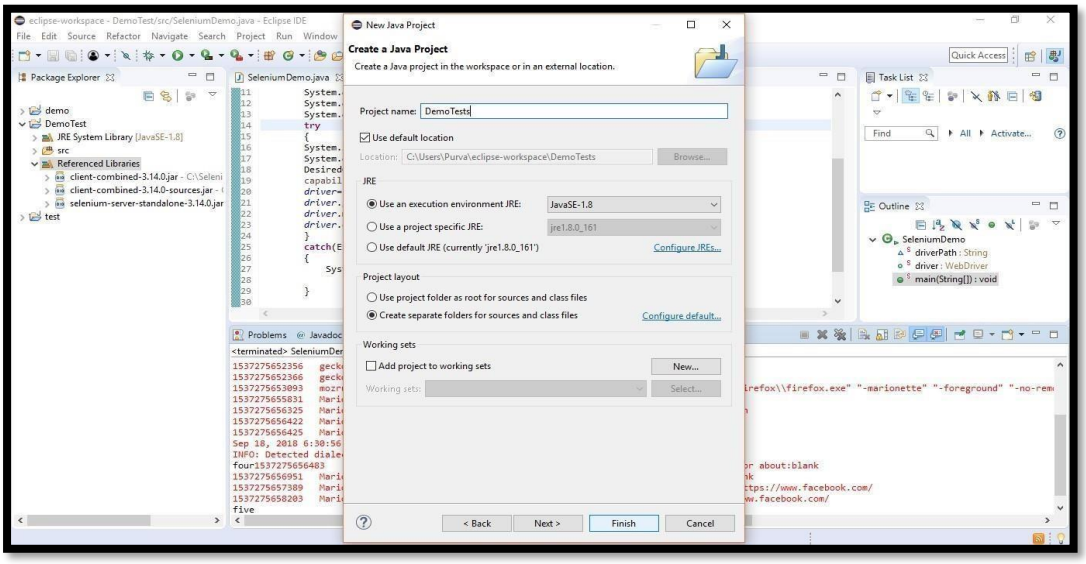
Go to Eclipse –> Click File –> New –> Project (from various options need to select just “project”)

In Select Wizard –> Click Java –> “Java Project”



Now we are done with creation of project and need to configure the Selenium Client driver to this Project

Right Click “DemoTests” project



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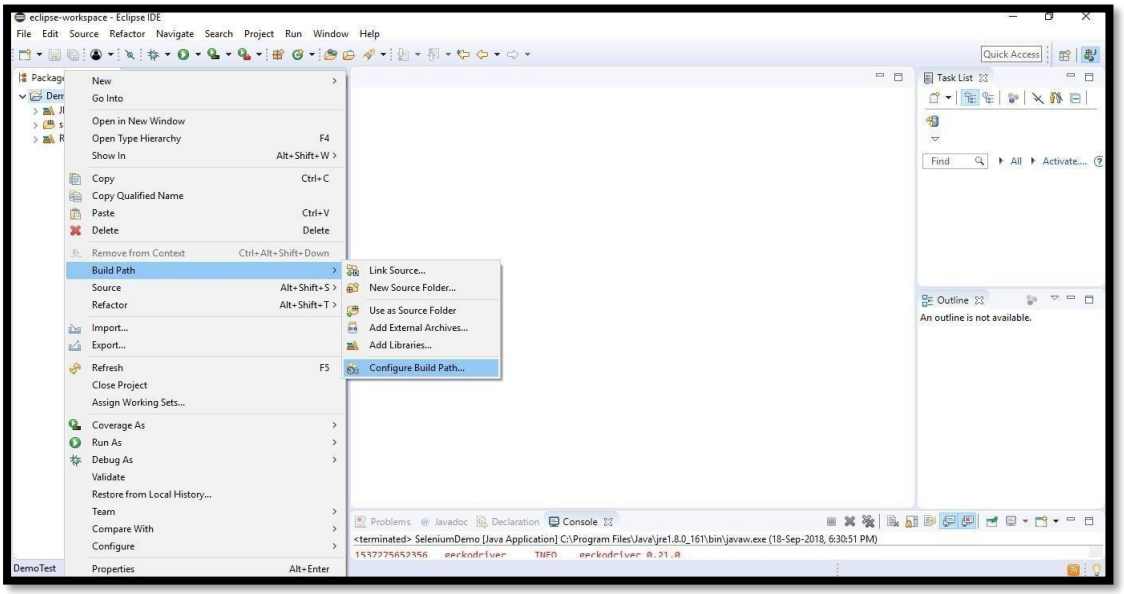
Click “Java Build Path”

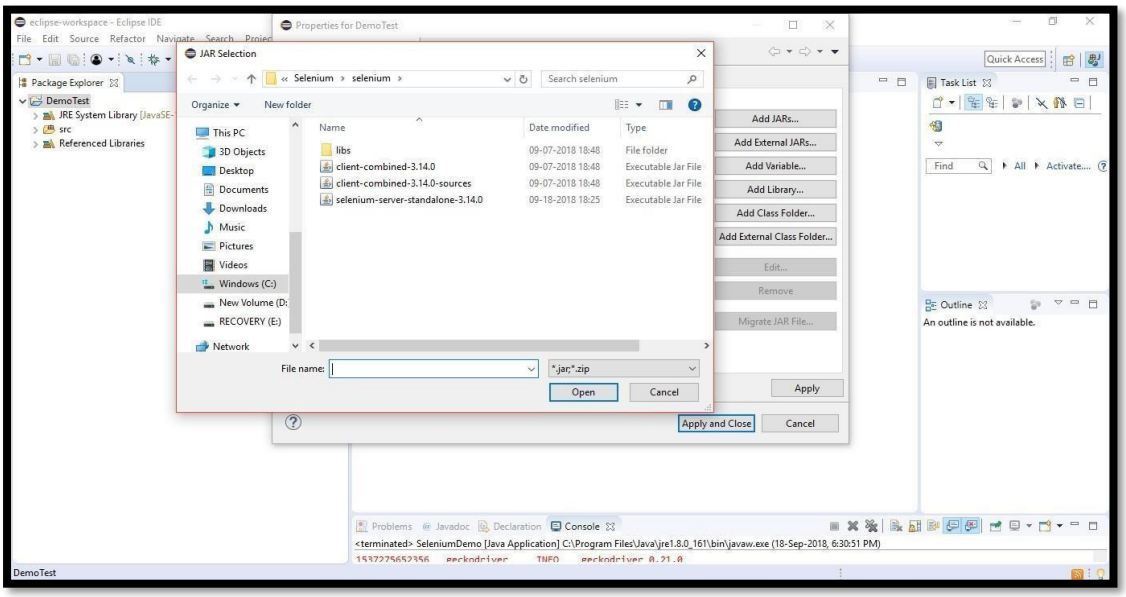
Click Libraries tab

Click “Add External JARs” button

Select “Selenium Client Drivers” unzipped in C:Selenium folder (Selenium

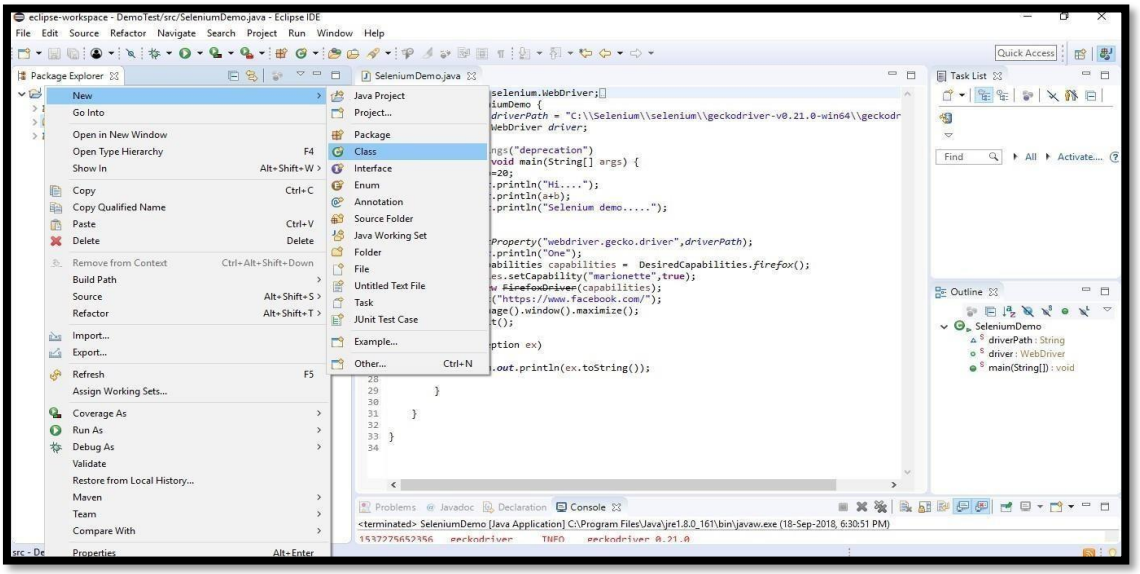
Server JAR file should not be added) Click OK





Create a new class file as “SeleniumDemo” in the “DemoTest” by right click on src folder.

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Code:

```
package practical3;


import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;

public class prac3 {

    public static WebDriver driver;
    static String driverPath = "C:\\Users\\Deepak\\Documents\\STQA\\geckodriver-v0.35.0-win-
64\\geckodriver.exe";

    public static void main(String[] args) {
        System.out.println("Starting Selenium demo...");

        // Set the path to the Geckodriver executable
        System.setProperty("webdriver.gecko.driver", "C:\\Users\\Deepak\\Documents\\STQA\\geckodriver-
v0.35.0-win-64\\geckodriver.exe");

        // Create FirefoxOptions instance
        FirefoxOptions options = new FirefoxOptions();
        // Add any desired options here, for example:
        // options.setHeadless(true); // Run Firefox in headless mode (no GUI)

        // Initialize FirefoxDriver with options
        driver = new FirefoxDriver(options);

        try {
            // Navigate to a webpage
            driver.get("https://www.facebook.com/");
```

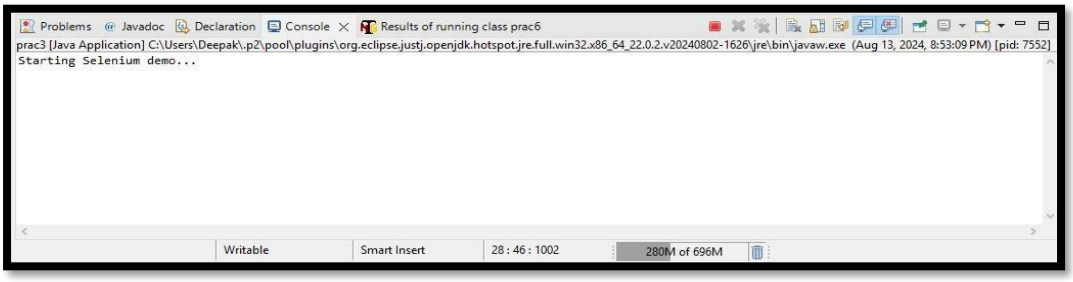
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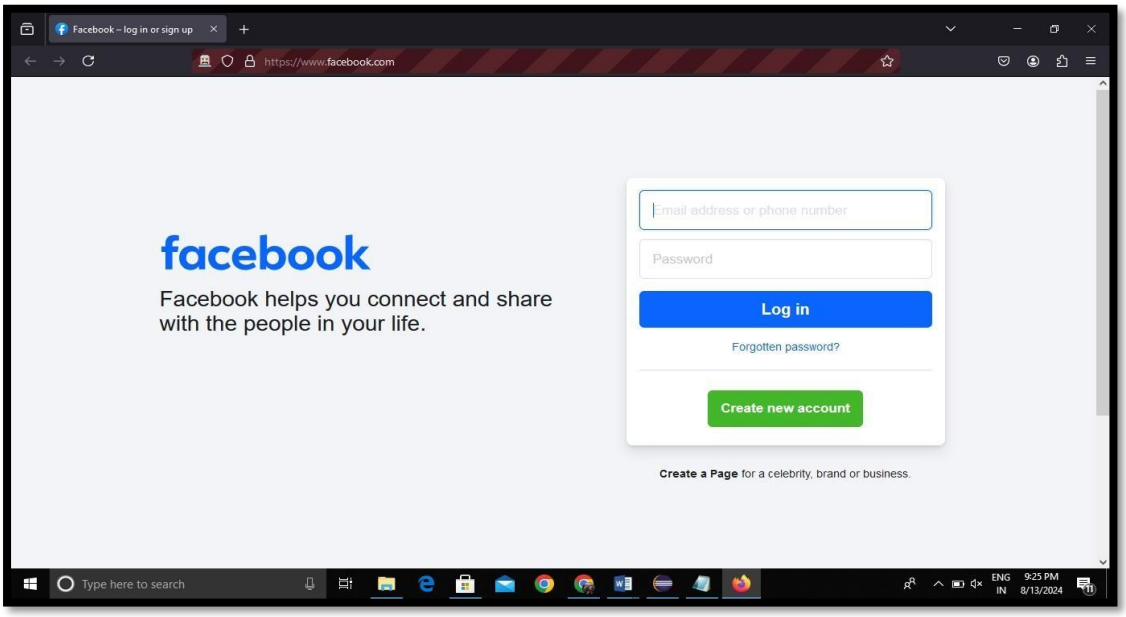
```
        driver.manage().window().maximize();

        // Perform any additional actions if needed

    } catch (Exception ex) {
        System.out.println("An error occurred: " + ex.toString());
    } finally {
        // Quit the driver to close all browser windows and end the session
        if (driver != null) {
            driver.quit();
        }
    }
  }
}
```

Output:





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PRACTICAL NO.4

AIM Write a program using Selenium WebDriver to automate the login process on a specific web page. Verify successful login with appropriate assertions.

"create a new file (file name " selenium_test.py ") and save it "

code :

```
from selenium import webdriver
from selenium.webdriver.chrome.options import Options
from selenium.webdriver.common.by import By

from time import sleep

options = Options()
options.add_experimental_option("excludeSwitches", ["enable-log

print("Testing started")
driver = webdriver.Chrome(options=options)
driver.get("https://www.instagram.com/taylorswift/")
sleep(5)

title = driver.title

driver.find_element(By.ID, "user-name").send_keys("standard_use
driver.find_element(By.ID, "password").send_keys("secret_sauce"
driver.find_element(By.ID, "login-button").click()
sleep(5)

assert "products" in title.lower()
print("TEST PASSED: Mr. Sai, your LOGIN is SUCCESSFUL")

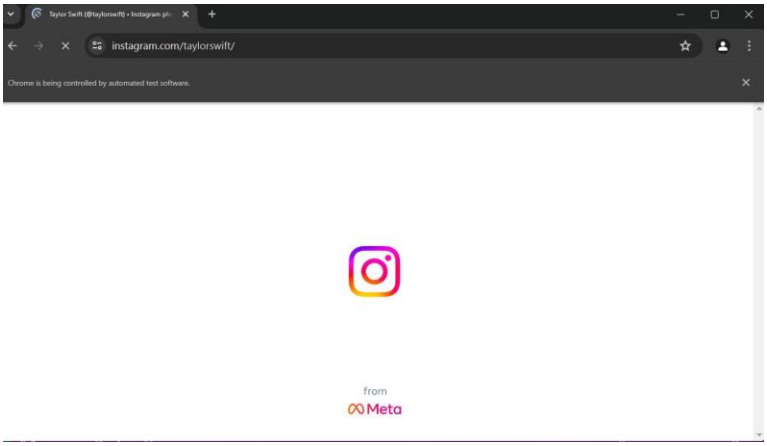
driver.quit()
```

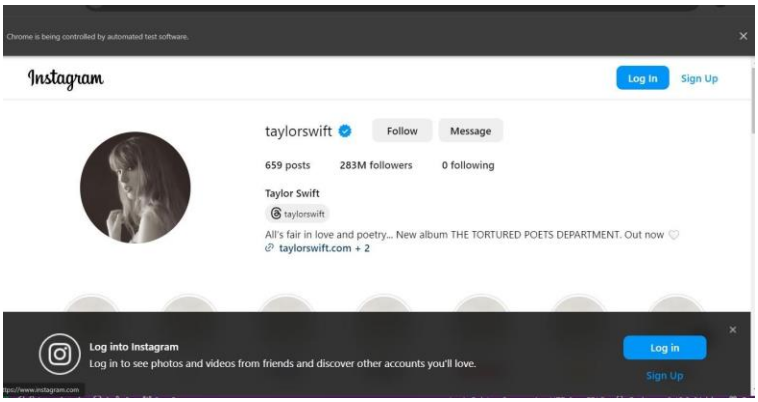
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Roll No : CS022075

write in terminal :

```
pip show selenium
" for run the code "
python selenium_test.py
```

OUTPUT





Name: Sujal Singh
Roll No : CS022075

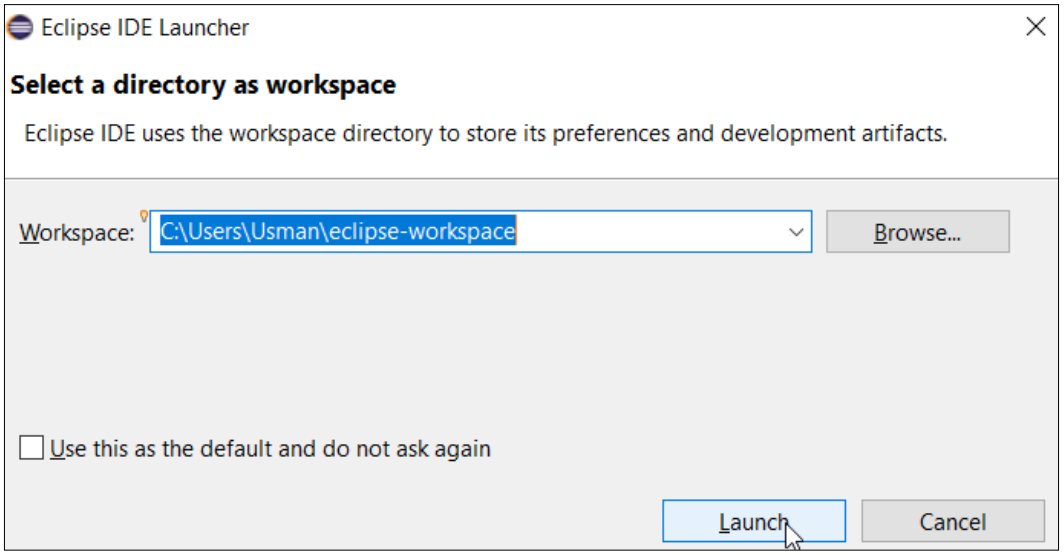
AIM : Write a program using Selenium WebDriver to update 10 student records in an Excel file. Perform data manipulation and verification.

PRE-REQUISITES:

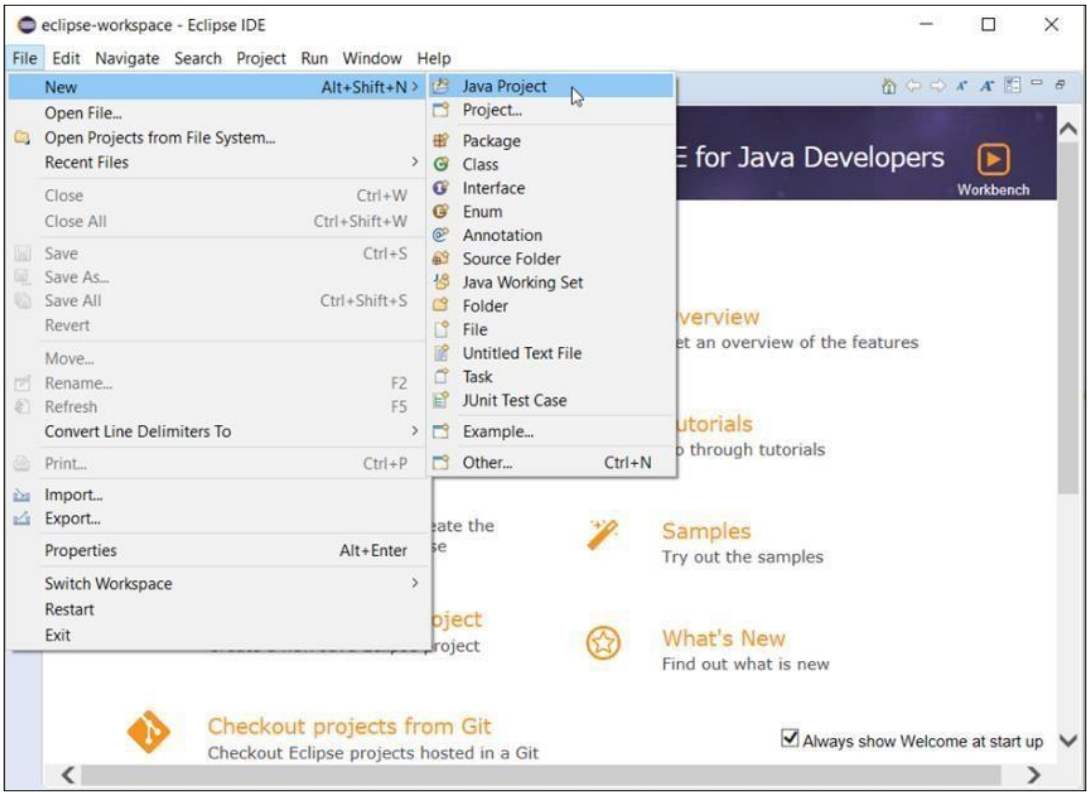
1) Check that you have **Eclipse IDE**.
2) To Download "**JXL.JAR**":
   - **Visit** http://www.java2s.com/Code/Jar/j/Downloadjxl26jar.htm
   - **Download** this file: "jxl/jxl-2.6.jar.zip(603 k)" and **extract it**.(you'll get the .jar file)

STEPS:

1) **Open Eclipse**. Select your workspace directory. Click **Launch:**

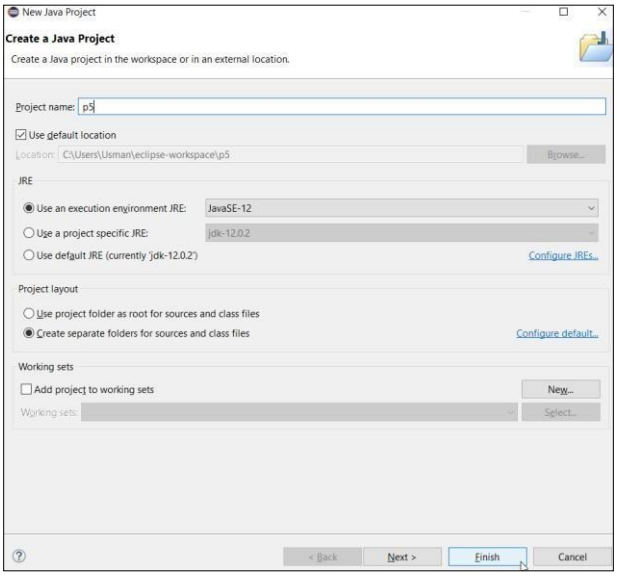


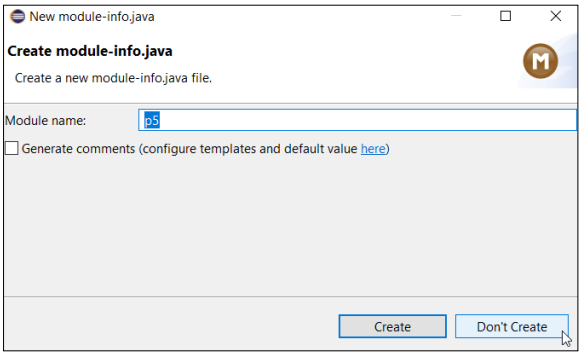
2) **Create** a Project(**File > New > Java Project**):



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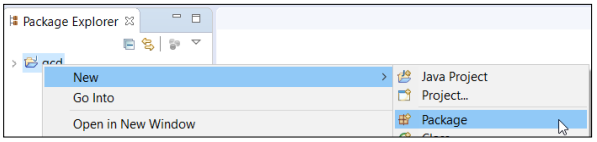
3) **Name the project** as “p5” **>** click **Finish >** click **Don’t Create** module:

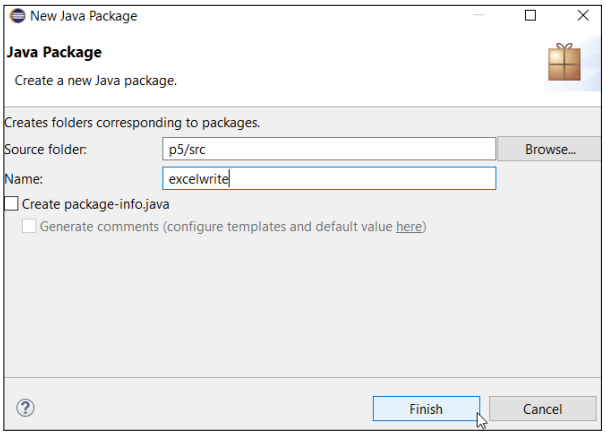




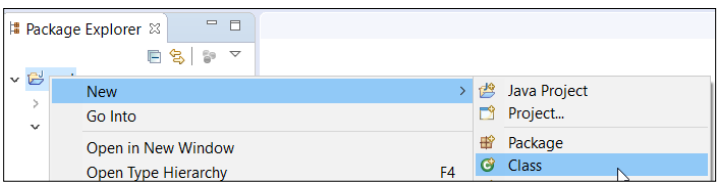
4) **Close the “Welcome” tab**.

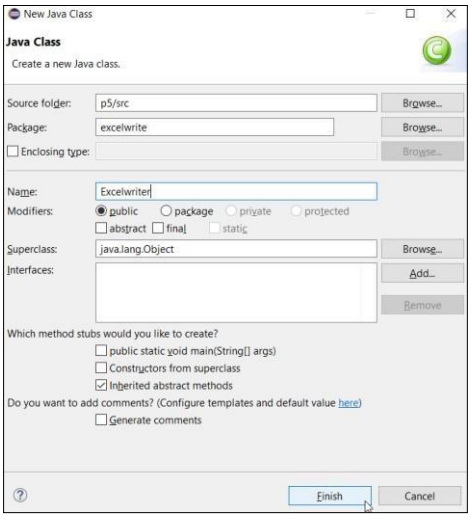
5) **Create** a Package(**right-click on Project Name > New > Package > Name** it **> Finish**):





6) **Create** a Class(**right-click on Project Name > New > Class > Name** it **> Finish**):

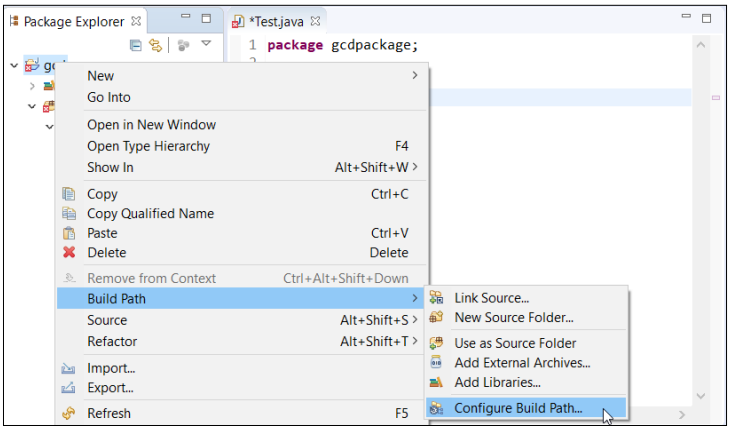




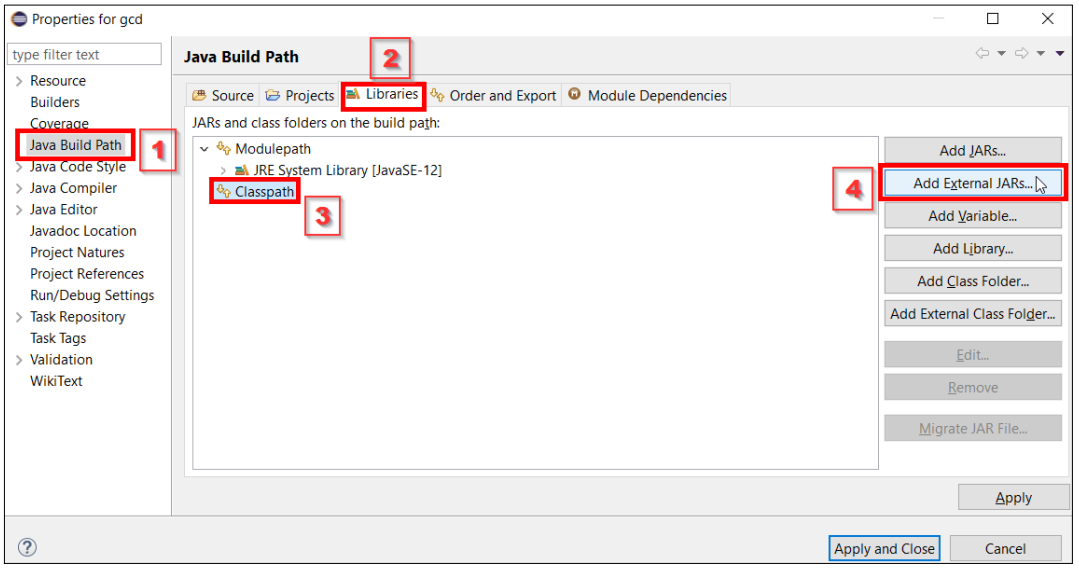
Name: Sujal Singh
Roll No : CS022075

7) Adding "**JXL(JAR file)"** in Eclipse IDE:

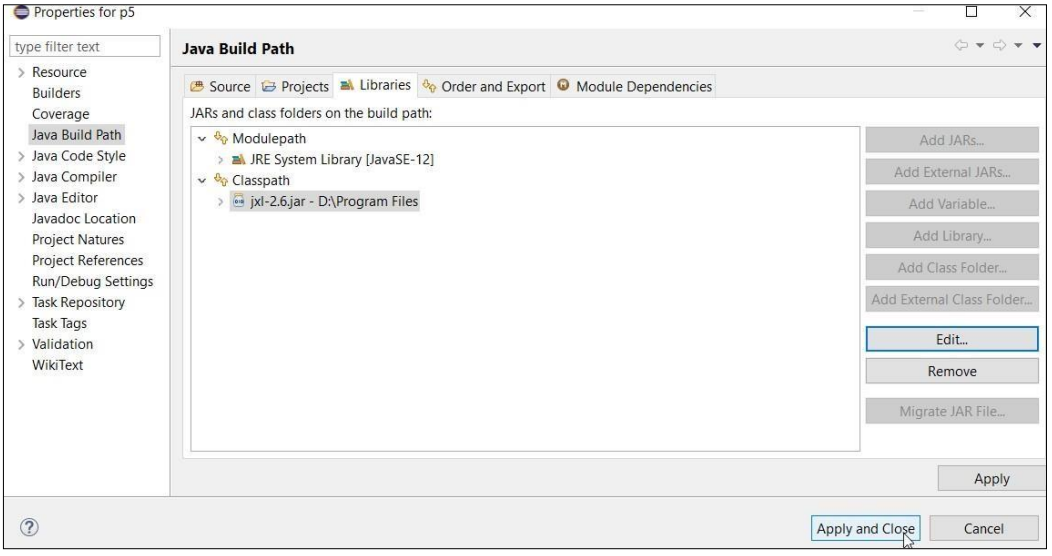
- **right-click on Project Name > Build Path > Configure Build Path…**



- now go under: **Java Build Path > Libraries > Classpath >** click **Add External JARs…**



- Browse and **add JAR file >** click **Apply and Close :**



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Roll No : CS022075

8) Creating the ***script*** *in JAVA*:

(NOTE that this **script** will be run by Eclipse IDE)
(In simple words, it's like we are
-ordering Eclipse to run a script or to do a job
-of creating and opening .xls file
-and putting the values in the cells with the help of jxl.jar
-and to show the result.
-Hence automating the work in a local system(PC)).

---(Excelwriter.java)---

```
package excelwrite;

import jxl.*; //used for WorkbookSettings,Workbook
import jxl.write.*; //used for WriteException,WritableWorkbook,WritableSheet,Label
import jxl.write.Number; //used for Number
import java.io.*; //used for IOException,File
import java.util.Locale; //used for Locale

public class Excelwriter {
	public static void main(String[] args) throws IOException,WriteException {
		// TODO Auto-generated method stub
		int r=0,c=0;
		String header[]={"Student
Name","Subject1","Subject2","Subject3","Total"};
		String
sname[]={"Carls","James","Paul","Philip","Smith","Thomson","Rhodey","Stark","Gary"
,"AnneMarie"};
		int marks[]={50,45,60,55,70,45,67,78,89,90,30};

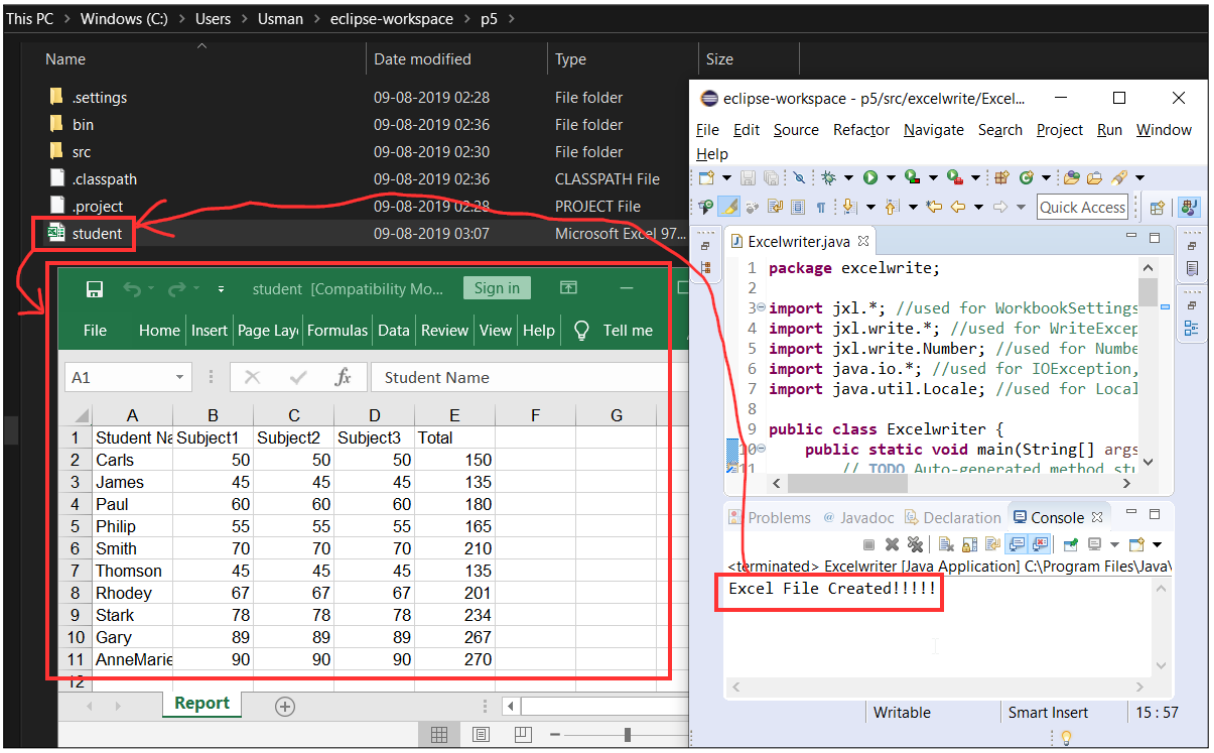
		File file = new File("student.xls");
		WorkbookSettings wbSettings = new WorkbookSettings();
		wbSettings.setLocale(new Locale("en", "EN"));
		WritableWorkbook workbook = Workbook.createWorkbook(file,
wbSettings);
		workbook.createSheet("Report",  0);
		WritableSheet excelSheet = workbook.getSheet(0);
		//creating header row
		for(r=0;r<1;r++) {
			for(c=0;c<header.length;c++) {
				Label l=new Label(c,r,header[c]);
				excelSheet.addCell(l);
			}
		}
		//filling name in column1
		for(r=1;r<=sname.length;r++) {
			for(c=0;c<1;c++) {
				Label l=new Label(c,r,sname[r-1]);
				excelSheet.addCell(l);
			}
		}
		//filling name in column2,3,4
		for(r=1;r<=sname.length;r++) {
			for(c=1;c<4;c++) {
				Number num = new Number(c, r, marks[r-1]);
				excelSheet.addCell(num);
			}
		}
```

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```
                //filling name in total
                for(r=1;r<=sname.length;r++) {
                        for(c=4;c<5;c++) {
                                int total=marks[r-1]+marks[r-1]+marks[r-1];
                                Number num = new Number(c, r, total);
                                excelSheet.addCell(num);
                        }
                }
                workbook.write();
                workbook.close();
                System.out.println("Excel File Created!!!!!");
        }
}
```

9) Run the file from Eclipse IDE:

- **OUTPUT:**



10) Finish!

What is JXL.JAR:

-Java libraries are distributed as a ".jar" file

-jxl.jar is the library of JExcelApi , which is open source java API to read, write, and modify Excel spreadsheets dynamically. It contains all the compiled *.class files, associated metadata and resources that are used by the Java Excel API internally .

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PRACTICAL NO.6

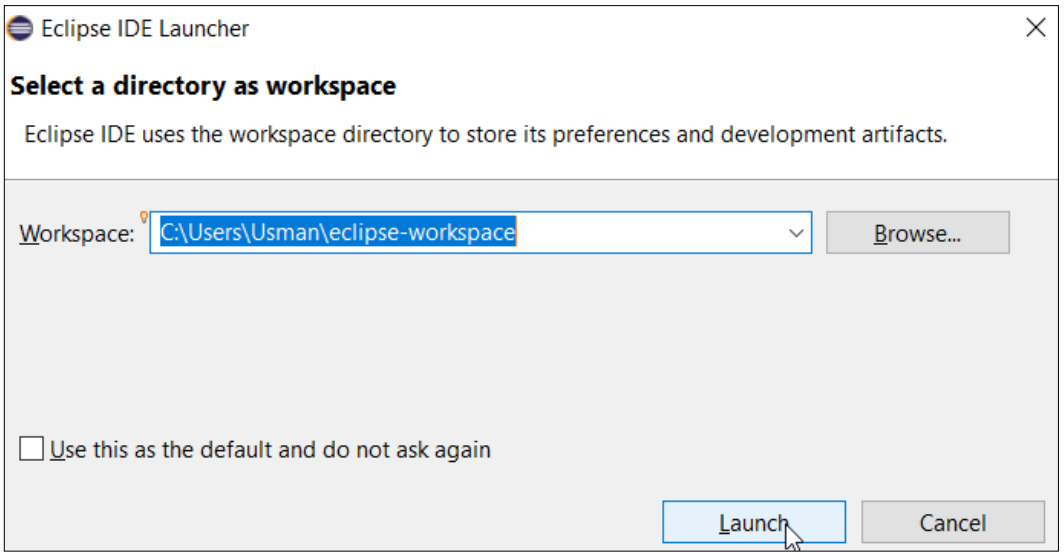
AIM : Write a program using Selenium WebDriver to select the number of students who have scored more than 60 in any one subject (or all subjects). Perform data extraction and analysis.

PRE-REQUISITES:

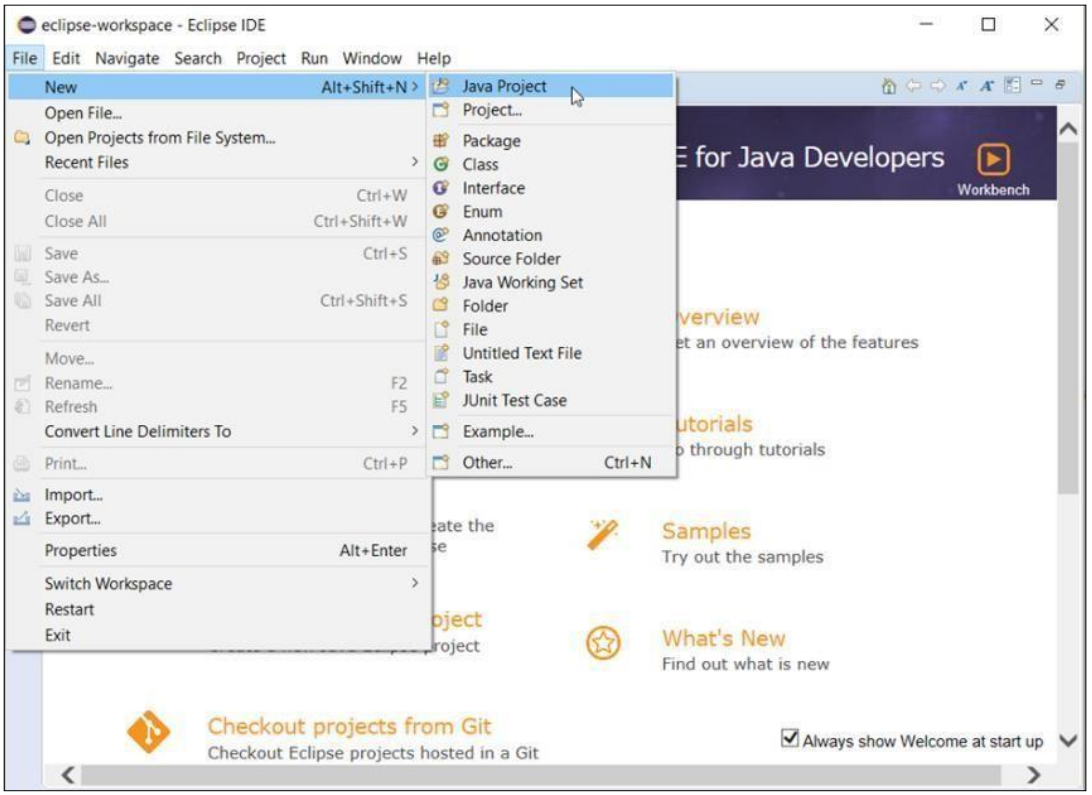
1) Check that you have **Eclipse IDE**.
2) Check that you have **JXL.JAR**.
3) Check that you've the ***Excel file****("student.xls")* that we'll be working on for reading data.

STEPS:

1) **Open Eclipse**. Select your workspace directory. Click **Launch:**

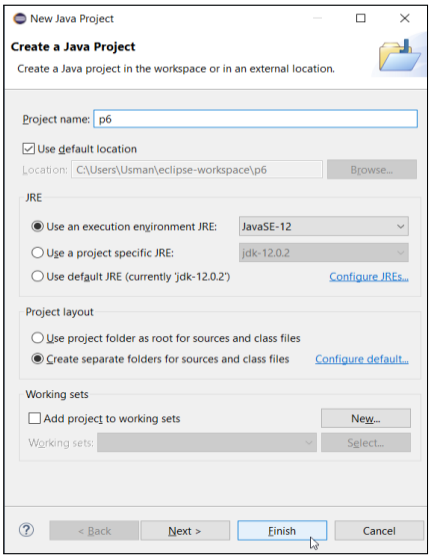


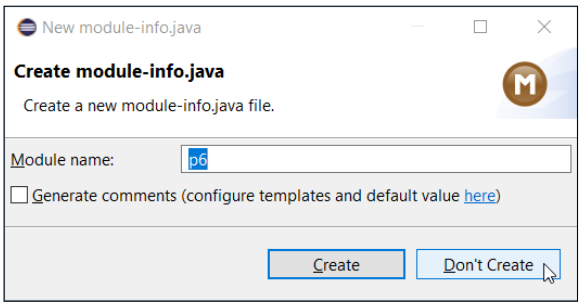
2) **Create** a Project(**File > New > Java Project**):



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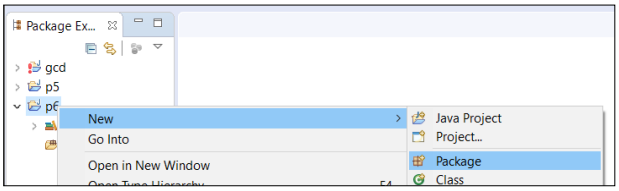
3) **Name the project** as “p6” **>** click **Finish >** click **Don’t Create** module:

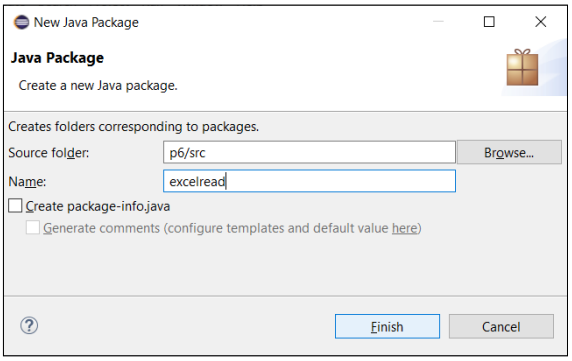




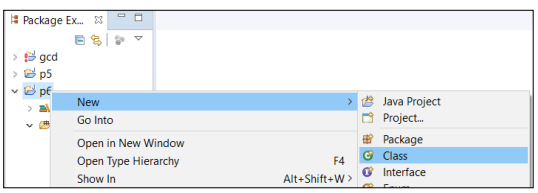
4) **Close the “Welcome” tab**.

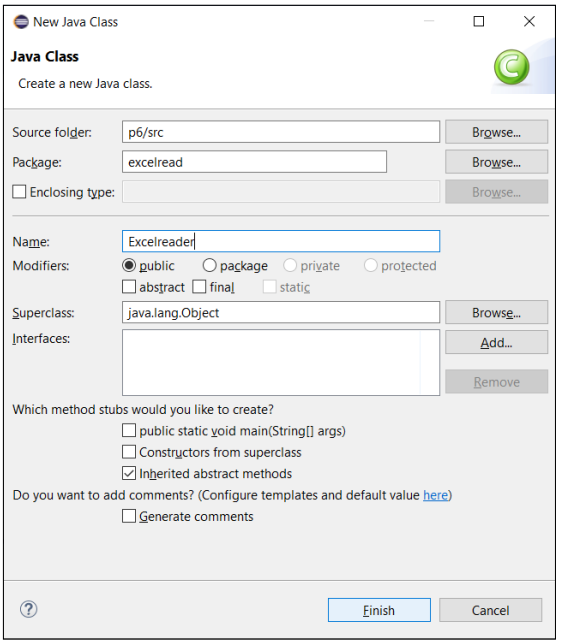
5) **Create** a Package(**right-click on Project Name > New > Package > Name** it **> Finish**):





6) **Create** a Class(**right-click on Project Name > New > Class > Name** it **> Finish**):

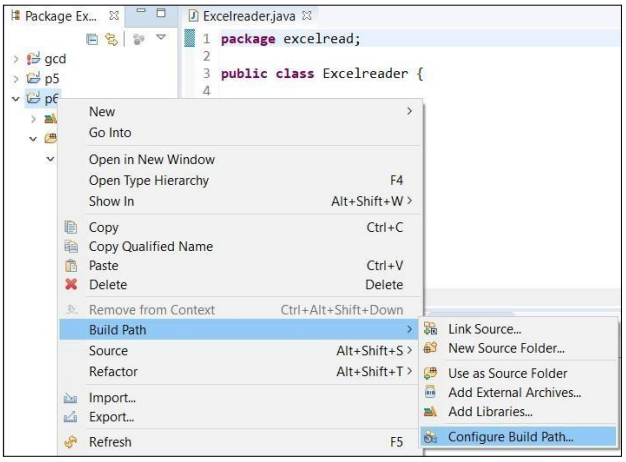




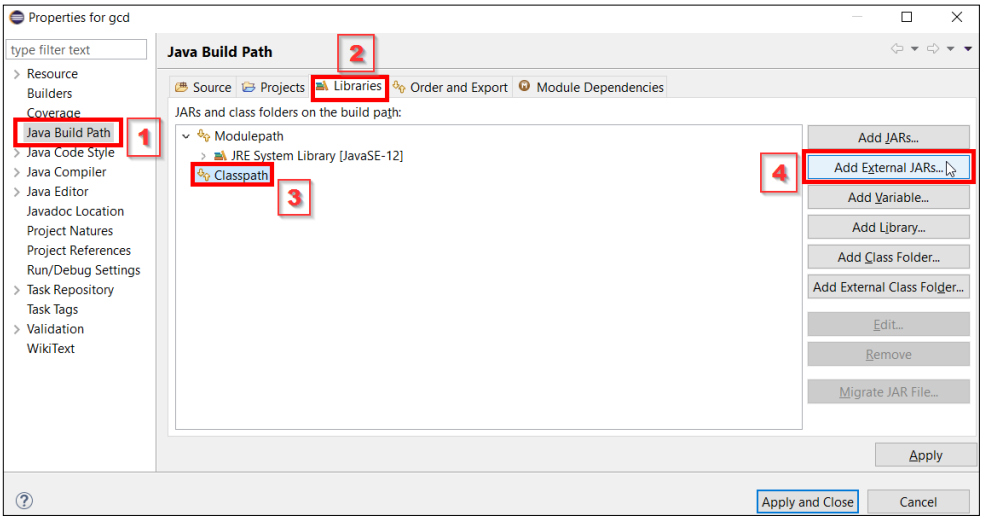
Name: Sujal Singh
Roll No : CS022075

7) Adding "**JXL(JAR file)"** in Eclipse IDE:

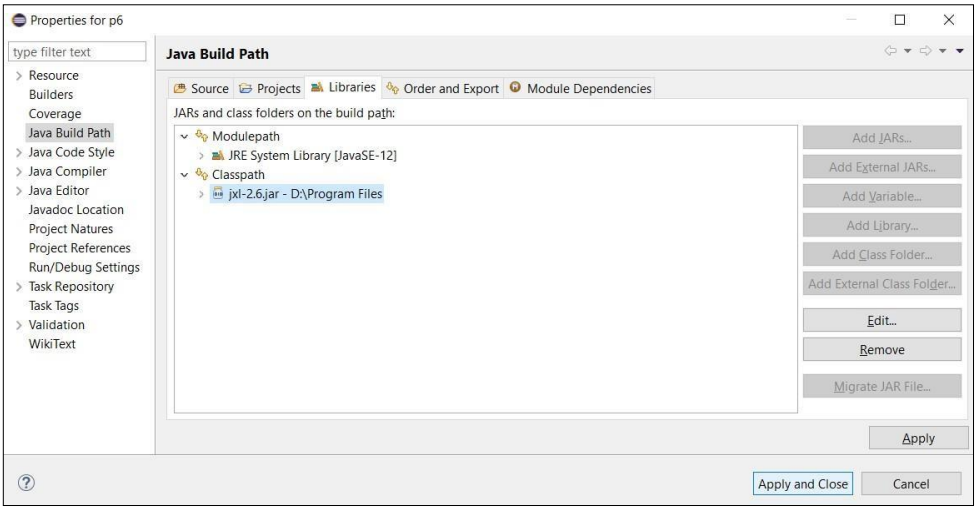
- **right-click on Project Name > Build Path > Configure Build Path…**



- now go under: **Java Build Path > Libraries > Classpath >** click **Add External JARs…**



- Browse and **add JAR file >** click **Apply and Close :**



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Roll No : CS022075

8) Creating the ***script*** *in JAVA*:

(NOTE that this **script** will be run by Eclipse IDE)
(In simple words, it's like we are
-ordering Eclipse to run a script or to do a job
-of opening .xls file
-and fetching the marks count>=60 from the cells with the help of jxl.jar
-and to show the result.
-Hence automating the work in a local system(PC)).

---(Excelreader.java)---

```
package excelread;

import  java.io.File;
import java.io.IOException;
import jxl.Cell;
import jxl.CellType;
import jxl.Sheet;
import jxl.Workbook;
import jxl.read.biff.BiffException;

public class Excelreader {
	private String inputFile;
	public void setInputFile(String inputFile) {this.inputFile = inputFile;}

	public void read() throws IOException {
		File inputWorkbook = new File(inputFile);
		Workbook w;
		boolean flag=false;
		int count=0;
		try {
			w = Workbook.getWorkbook(inputWorkbook);
			// Get the first  sheet
			Sheet sheet = w.getSheet(0);
			// Loop over first 10 column and lines
			for (int j = 0; j < sheet.getRows(); j++) {
				for (int i = 0; i < sheet.getColumns()-1; i++) {
					Cell cell = sheet.getCell(i, j);
					if (cell.getType() == CellType.NUMBER) {

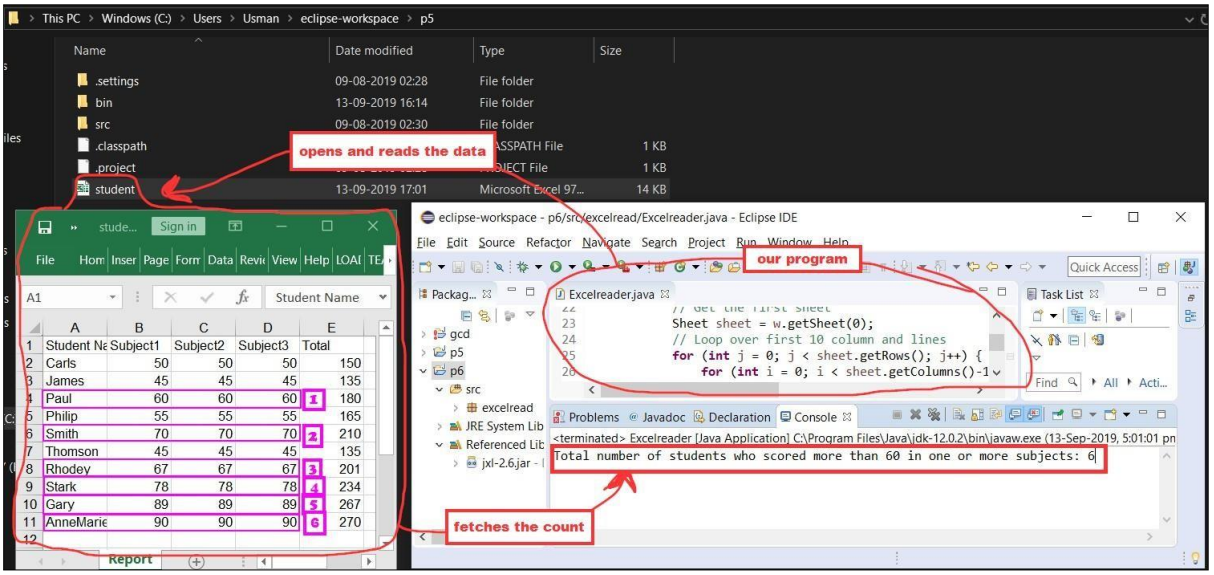
	if(Integer.parseInt(cell.getContents())>=60){
							flag = true;
							if(flag == true){
								count++;
								flag=false;
							}
							break;
						}
					}
				}
			}
			System.out.println("Total number of students who scored more
than 60 in one or more subjects: " +count);
		}
		catch (BiffException e) {e.printStackTrace();}
	}
```

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Roll No : CS022075

```
	public static void main(String[] args) throws IOException {
		Excelreader test = new Excelreader();
		test.setInputFile("C:\\Users\\Usman\\eclipse-
workspace\\p5\\student.xls");
		test.read();
	}
}
```

9) **Run** the file from Eclipse IDE:

- **OUTPUT:**



10) **Finish!**

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PRACTICAL NO.7

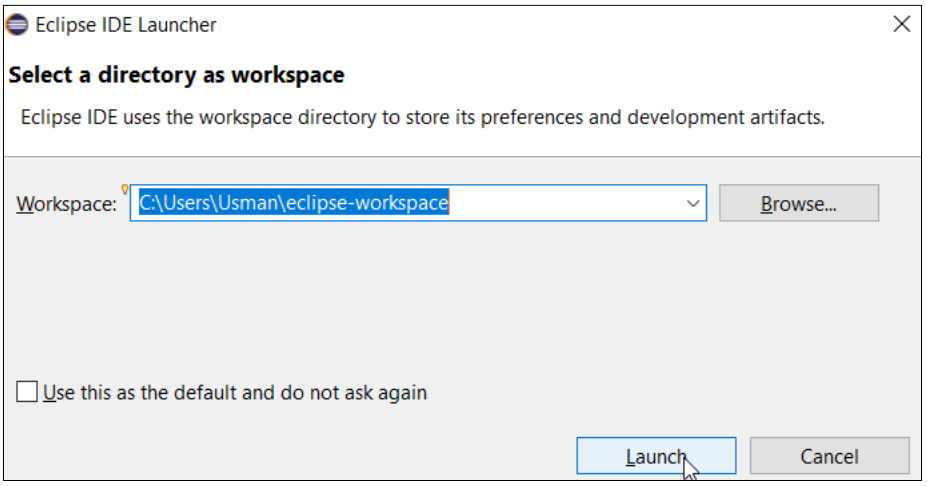
AIM : Write a program using Selenium WebDriver to provide the total number of objects present or available on a web page. Perform object identification and counting.

PRE-REQUISITES:

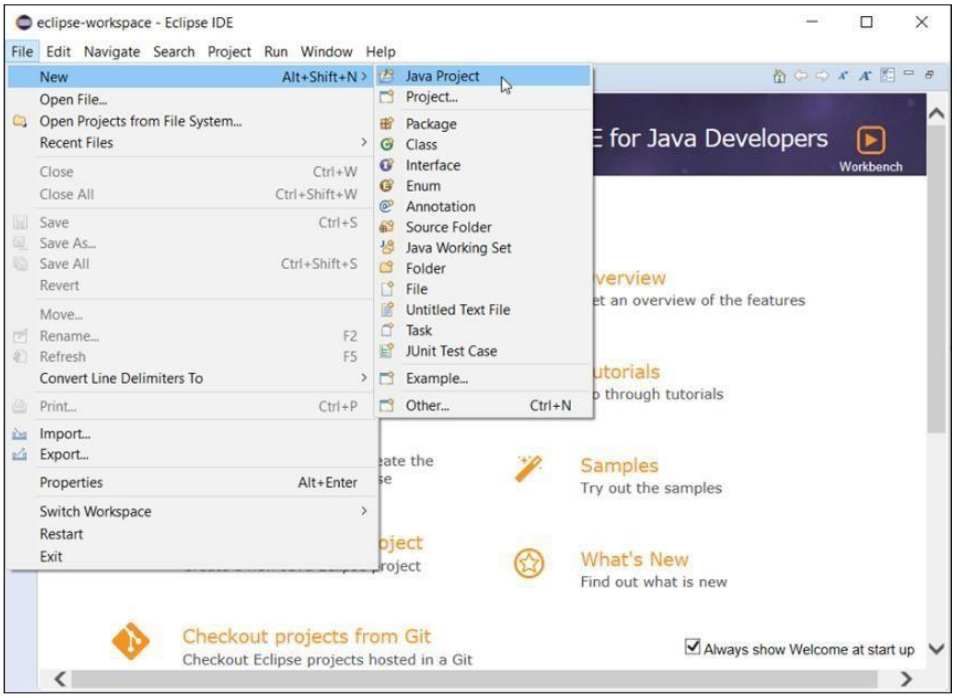
1) Check that you have **JDK**.
2) Check that you have **Eclipse IDE**.
3) Check that you have **Selenium *Server Driver* and *Client Driver*(JAR files)**.
4) Check that you have **Gecko Driver**.
5) Check that you have a stable Internet connection.

STEPS:

1) **Open Eclipse**. Select your workspace directory. Click **Launch:**

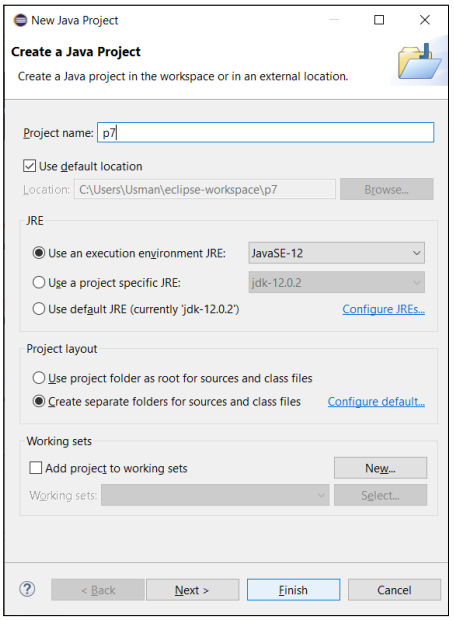


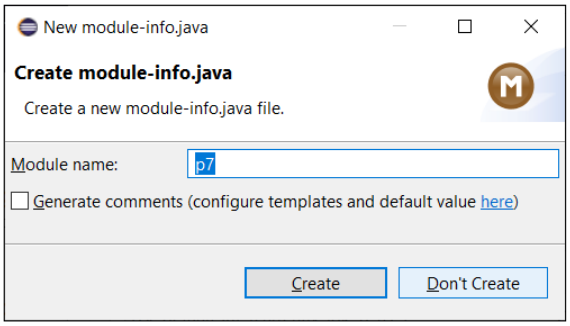
2) **Create** a Project(**File > New > Java Project**):



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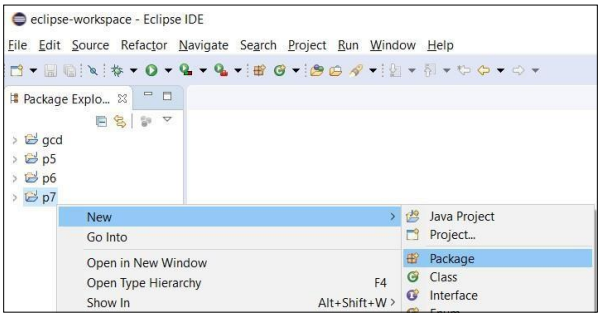
3) **Name the project** as “p7” **>** click **Finish >** click **Don’t Create** module:

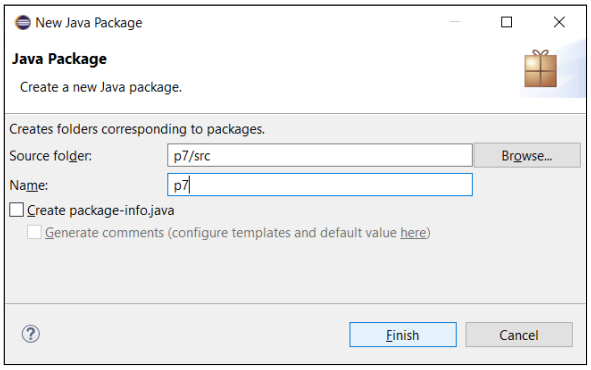




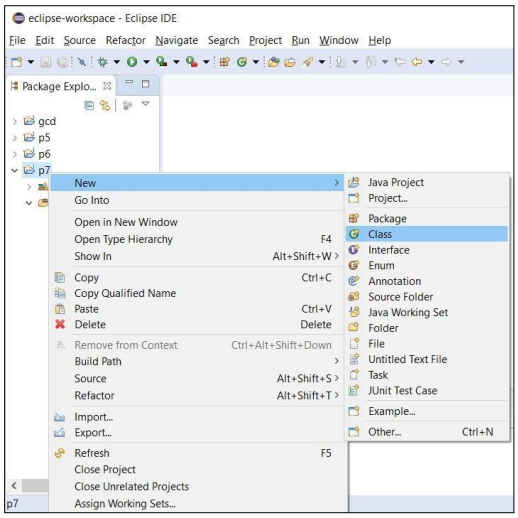
4) **Close the “Welcome” tab**.

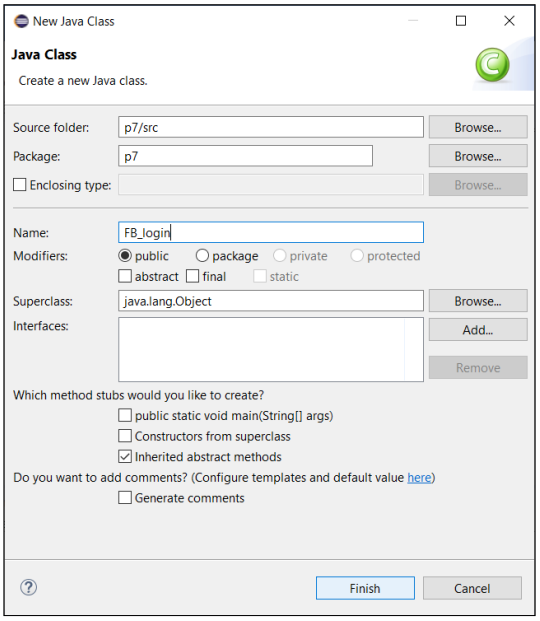
5) **Create** a Package(**right-click on Project Name > New > Package > Name** it **> Finish**):





6) **Create** a Class(**right-click on Project Name > New > Class > Name** it **> Finish**):

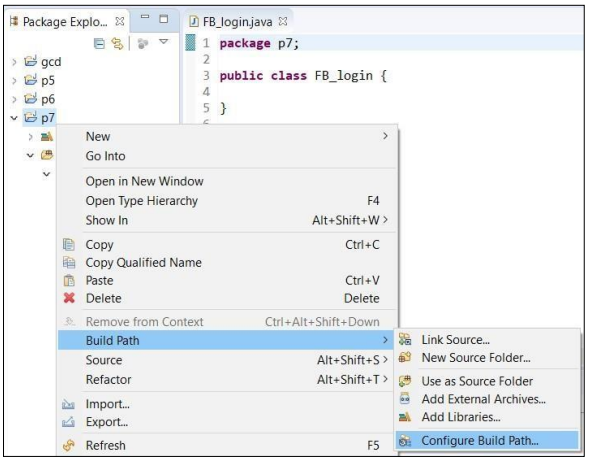




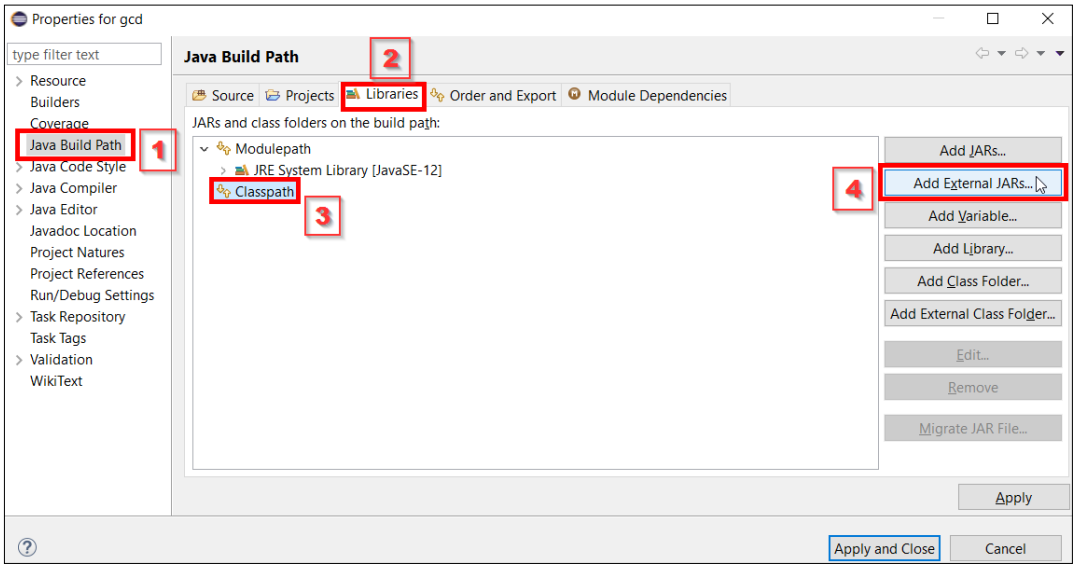
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Roll No : CS022075

7) Adding **"Selenium *Server Driver* and *Client Driver*(JAR files)"** in Eclipse IDE:

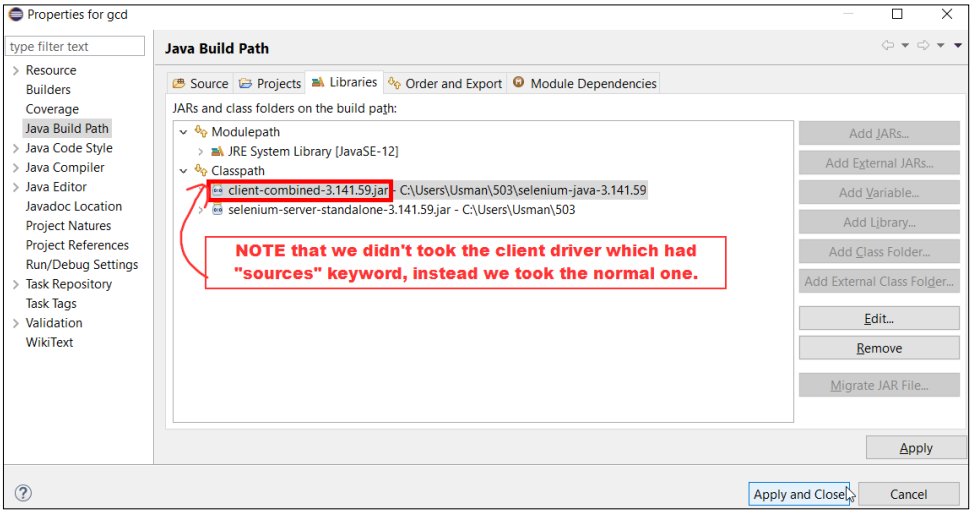
- **right-click on Project Name > Build Path > Configure Build Path…**



- now go under: **Java Build Path > Libraries > Classpath >** click **Add External JARs…**

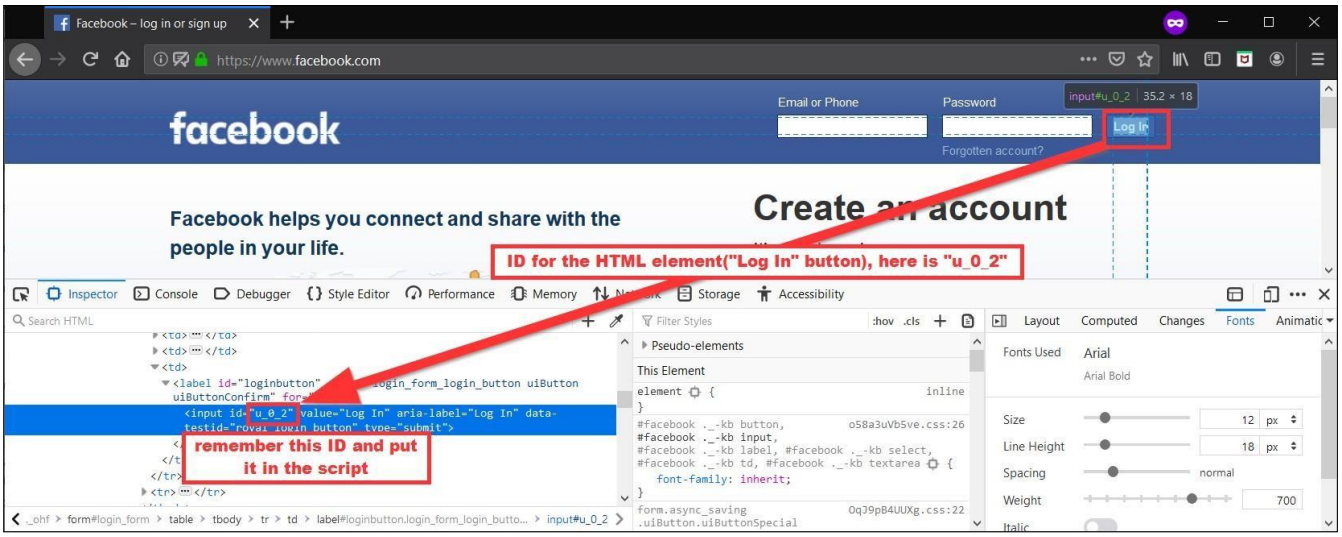


- Browse and **add JAR files >** click **Apply and Close :**



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8) Fetching ID's of the HTML attributes of the elements from the facebook webpage.
For instance we looking for the ID behind "Log In" button HTML element
(right-click over element(button/textbox) **>** click **Inspect Element):**
(do the same for EMAIL TEXTBOX & PASSWORD TEXTBOX)



9) Creating the ***script*** *in JAVA*:
(NOTE that this **script** will be run by Eclipse IDE)
(In simple words, it's like we are
-ordering Eclipse to run a script or to do a job
-of opening the browser, visiting the URL
-and putting credentials in textboxes and clicking button with the help of Selenium Drivers
-and to show the result.
-Hence automating the work in browser for logging in)

- Now we'll put the path of "geckodriver" in a String ***driverPath***
- And in here, we'll **put the ID's in the *.id()* method respectively**.

---(FB_login.java)---

```
package p7;

import   org.openqa.selenium.By;
import org.openqa.selenium.Keys;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.*;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.firefox.FirefoxProfile;
import org.openqa.selenium.firefox.internal.ProfilesIni;

public class FB_login {
      static String driverPath = "C:\\Users\\Usman\\503\\geckodriver.exe";

    public static void main(String[] args) {
       System.setProperty("webdriver.gecko.driver", driverPath);
             //DesiredCapabilities capabilities = DesiredCapabilities.firefox();
             //capabilities.setCapability("marionette",true);
             ProfilesIni allProfiles = new ProfilesIni();
             FirefoxProfile fp = new FirefoxProfile();
```

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```
            fp.setPreference(FirefoxProfile.PORT_PREFERENCE,"7055");
            FirefoxOptions options = new FirefoxOptions();
            options.setProfile(fp);

            //objects and variables instantiation
        WebDriver driver = new FirefoxDriver(options);
        String appUrl = "https://www.facebook.com/";

        //launch the firefox browser and open the application url
        driver.get(appUrl);

        //maximize the browser window
        driver.manage().window().maximize();

        //declare and initialize the variable to store the expected title of the
webpage.
        String expectedTitle = "Facebook – log in or sign up";

        //fetch the title of the web page and save it into a string variable
        String actualTitle = driver.getTitle();

        //compare the expected title of the page with the actual title of the page
and print the result
        if (expectedTitle.equals(actualTitle)) {
            System.out.println("Verification Successful - The correct title is
displayed on the web page.");
            }
        else {
            System.out.println("Verification Failed - An incorrect title is
displayed on the web page.");
            }

        //enter a valid username in the email textbox
        WebElement username = driver.findElement(By.id("email"));
        username.clear();
        username.sendKeys("your email id");

        //enter a valid password in the password textbox
        WebElement password = driver.findElement(By.id("pass"));
        password.clear();
        password.sendKeys("your password");

        password.sendKeys(Keys.ENTER);

        //click on the Sign in button
        WebElement LogInButton = driver.findElement(By.id("u_0_2"));
        LogInButton.click();

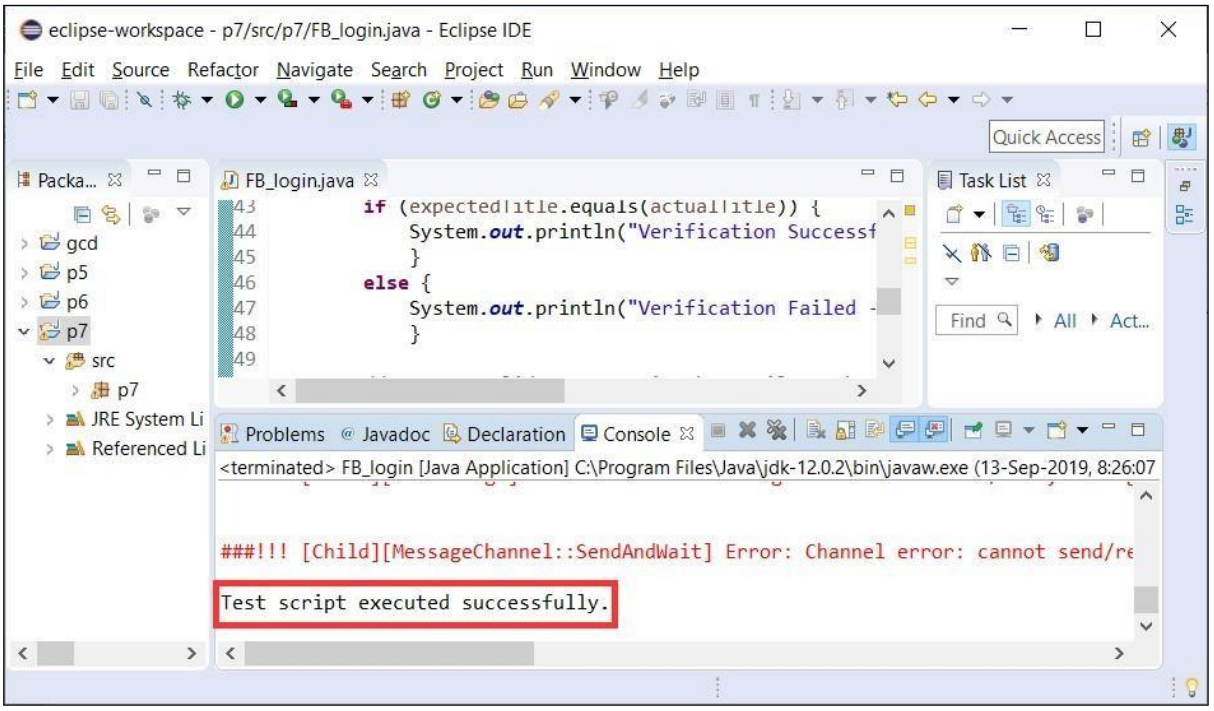
        //close the web browser
        driver.close();
        System.out.println("Test script executed successfully.");

        //terminate the program
        System.exit(0);
    }
}
```

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10) Run the file from Eclipse IDE:

- **OUTPUT:**



11) Finish!

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PRACTICAL NO.8

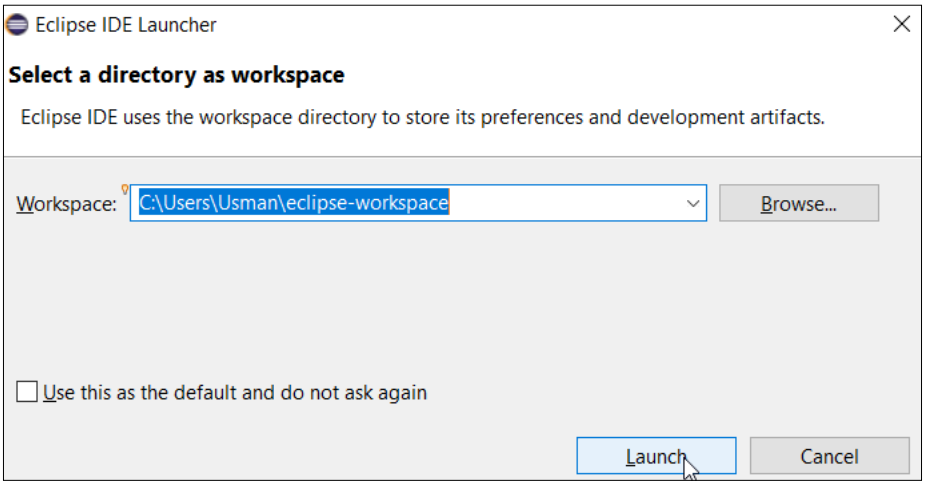
AIM : Write a program using Selenium WebDriver to get the number of items in a list or combo box on a web page. Perform element identification and counting.

PRE-REQUISITES:

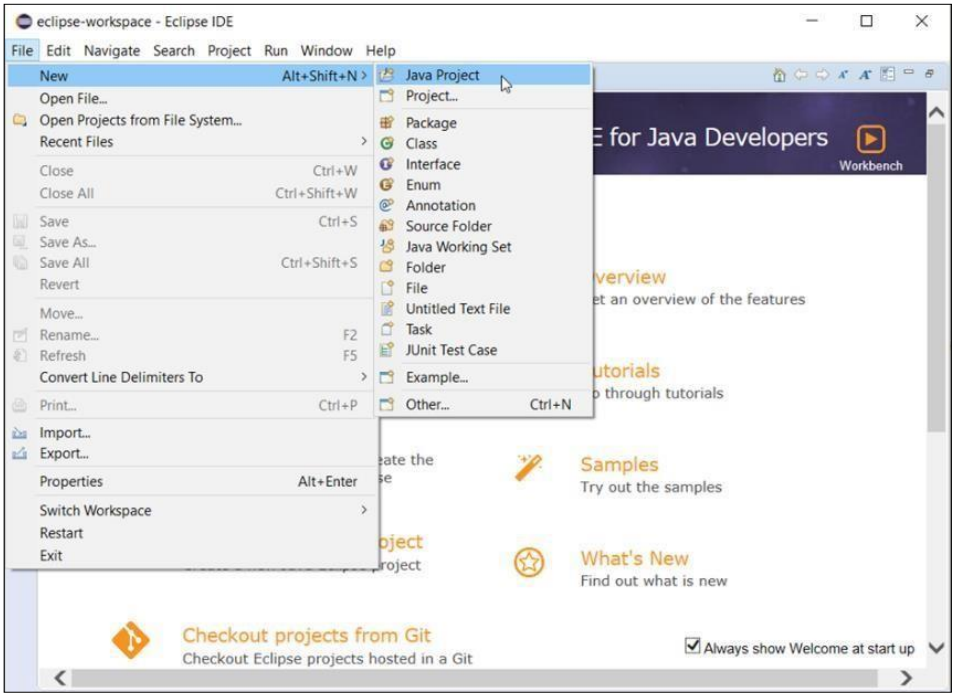
1) Check that you have **JDK**.
2) Check that you have **Eclipse IDE**.
3) Check that you have **Selenium *Server Driver* and *Client Driver*(JAR files)**.
4) Check that you have **Gecko Driver**.
5) Check that you have a stable Internet connection.

STEPS:

1) **Open Eclipse**. Select your workspace directory. Click **Launch:**

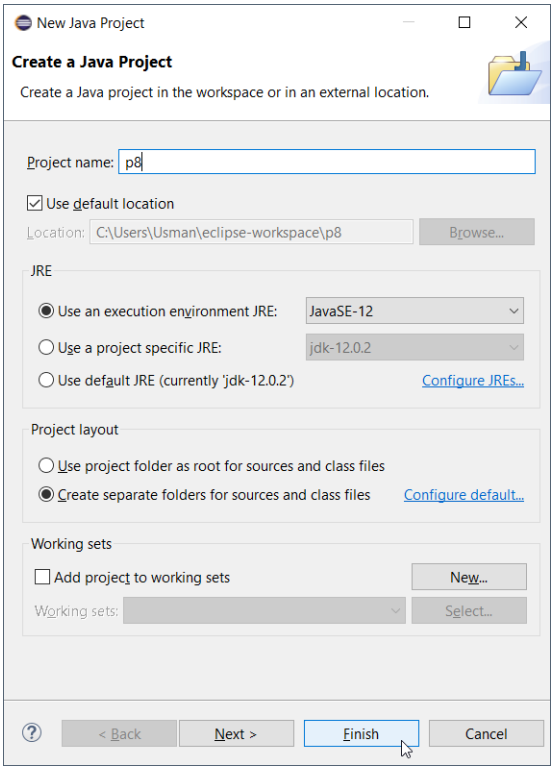


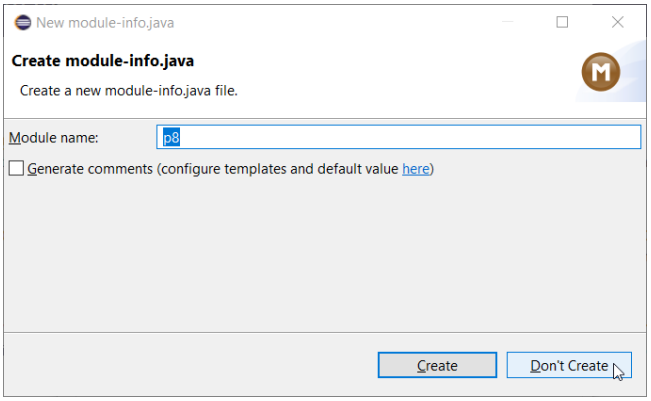
2) **Create** a Project(**File > New > Java Project**):



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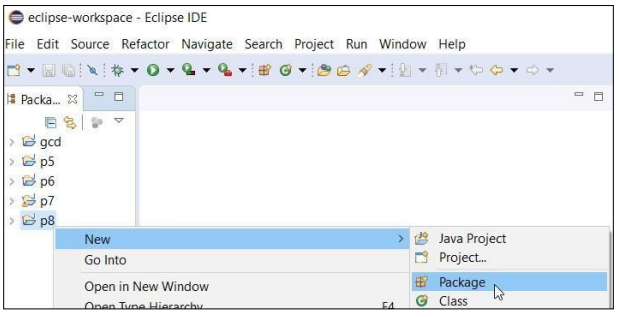
3) **Name the project** as “p8” **>** click **Finish >** click **Don’t Create** module:

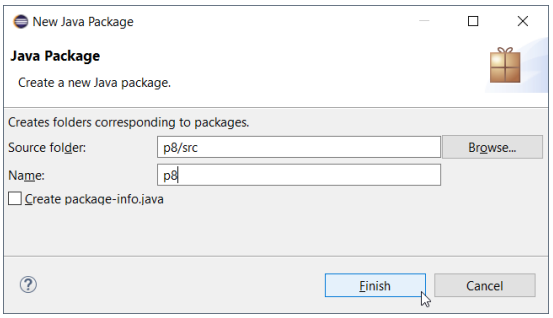




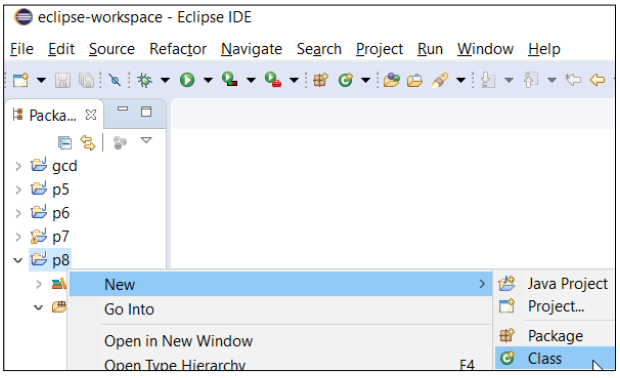
4) **Close the “Welcome” tab**.

5) **Create** a Package(**right-click on Project Name > New > Package > Name** it **> Finish**):





6) **Create** a Class(**right-click on Project Name > New > Class > Name** it **> Finish**):

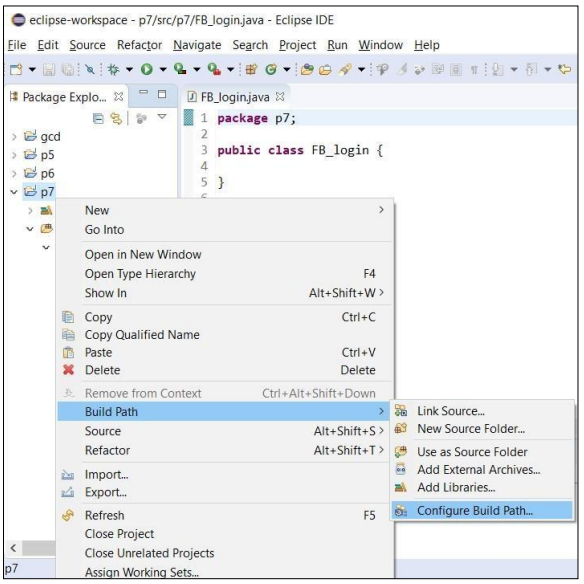




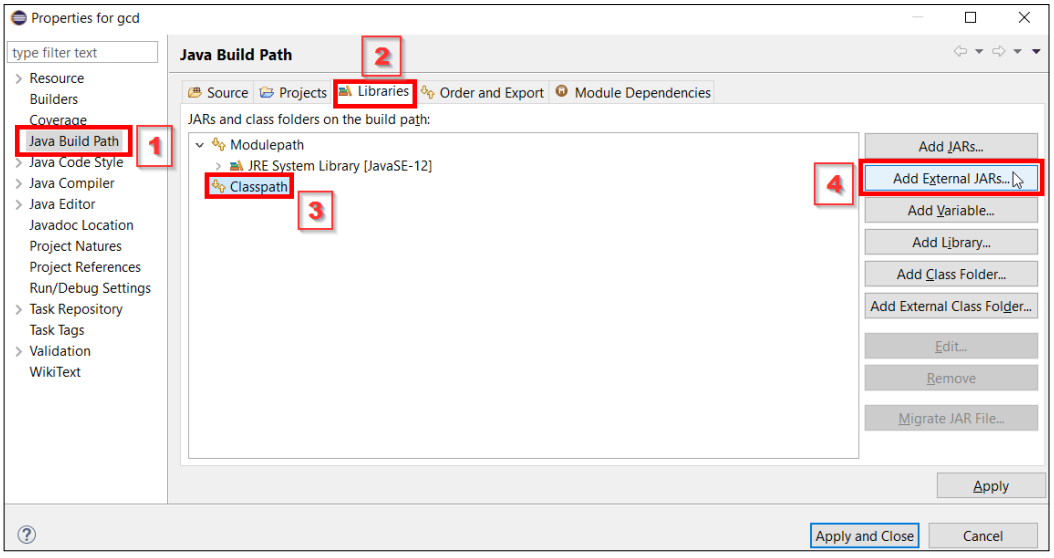
Name: Sujal Singh
Roll No : CS022075

7) Adding **"Selenium *Server Driver* and *Client Driver*(JAR files)"** in Eclipse IDE:

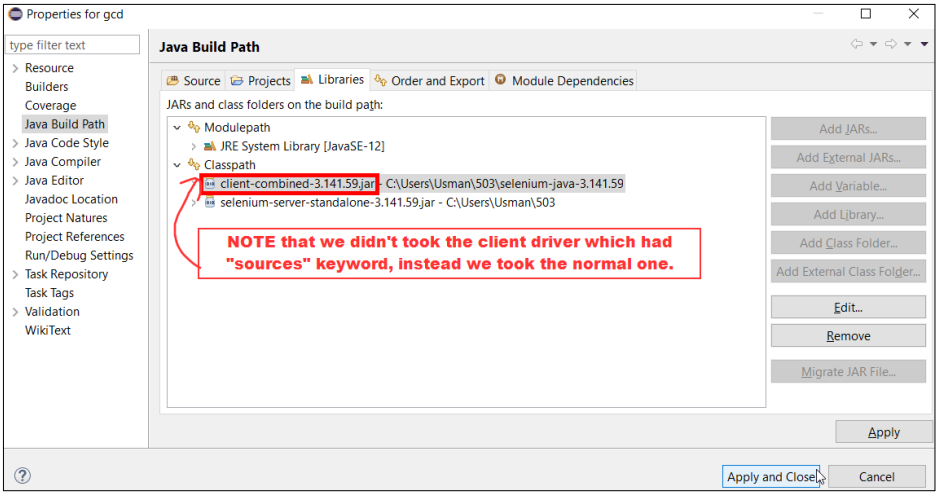
- **right-click on Project Name > Build Path > Configure Build Path…**



- now go under: **Java Build Path > Libraries > Classpath >** click **Add External JARs…**



- Browse and **add JAR files >** click **Apply and Close :**



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Roll No : CS022075

8) Creating the ***script*** *in JAVA*:

(NOTE that this **script** will be run by Eclipse IDE)
(In simple words, it's like we are
-ordering Eclipse to run a script or to do a job
-of opening the browser, visiting the URL
-and finding the <a> tag WebElements with the help of Selenium Drivers
-and to show the result.
-Hence automating the work in browser)

- Now we'll put the path of "geckodriver" in a String ***driverPath***

---(FindAllLinks.java)---

```
package p8;

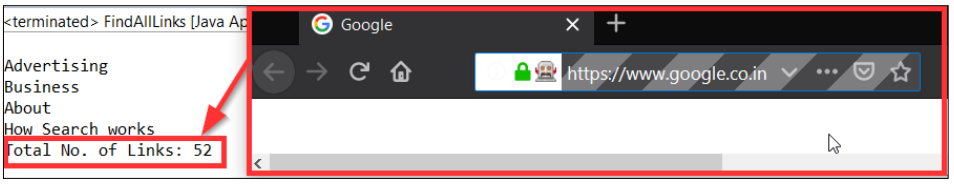
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.*;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.firefox.FirefoxProfile;
import org.openqa.selenium.firefox.internal.ProfilesIni;

public class FindAllLinks {
	static String driverPath = "C:\\Users\\Usman\\503\\geckodriver.exe";
	public static void main(String[] args) {
		System.setProperty("webdriver.gecko.driver",driverPath);
		//NOTE THAT: following commented lines are required for old machines
		//DesiredCapabilities capabilities = DesiredCapabilities.firefox();
		//capabilities.setCapability("marionette",true);
		//ProfilesIni allProfiles = new ProfilesIni();
		//FirefoxProfile fp = new FirefoxProfile();
		//fp.setPreference(FirefoxProfile.PORT_PREFERENCE,"7055");
		//FirefoxOptions options = new FirefoxOptions();
		//options.setProfile(fp);
		WebDriver driver = new FirefoxDriver();
		String appUrl ="https://www.google.co.in/";
		driver.get(appUrl);

		java.util.List<WebElement> links =
driver.findElements(By.tagName("a"));

		for (int i = 1; i<links.size(); i=i+1)
		{
			System.out.println(links.get(i).getText());
		}
		System.out.println("Total No. of Links: "+links.size());
		//driver.quit();
	}
}
```

9) **Run the file from Eclipse IDE: OUTPUT:**



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PRACTICALNO.9

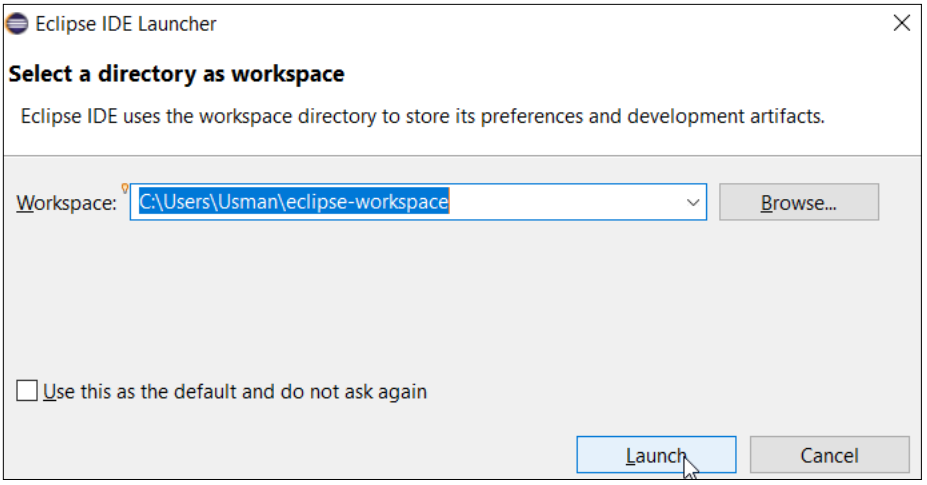
AIM : Write a program using Selenium WebDriver to count the number of checkboxes on a web page, including checked and unchecked counts. Perform checkbox identification and counting.

PRE-REQUISITES:

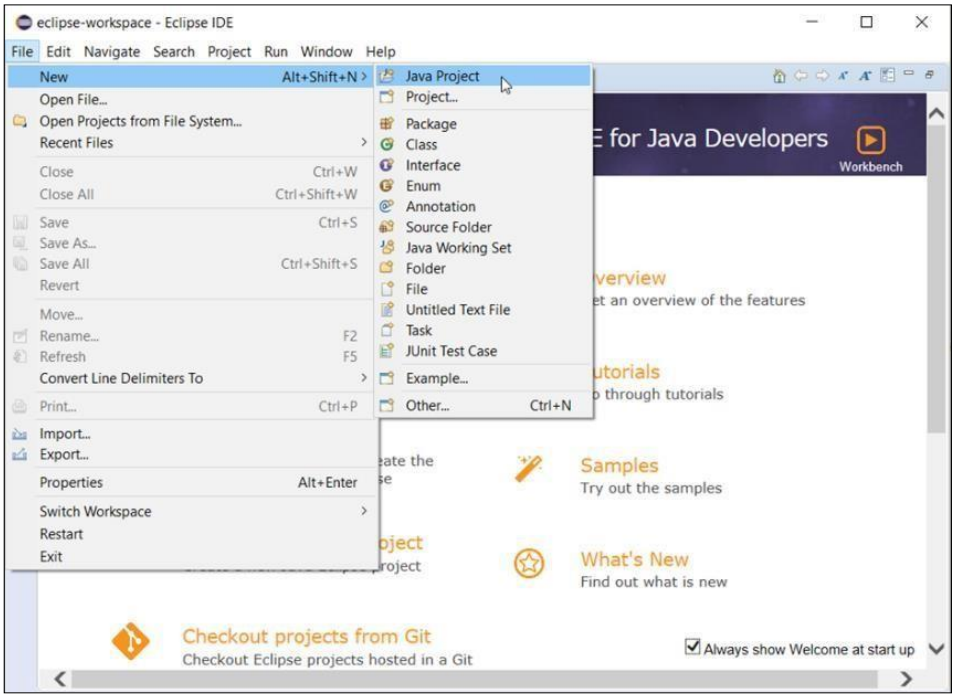
1) Check that you have **JDK**.
2) Check that you have **Eclipse IDE**.
3) Check that you have **Selenium *Server Driver* and *Client Driver*(JAR files)**.
4) Check that you have **Gecko Driver**.
5) Check that you have a stable Internet connection.

STEPS:

1) **Open Eclipse**. Select your workspace directory. Click **Launch:**

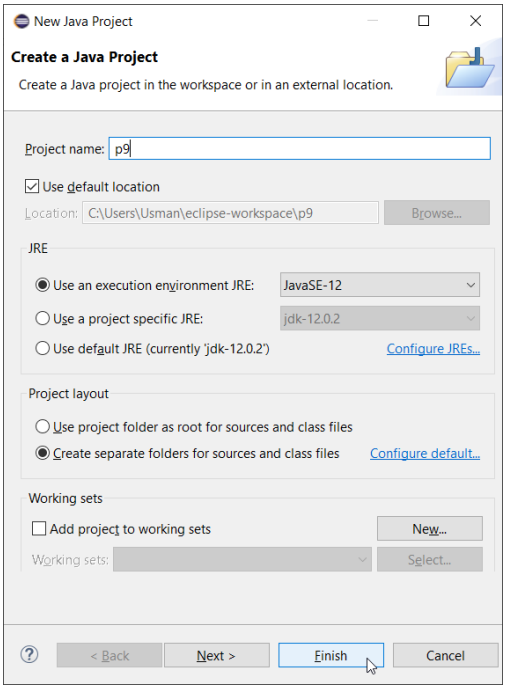


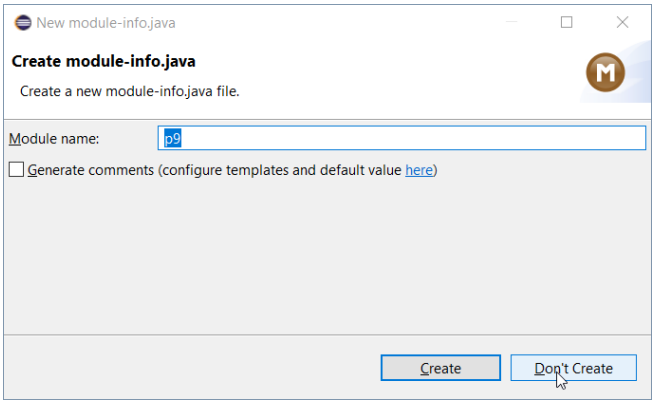
2) **Create** a Project(**File > New > Java Project**):



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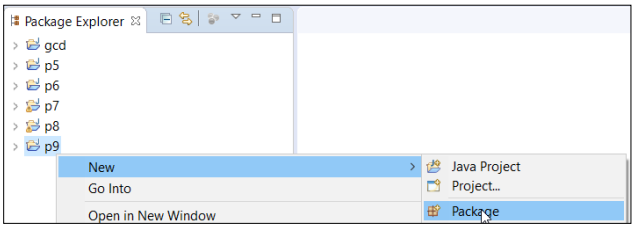
3) **Name the project** as “p9” **>** click **Finish >** click **Don’t Create** module:

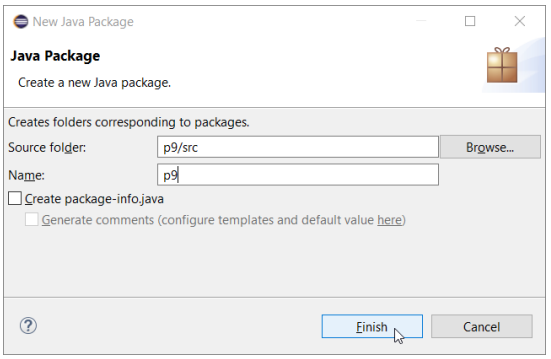




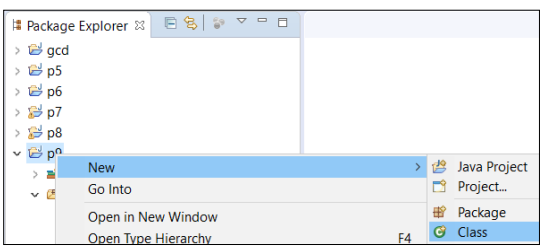
4) **Close the “Welcome” tab**.

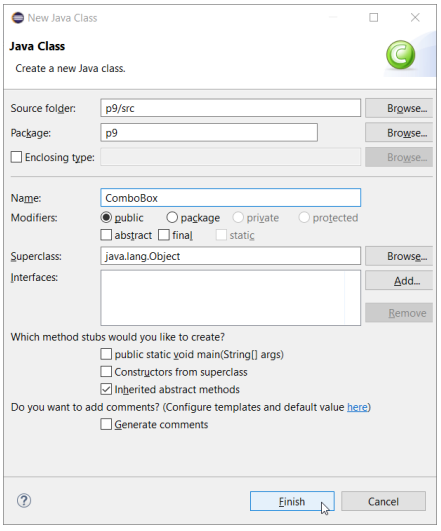
5) **Create** a Package(**right-click on Project Name > New > Package > Name** it **> Finish**):





6) **Create** a Class(**right-click on Project Name > New > Class > Name** it **> Finish**):

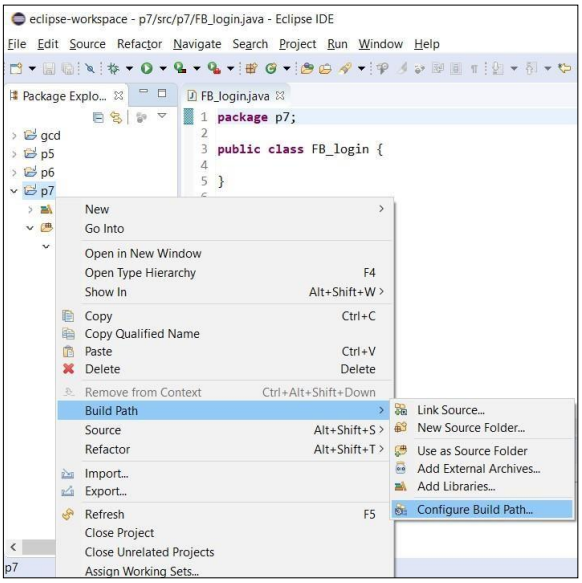




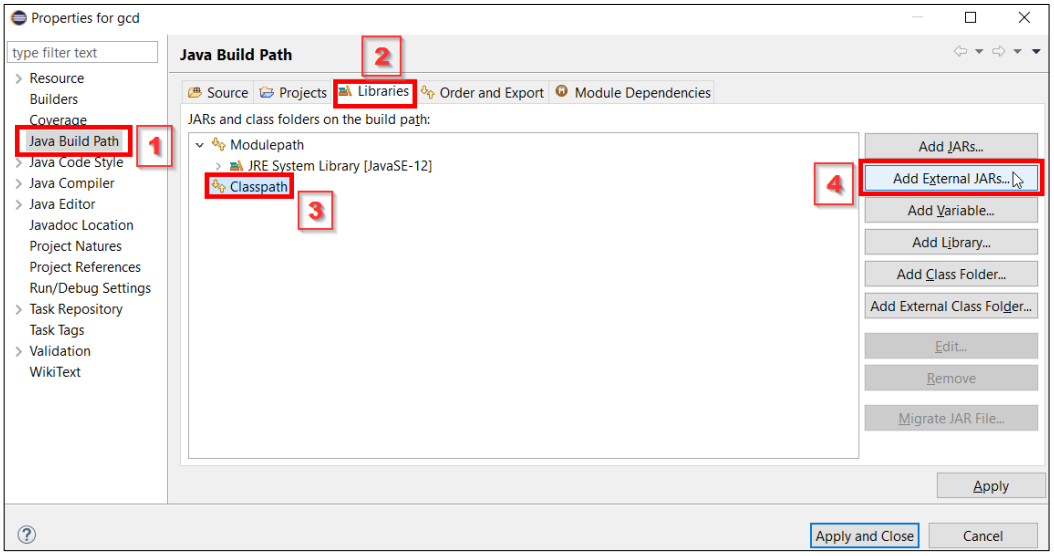
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Roll No : CS022075

7) Adding **"Selenium *Server Driver* and *Client Driver*(JAR files)"** in Eclipse IDE:

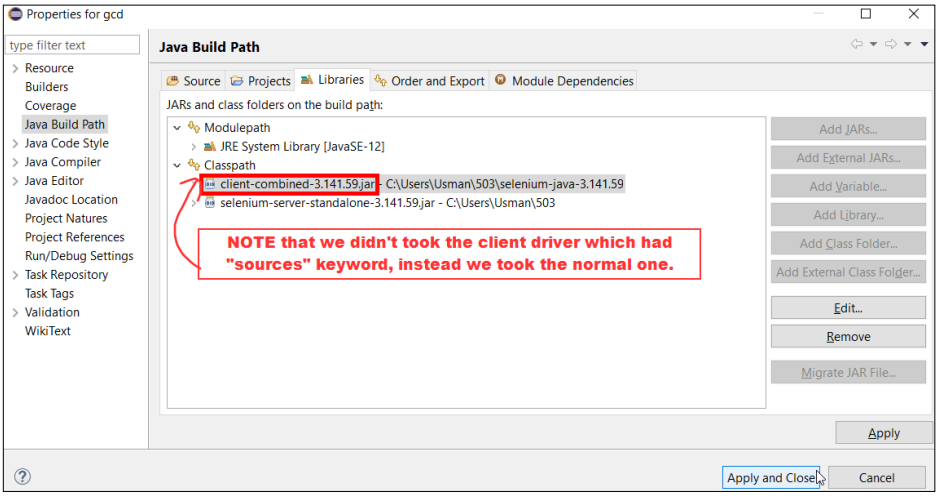
- **right-click on Project Name > Build Path > Configure Build Path…**



- now go under: **Java Build Path > Libraries > Classpath >** click **Add External JARs…**



- Browse and **add JAR files >** click **Apply and Close :**

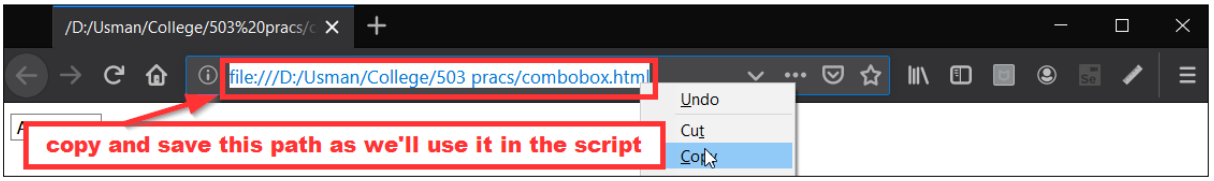


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Roll No : CS022075

8) **Create** HTML file in a notepad **> Save it > Open it in browser > Copy the URL:**
(NOTE THAT: as this is our LOCAL FILE, this file will be opened thru STATIC URL in script)

---(combobox.html)---

```
<select id="continents">
        <option value="Asia">Asia</option>
        <option value="Europe">Europe</option>
        <option  value="Africa">Africa</option>
</select>
```



9) Creating the ***script*** *in JAVA*:
(NOTE that this **script** will be run by Eclipse IDE)
(In simple words, it's like we are
-ordering Eclipse to run a script or to do a job
-of opening the browser, visiting the URL
-and finding the WebElement By "ID" with the help of "Select" class and Selenium Drivers
-and to show the result.
-Hence automating the work in browser)

- Now we'll put the path of our LOCAL FILE(combobox.html) in a string
- Also put the path of "geckodriver" in a String ***driverPath***

---(ComboBox.java)---

```
package p9;

import java.util.List;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.*;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.firefox.FirefoxProfile;
import org.openqa.selenium.firefox.internal.ProfilesIni;
import org.openqa.selenium.support.ui.Select;

public class ComboBox {
	static String driverPath = "C:\\Users\\Usman\\503\\geckodriver.exe";
	public static void main(String[] args) {
	    System.setProperty("webdriver.gecko.driver",driverPath);
	    //NOTE THAT: following commented lines are required for old machines
	    //DesiredCapabilities capabilities = DesiredCapabilities.firefox();
	      //capabilities.setCapability("marionette",true);
	      //ProfilesIni allProfiles = new ProfilesIni();
	      //FirefoxProfile fp = new FirefoxProfile();
	      //fp.setPreference(FirefoxProfile.PORT_PREFERENCE,"7055");
	      //FirefoxOptions options = new FirefoxOptions();
```

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```
            //options.setProfile(fp);
            WebDriver driver = new FirefoxDriver();
            String appUrl ="https://www.toolsqa.com/automation-practice-form/";
//DYNAMIC URL(WEBSITE)
            //String appUrl = "
file:///D:/Usman/College/503%20pracs/combobox.html"; //STATIC URL(LOCAL FILE)
            driver.get(appUrl);

            Select oSelect = new
Select(driver.findElement(By.id("continents")));
            //Select oSelect = new
Select(driver.findElement(By.tagName("select"))); //this works too

            List<WebElement> oSize = oSelect.getOptions();
            int iListSize = oSize.size();

            for(int i =0; i < iListSize ; i++)
            {
                  // Storing the value of the option
                  String sValue = oSelect.getOptions().get(i).getText();
                  // Printing the stored value
                  System.out.println(sValue);
            }
            System.out.println("Total No. Items in Dropdown: "+iListSize);
            //driver.quit();
      }
}
```

10) Run the file from Eclipse IDE:

- **OUTPUT:**

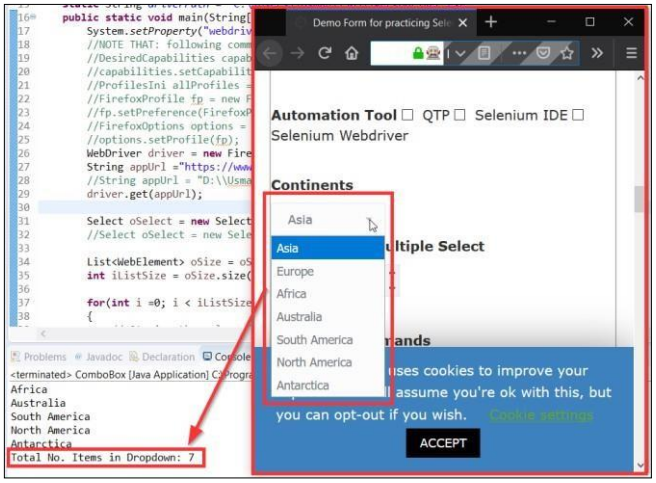


Figure 1: by using the DYNAMIC URL(website)

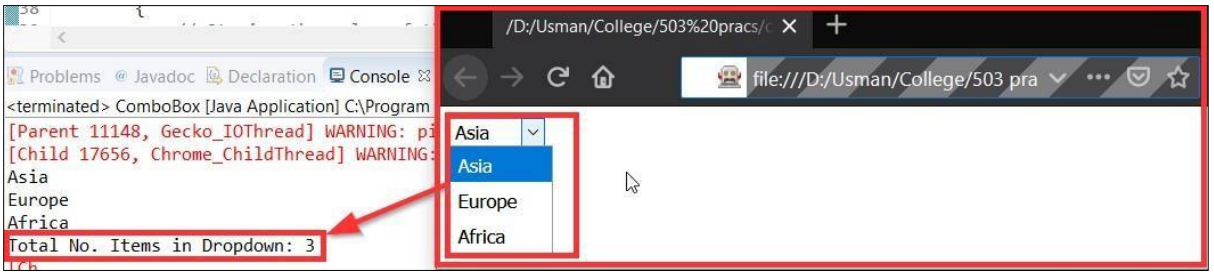


Figure 2: by using the STATIC URL(local file)

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PRACTICAL NO.10

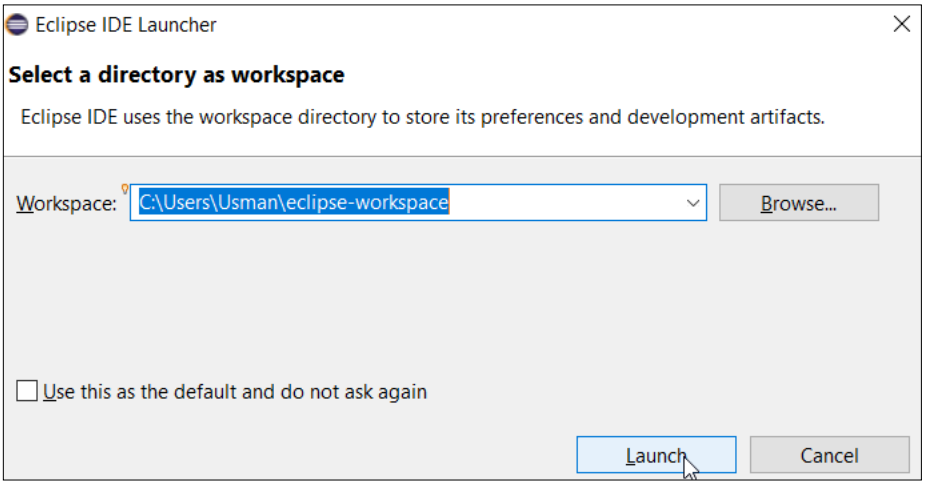
AIM : Perform load testing on a web application using JMeter. Generate and analyze load scenarios. Additionally, explore bug tracking using Bugzilla as a tool for logging and tracking software defects.

PRE-REQUISITES:

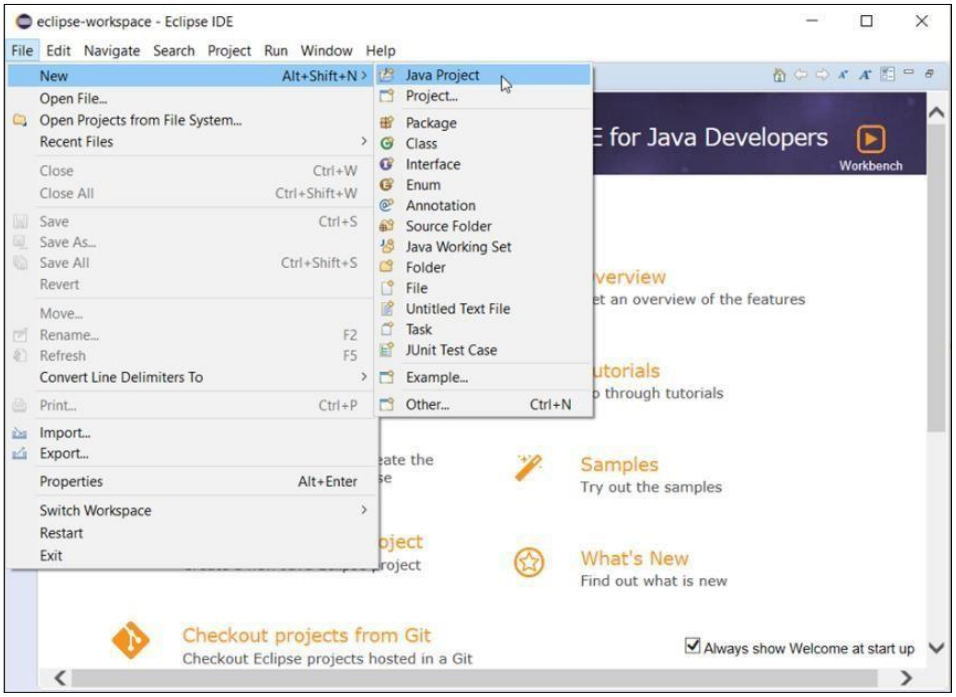
1) Check that you have **JDK**.
2) Check that you have **Eclipse IDE**.
3) Check that you have **Selenium *Server Driver* and *Client Driver*(JAR files)**.
4) Check that you have **Gecko Driver**.
5) Check that you have a stable Internet connection.

STEPS:

1) **Open Eclipse**. Select your workspace directory. Click **Launch:**

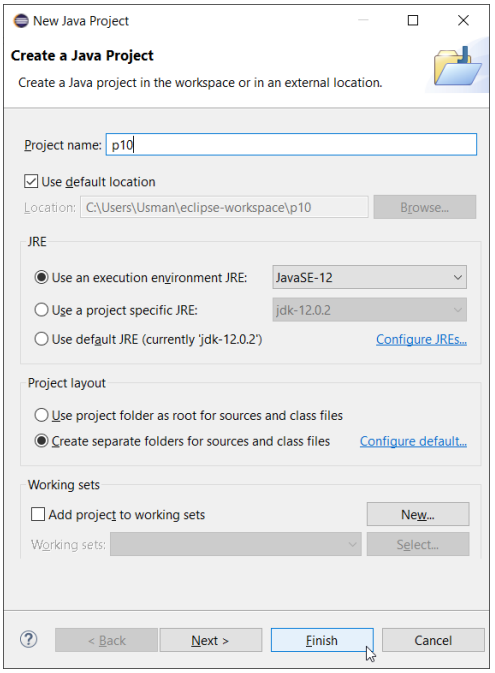


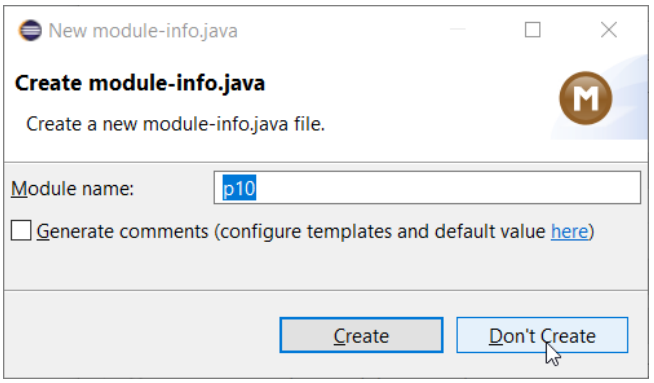
2) **Create** a Project(**File > New > Java Project**):



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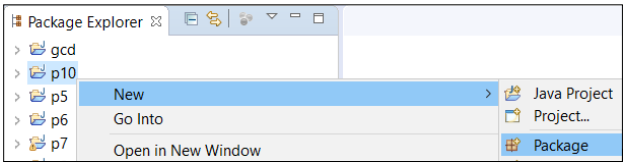
3) **Name the project** as “p10” **>** click **Finish >** click **Don’t Create** module:

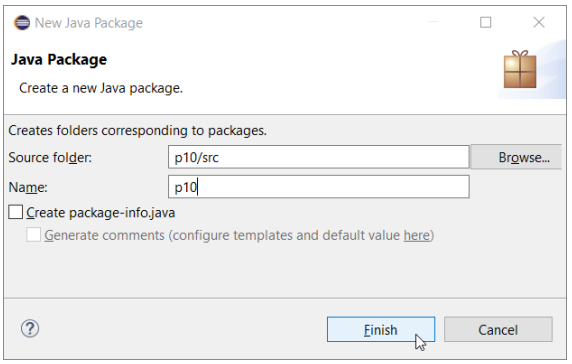




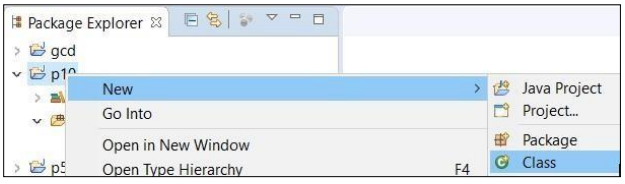
4) **Close the “Welcome” tab**.

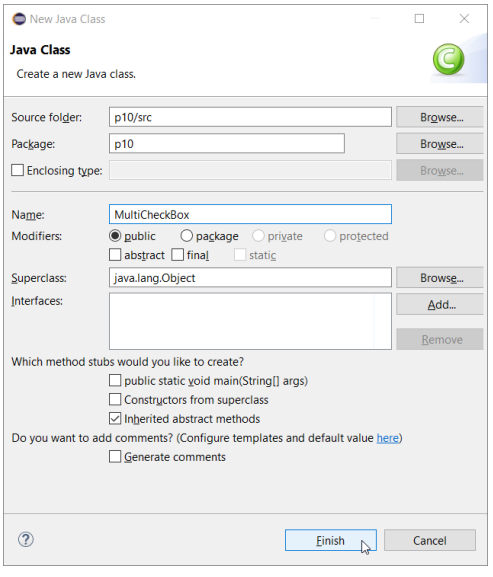
5) **Create** a Package(**right-click on Project Name > New > Package > Name** it **> Finish**):





6) **Create** a Class(**right-click on Project Name > New > Class > Name** it **> Finish**):

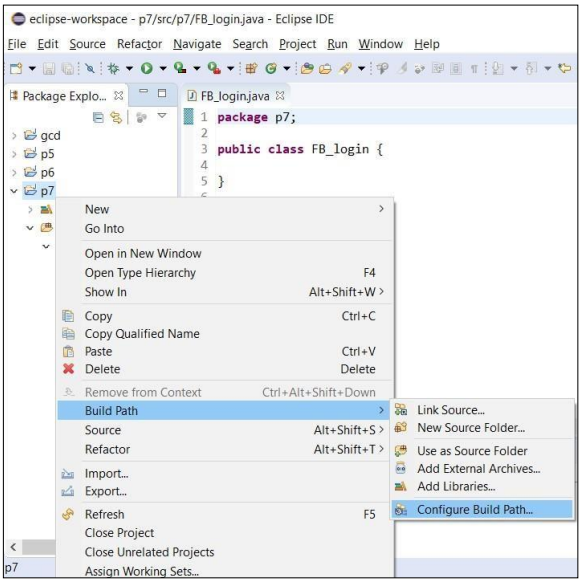




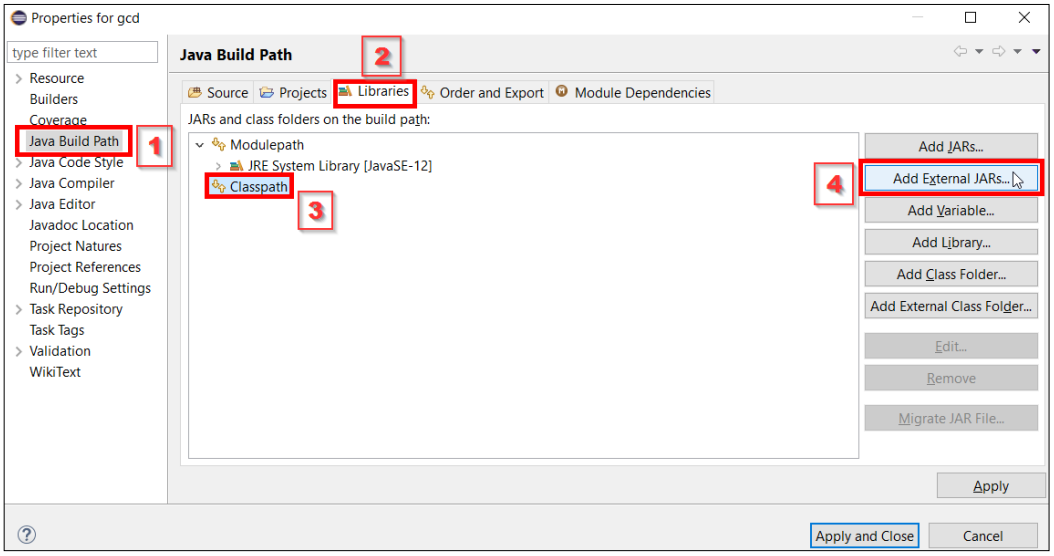
Name: Sujal Singh
Roll No : CS022075

7) Adding **"Selenium *Server Driver* and *Client Driver*(JAR files)"** in Eclipse IDE:

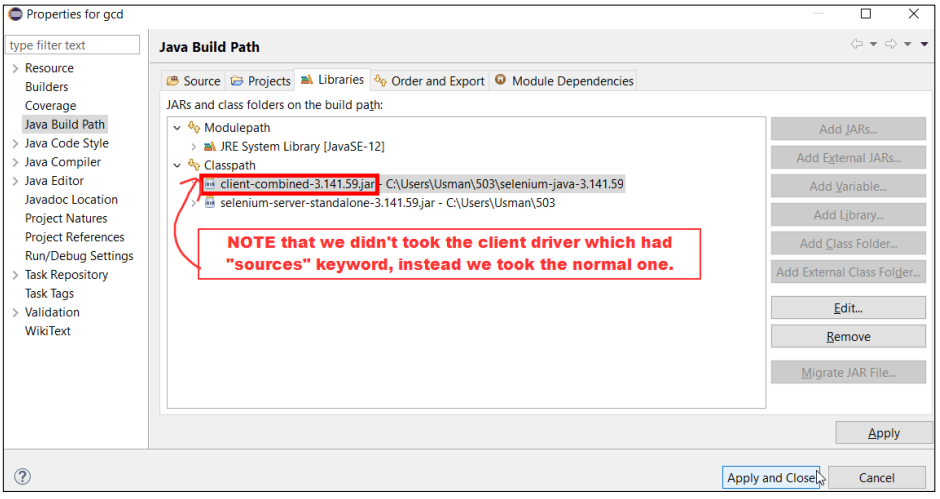
- **right-click on Project Name > Build Path > Configure Build Path…**



- now go under: **Java Build Path > Libraries > Classpath >** click **Add External JARs…**



- Browse and **add JAR files >** click **Apply and Close :**

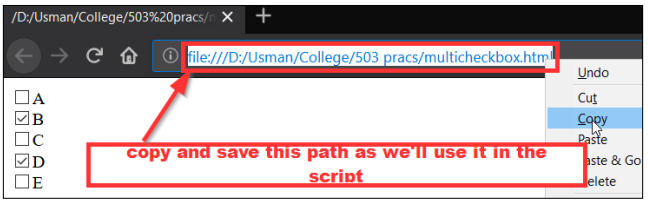


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8) **Create** HTML file in a notepad **> Save it > Open it in browser > Copy the URL:**
(NOTE THAT: as this is our LOCAL FILE, this file will be opened thru STATIC URL in script)

---(multicheckbox.html)---

```
<input type="checkbox" value="A">A<br>
<input type="checkbox" value="B" CHECKED>B<br>
<input type="checkbox" value="C">C<br>
<input type="checkbox" value="D" CHECKED>D<br>
<input type="checkbox" value="E">E<br>
```



9) Creating the ***script*** *in JAVA*:
(NOTE that this **script** will be run by Eclipse IDE)
(In simple words, it's like we are
-ordering Eclipse to run a script or to do a job
-of opening the browser, visiting the URL
-and finding the WebElement By "xpath" with the help of Selenium Drivers
-and to show the result.
-Hence automating the work in browser)

- Now we'll put the path of our LOCAL FILE(multicheckbox.html) in a string
- Also put the path of "geckodriver" in a String ***driverPath***

---(MultiCheckBox.java)---

```
package p10;

import java.util.List;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.*;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.firefox.FirefoxProfile;
import org.openqa.selenium.firefox.internal.ProfilesIni;

public class MultiCheckBox {
	static String driverPath = "C:\\Users\\Usman\\503\\geckodriver.exe";
	public static void main(String[] args) {
		System.setProperty("webdriver.gecko.driver",driverPath);
		//DesiredCapabilities capabilities = DesiredCapabilities.firefox();
		//capabilities.setCapability("marionette",true);
		//ProfilesIni allProfiles = new ProfilesIni();
		//FirefoxProfile fp = new FirefoxProfile();
		//fp.setPreference(FirefoxProfile.PORT_PREFERENCE,"7055");
		//FirefoxOptions options = new FirefoxOptions();
		//options.setProfile(fp);
		WebDriver driver = new FirefoxDriver();
```

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```
            String appUrl = "https://www.toolsqa.com/automation-practice-form/";
//DYNAMIC URL(WEBSITE)
            //String appUrl =
"file:///D:/Usman/College/503%20pracs/multicheckbox.html"; //STATIC URL(LOCAL
FILE)
            driver.get(appUrl);

            List<WebElement> checkBoxes =
driver.findElements(By.xpath("//input[@type='checkbox']"));

            int checkedCount=0, uncheckedCount=0;
            for(int i=0; i<checkBoxes.size(); i++){
                  System.out.println(i+" checkbox is selected
"+checkBoxes.get(i).isSelected());
                  if(checkBoxes.get(i).isSelected()) {checkedCount++;}
                  else {uncheckedCount++;}
            }
            //driver.quit();
            System.out.println("No. of selected checkbox: "+checkedCount);
            System.out.println("No. of unselected checkbox: "+uncheckedCount);
      }
}
```

10) Run the file from Eclipse IDE:

- **OUTPUT:**

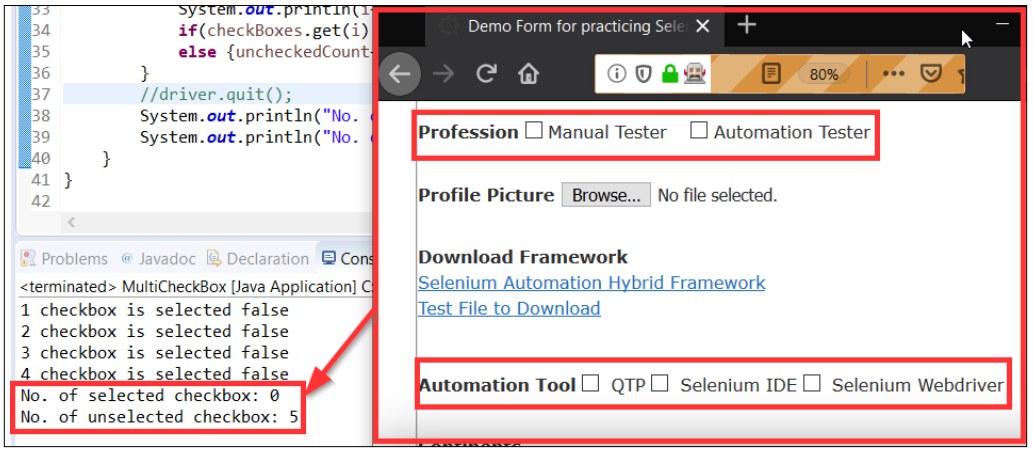


Figure 1: by using the DYNAMIC URL(website)

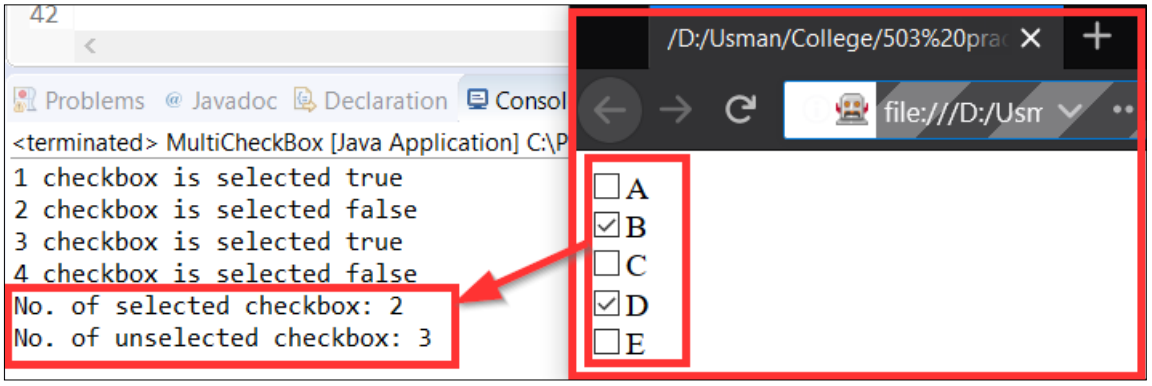


Figure 2: by using the STATIC URL(local file)